SOFTWARE DEFINED RADIO: GENERAL OVERVIEW, AFE, DFE, MIXED-SIGNAL, AND DBB

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OUTLINE

- **PART I:** Software Defined Radio
- **PART II:** Link-power Budget Calculation
- **PART III:** Analog Front-End
- **PART IV:** Mixed-Signal Front-End
- **PART V:** Digital Front-End
- **PART VI:** Digital Base-Band
- **PART VII:** SDR Hardware Platforms
- **PART VIII:** SDR Software Platforms

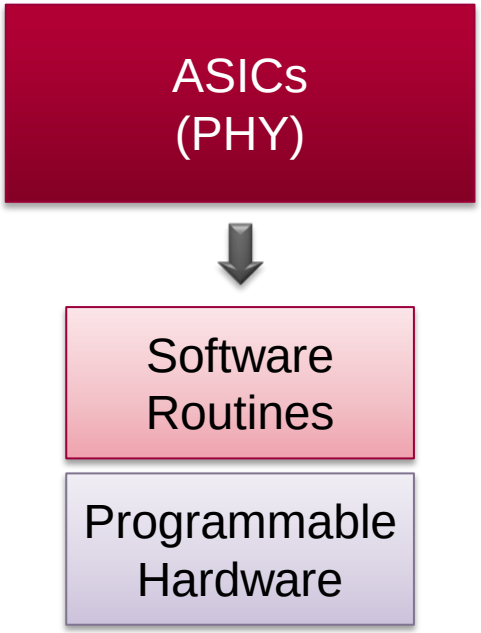
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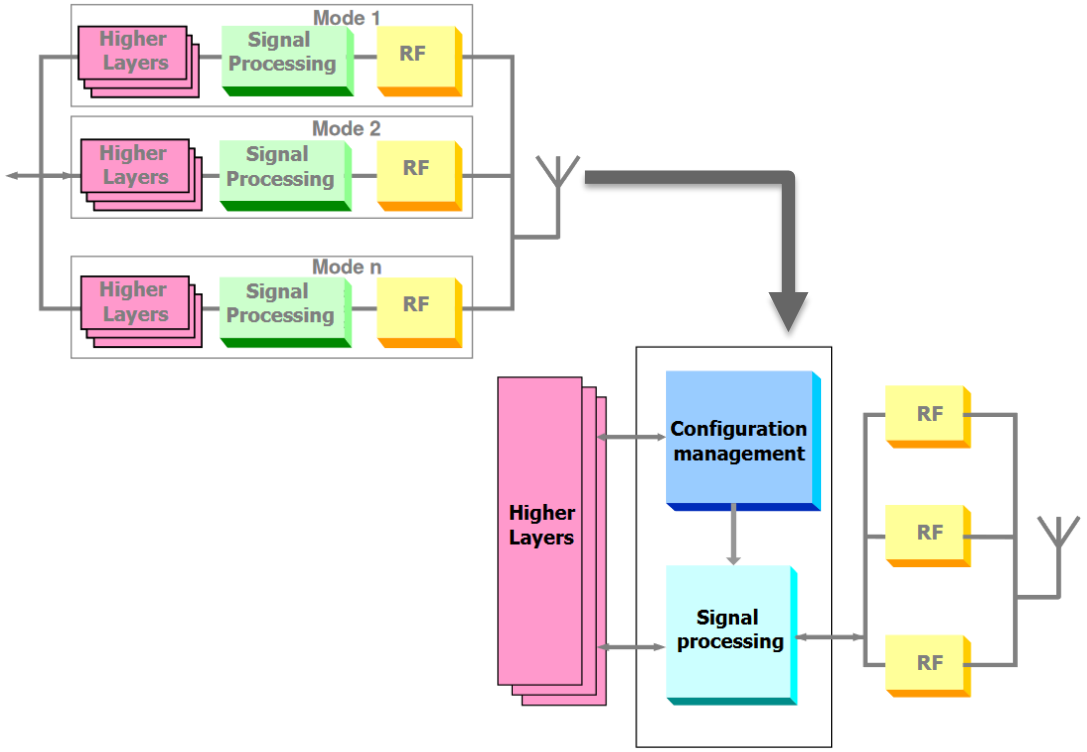
PART I:

SOFTWARE DEFINED RADIO

SOFTWARE DEFINED RADIO (SDR)

- Use software routines instead of ASICs for the physical layer operations of wireless communication system



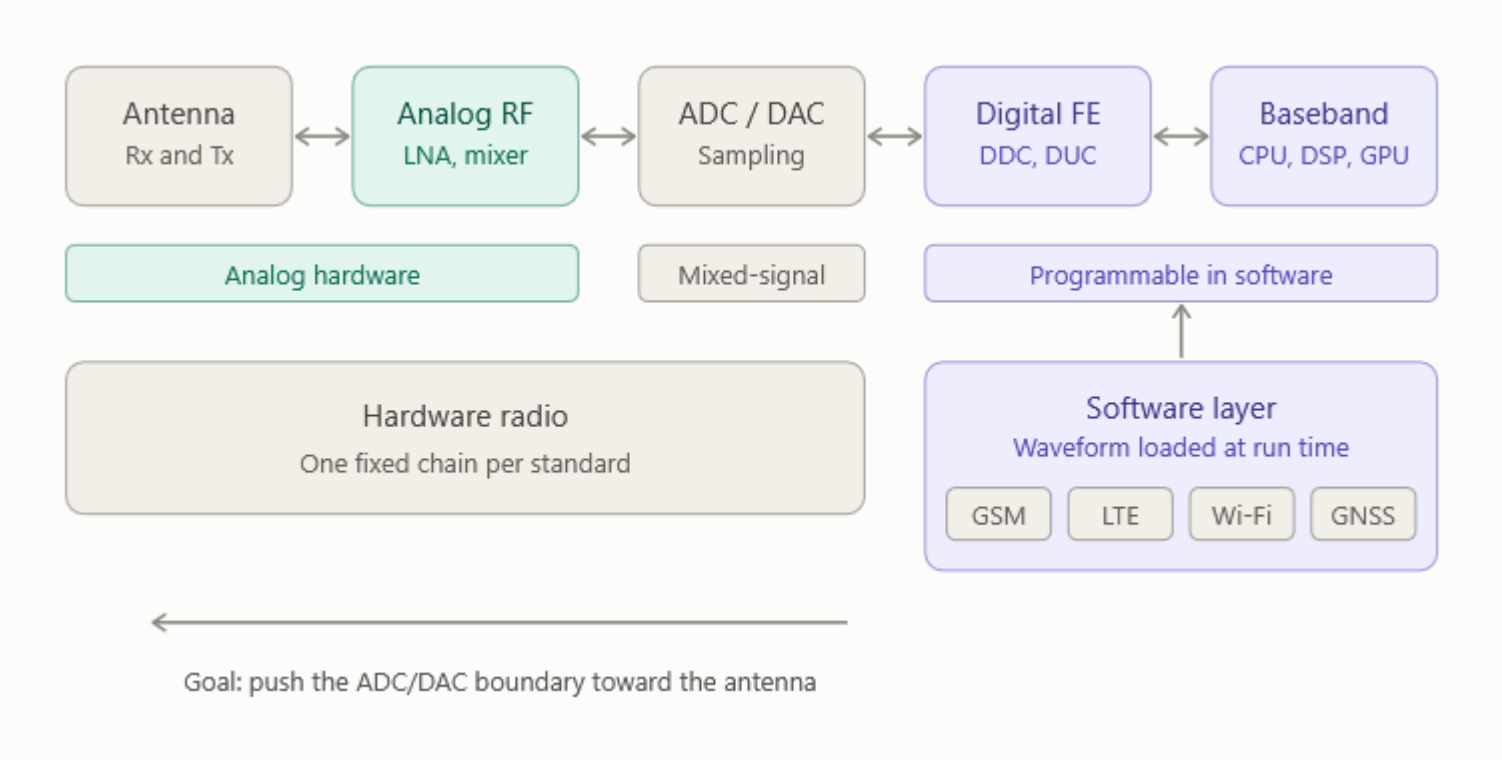


HDR C. Moy

- Analog Front-End (AFE), Digital Front-End (DFE), and Digital Base-Band (DBB) are the scope of SDR.

SDR

According to the SDR concept, “the boundary between analog and digital sections of the front-end has to be pushed as much as possible towards the antenna, to minimize the presence of analog components.”



<source: http://www.sdrforum.org>

WHY & WHERE?

Why we need SDR:

- Widely different wireless protocols: GSM, W-CDMA, LTE, IEEE 802.11b IEEE 802.11a/g/n, WiMAX ….
- Needs a terminal that can support multiple wireless protocols
- Easy infrastructure upgrade
- Wireless protocols evolve continuously
  - "3GPP (the standards body responsible for specifying the LTE cellular standard) expects to release major updates to the basic LTE standard once every **18 months**, and minor updates **continuous**" *
- Time to market == >Reduce hardware development time and cost

Where can we use SDR:

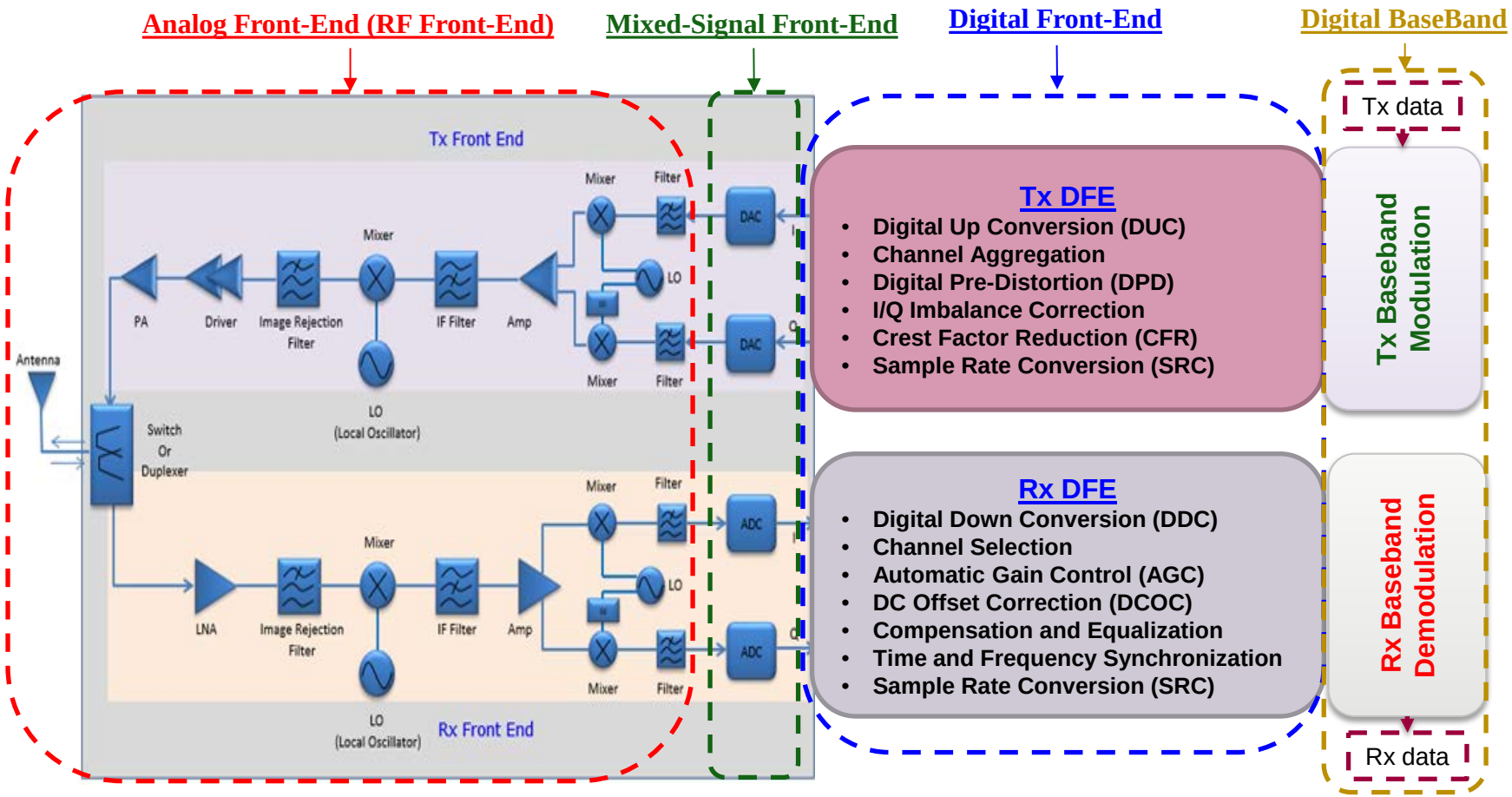
- **Base stations**: weak constraint on power and area, support several hundred subscribers
- **Wireless terminals**: strong constraint on power and area



* M. Bansal and *al.*, "Openradio: a programmable wireless dataplane," in Proceedings of the first workshop on Hot topics in software defined networks, NY, 2012, pp. 109–114

GENERAL TRANSCEIVER ARCHITECTURE

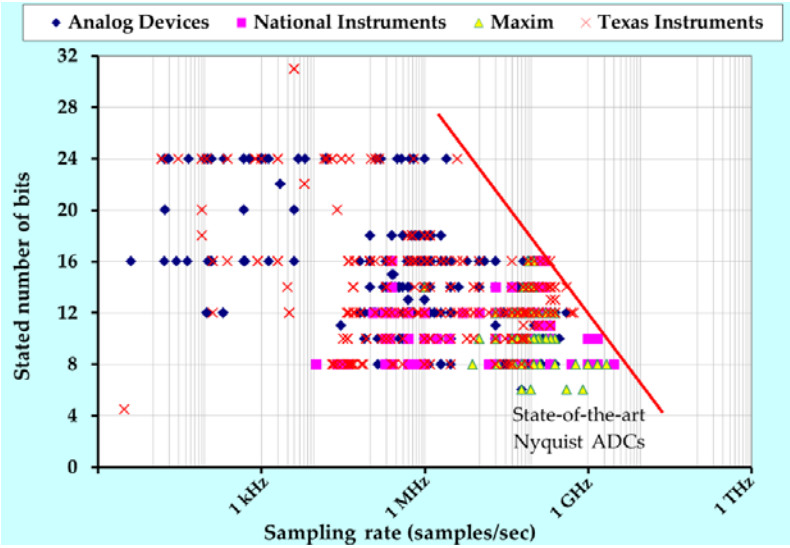
Main parts:



WHY SDR IS CHALLENGING ?

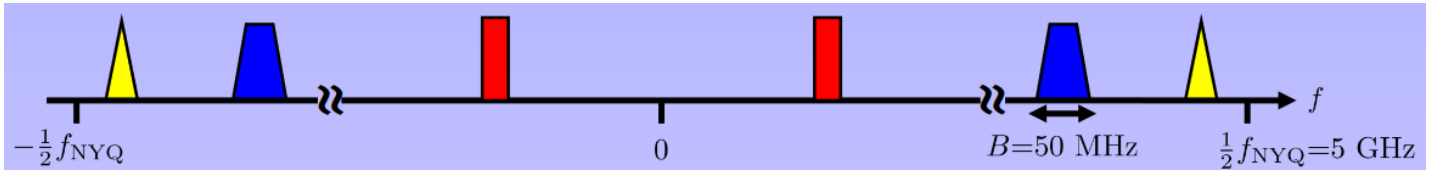
Analog Front-End (AFE):

- Must be tunable across a range of carrier frequencies and bandwidth
- Achieving adequate dynamic range to accommodate high power background signal to low power desired signals
- Amplification of the desired signal to the range of analog-to-digital converter with minimal distortion



Mixed-Signal Front-End (M-SFE)

- Increase sampling rate (Gsamples/s)
- Increase front-end bandwidth
- Increase (effective) number of bits
- Low power consumption
- Exploit the compressive sensing theory + low speed ADCs



WHY SDR IS CHALLENGING ?

Digital Front-End (DFE):

- Multi-rate signal processing ➔ variable/fractional sampling rate
- Multimode Rx with same ADCs
- Increase sampling rate before DAC to relax interpolation filter
- Separation of all channels from a certain frequency band using frequency domain filtering, polyphase FFT,…
- Reconfigurable digital IF-to-BaseBand conversion

Digital BaseBand (DBB)

- Super computer level computation power
- Smart algorithms
- Tight power budget
- High level of programmability:
  - Hardware: heterogeneous platforms (GPU, GPP, FPGA), parameterization, common operators, FPGA dynamic partial reconfiguration.
  - Signal processing: reconfigurable signal processing algorithms,…

SUMMARY: WHAT IS SDR

- A radio in which some or all of the physical-layer functions are implemented in software running on a microprocessor, and/or on an FPGA
- Digital signal processing algorithms running as real-time software on a CPU and/or FPGA
- Software can be implemented on embedded CPU (e.g. ARM Cortex-M…)
- Joe Mitola first coined the term “software-defined radio” in 1991

- SDR provides:
  - Flexibility
  - Upgradability
  - Reconfigurability
  - Lower Cost



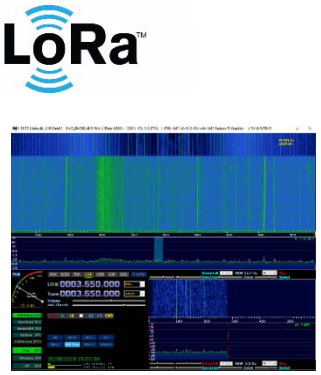
Visite de Joe Mitola à SCEE du 10 au 12 mai 2011

SDR: APPLICATIONS

- Cellular handsets (baseband processors such as Qualcomm Snapdragon,..)
- Cellular basestations (OpenBTS, Osmocom, srsLTE, OpenLTE, Eurecom OAI)
- Cellular protocol stack emulation (GSM, WCDMA, LTE, 5G NR, UE/eNodeB)
- GPS Receivers and Simulators (GNSS-SDR, GPS-SDR-Sim,..)
- Adaptive Radio and Cognitive Radio
- Satellite Communications (Ground Stations)
- Wireless Security Research
- Spectrum Monitoring
- Waveform Prototyping
- Wireless Systems Testing / Wireless Testbeds
- Public safety radio
- Drone Communications, Drone Detection, Drone Defense
- Direction Finding / Angle-of-Arrival
- Radio Astronomy
- MIMO and massive-MIMO Systems

SDR: SOFTWARE TOOLCHAINS

- Processing can be either real-time or off-line / post-processing
- C++ and the USRP Hardware Driver (UHD) API (open-source)
- GNU Radio (Python, NumPy, SciPy, Matplotlib, etc.) (open-source)
- LabVIEW (National Instruments)
- Matlab-Simulink (MathWorks)
- Application-Specific:
  - Cellular: OpenBTS (GSM), OpenBTS-UMTS, OpenLTE, srsLTE, Amarisoft
  - GPS: GNSS-SDR, GPS-SDR-Sim,…
  - Amateur Radio: HDSDR, SDR#, SDR-Console
  - Open-Source LoRa Physical Layer Prototype (https://www.epfl.ch/labs/tcl/wp-content/uploads/2020/02/gr-lora_sdr.zip)
  - DVB-T, DVB-T2, DVB-S, DVB-S2 …



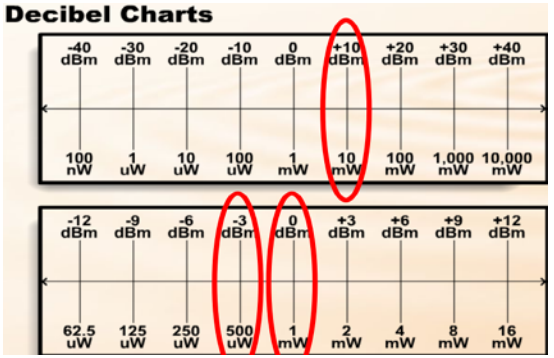
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PART II:

LINK-POWER BUDGET CALCULATION

UNITS IN RF

Units in RF Design



- The voltage gain and power gain are expected in decibels (dB)

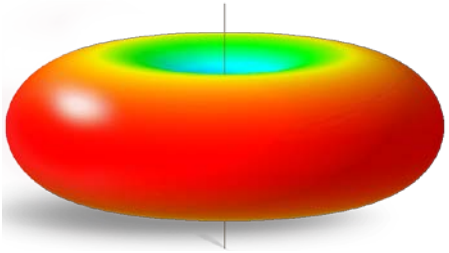
$$A_V|_{\text{dB}} = 20\log\frac{V_{out}}{V_{in}} \qquad A_P|_{\text{dB}} = 10\log\frac{P_{out}}{P_{in}}$$

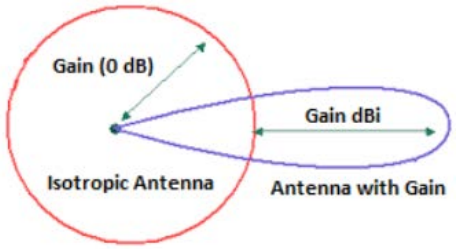
RMS values

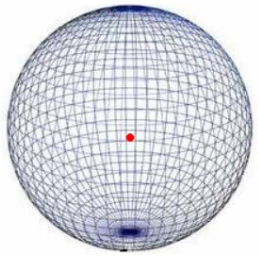
- The absolute signal levels are often expressed in *dBm* rather than in *watts* or *volts*.

$$P_{sig}|_{\text{dBm}} = 10\log\left(\frac{P_{sig}}{1\ \text{mW}}\right)$$

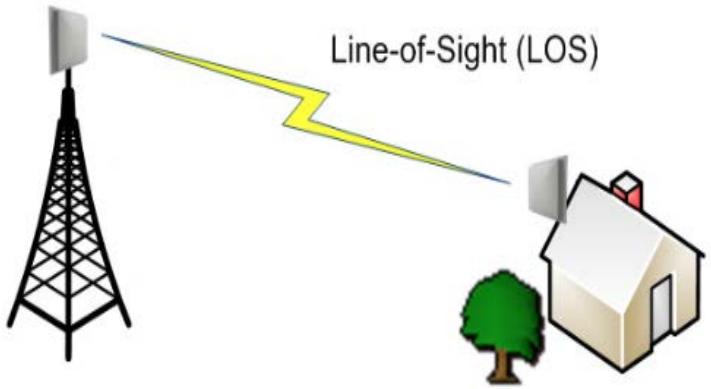
- A dipole antenna gain is 1.64 ➔ $10\log(1.64) = 2.15\ dBi$
- A half-wave dipole antenna has a power gain of 1.64 (2.15 dBi) over an isotropic source.







LINE-OF-SIGHT (LOS) LINK BUDGET



	FSPL (dB)		
Distance	900MHz	2.4GHz	5.8GHz
1km	91.53	100.05	107.72
2km	97.56	106.07	113.74
3km	101.08	109.60	117.26
4km	103.58	112.10	119.76
5km	105.51	114.03	121.70
10km	111.53	120.05	127.72
20km	117.56	126.07	133.74

- The **link-power budget** can be expressed as:

$$\mathrm{P}_{RX}(dBm) = \mathrm{P}_{TX}(dBm) + \mathrm{G}(dB) + \mathrm{L}(dB)$$

- The link-power budget is said to be sufficient if the estimated received power is sufficiently large.
- The amount by which the received power exceeds receiver sensitivity is called the **link margin**
- Losses are mainly due to free-space path loss (FSPL)

$$FSPL = \left(\frac{4\pi df}{C}\right)^2$$

$$L_{fs}(dB) = 32.45 + 20\log(d_{KM}) + 20\log(f_{MHz})$$

$$L_{fs}(dB) = 92.45 + 20\log(d_{KM}) + 20\log(f_{GHz})$$

$$L_{fs}(dB) = 20\log(d_m) + 20\log(f_{Hz}) - 147.55$$

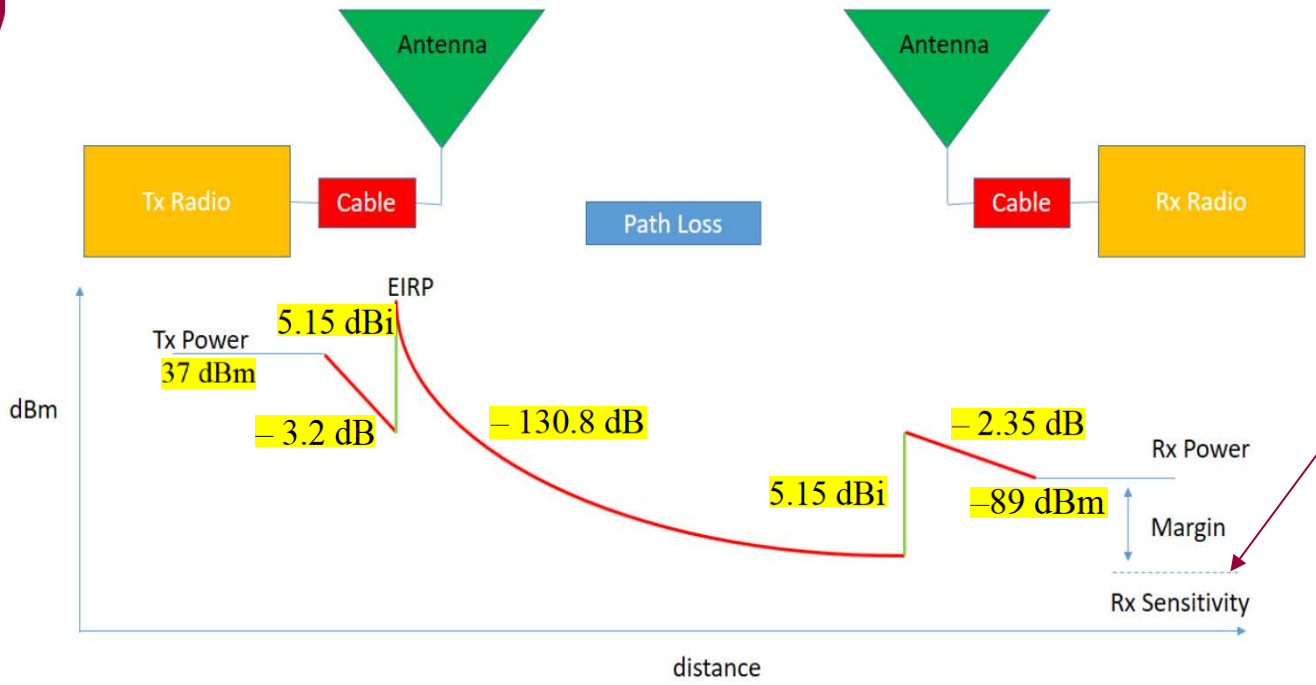
EXAMPLE: WIFI

- Free space loss versus distance for WiFi bands: 2.4 GHz and 5 GHz

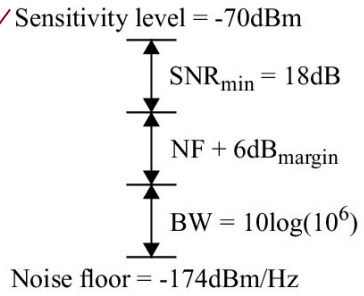


https://www.semfionetworks.com/

LINK-POWER BUDGET CALCULATION



Example: **Bluetooth**



$$P_{rx} = P_{tx} + G_{tx} - L_{tx} - L_{fs} - L_m + G_{rx} - L_{rx}$$

- P_{rx}, P_{tx} : received and transmitted power (dBm)
- G_{tx} , G_{rx}: received and transmitted antennas gains (dBi)
- L_{tx} , L_{rx}: transmitter and receiver losses (coax, connector) (dB)
- L_{fs}: path loss, usually free space loss (dB)
- L_m: various others attenuations (fading margin, body loss,...) (dB)

https://www.aero.iitb.ac.in/satelliteWiki/index.php/Link_Budget

EXAMPLE: LORA

- Main formula:

LoRa Receiver Sensitivity : -

$$S = -174 + 10 \log_{10} BW + NF + SNR$$

Where,
BW = Bandwidth in KHz
NF = Noise Figure of Receiver in dB
SNR = Signal to Noise Ratio from Table as per SF (Spreading Factor)
S = Sensitivity in dBm

SF, bitrate, sensitivity and SNR for a 125 kHz channel

Spreading factor	Bitrate (bit/sec)	Sensitivity (dBm)	LoRa demodulator SNR
7 (128)	5 469	-124 dBm	-7.5 dB
8 (256)	3 125	-127 dBm	-10 dB
9 (512)	1 758	-130 dBm	-12.5 dB
10 (1024)	977	-133 dBm	-15 dB
11 (2048)	537	-135 dBm	-17.5 dB
12 (4096)	293	-137 dBm	-20 dB

- Sensitivity: lowest input received power with acceptable link quality (*typically 1% PER*)

Bandwidth (kHz)	Spreading Factor	Nominal Rb (bps)	Sensitivity (dBm)
125	6	9380	-122
125	12	293	-137
250	6	18750	-119
250	12	586	-134
500	6	3750	-116
500	12	1172	-131

KEY PRODUCT FEATURES

- LoRa™ Modem
- 157 dB maximum link budget
- +20 dBm at 100 mW constant RF output vs. V supply
- +14 dBm high efficiency PA
- Programmable bit rate up to 300 kbps
- High sensitivity: down to -137 dBm

https://www.rfwireless-world.com/calculators/LoRa-Link-Budget-Calculator.html

EXAMPLE: OTHERS

UL link-power budget

Technologies	WAVIoT NB-Fi	LORA	Sigfox	LTE-M
TX power (dBm)	15	15	15	20
TX Cable loss (dB)	-1	-1	-1	-1
TX Antenna gain, dBi	0	0	0	0
TX subtotal (dBm)	14	14	14	19
RX Sensitivity (dBm)	-152	-137	-142	-129
Rx Environment noise (dB)	-10	-10	-10	-10
RX Antenna gain diversity (dBi)	10	10	10	10
RX SubTotal (dBm)	152	137	142	129
Maximum Allowable Pathloss (dB)	166	151	156	148

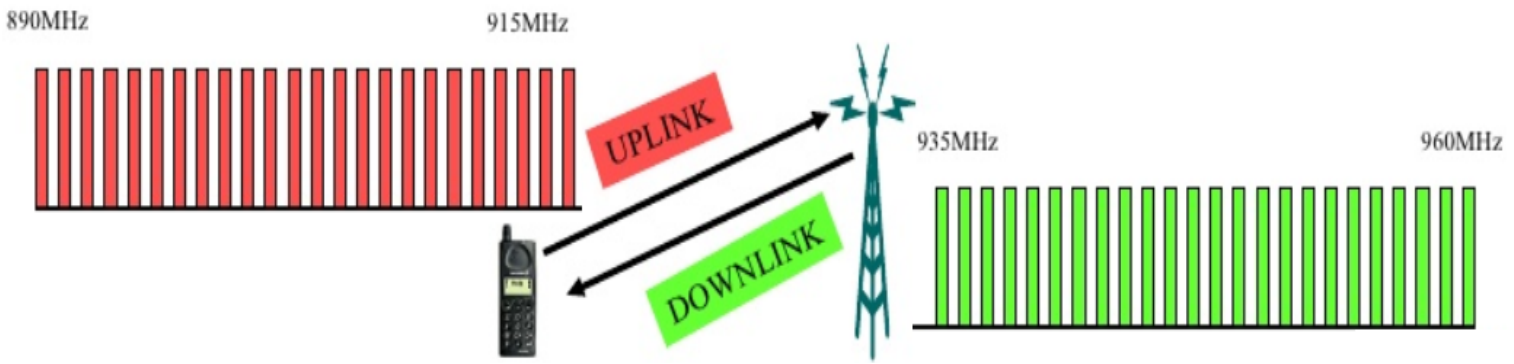
DL link-power budget

Technologies	WAVIoT NB-Fi	LORA	Sigfox	LTE-M
TX power (dBm)	30	21	24	40
TX Cable loss (dB)	-3	-3	-6	-3
TX Antenna gain, dBi	0	9	9	10
TX subtotal (dBm)	27	27	27	47
RX Sensitivity (dBm)	-147	-137	-129	-129
Rx Environment noise (dB)	0	0	0	0
RX Antenna gain diversity (dBi)	0	0	0	0
RX SubTotal (dBm)	-147	-137	-129	-129
Maximum Allowable Pathloss (dB)	174	164	156	176

GSM LINK BUDGET: EXERCISE

- The radio path (f = 900 MHz) between a mobile phone and the base station has a loss of 30 dB in addition to the free space loss. The base station measures the level of the signal transmitted by the phone, and it sends the phone a command to set the transmit power to a level, which gives a received power level of -90 dBm. The antenna gain of the phone is 2 dBi and the receive gain of the base station antenna is 7 dBi. The distance between the phone and the base station is 500 m.

  A. What will be the transmit power level of the phone?
  B. Same problem with f = 1800 MHz. All other values remain the same. (In real life the 30 dB attenuation value may rise by a few dB.)
  C. Make conclusions from the two answers bearing in mind that the frequencies used in future 5G networks will get higher (3.4-3.8 GHz).

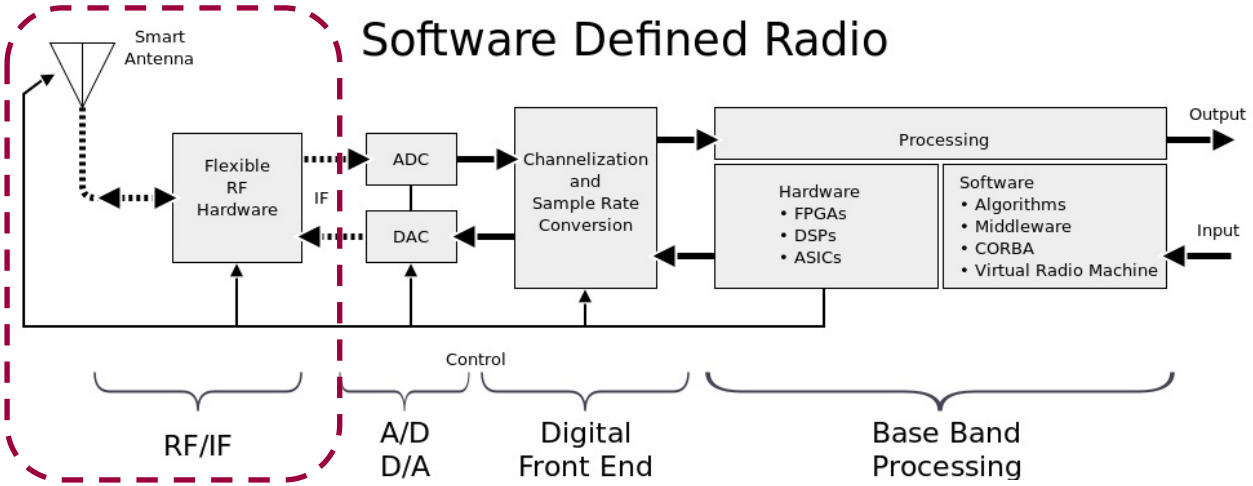


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PART III:

ANALOG FRONT-END (AFE)

SDR: AFE ARCHITECTURE

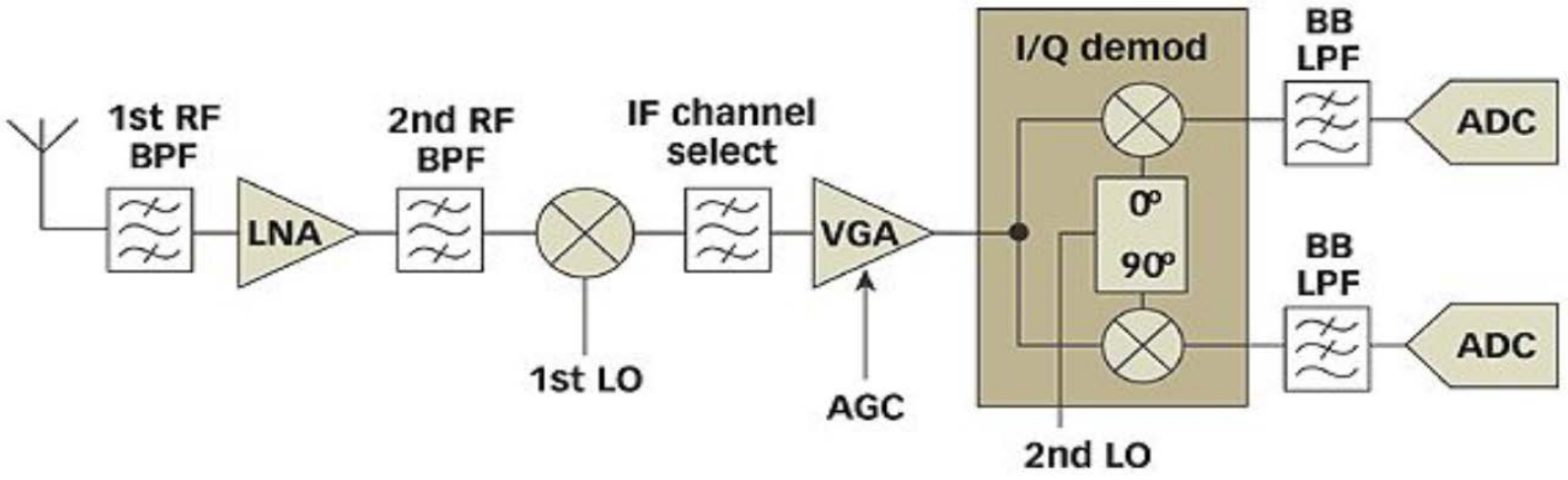


- The input signal to receiver is obtained from a receiving antenna
- Received signals are typically very weak
  - LoRa RX sensitivity : UL = -137 dBm, DL= -134 dBm
  - Sigfox RX sensitivity: UL = -142 dBm, DL= -126 dBm
- Receiver must perform a number of actions on the received signal. Ideally without impact the signal-to-noise ratio.
- The most known architectures are:
  - The super-heterodyne architecture
  - The direct-conversion architecture (homodyne, zero-IF)

Source: University of California/Berkeley

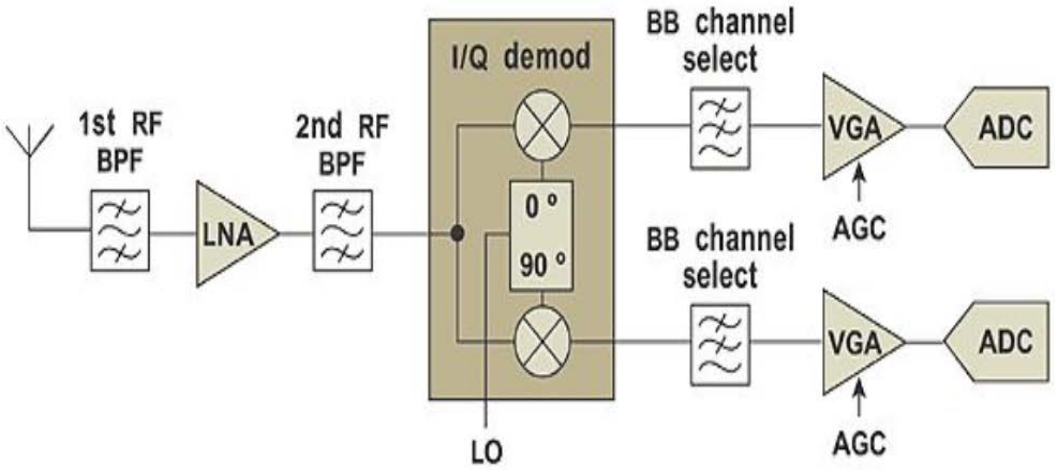
AFE ARCHITECTURE

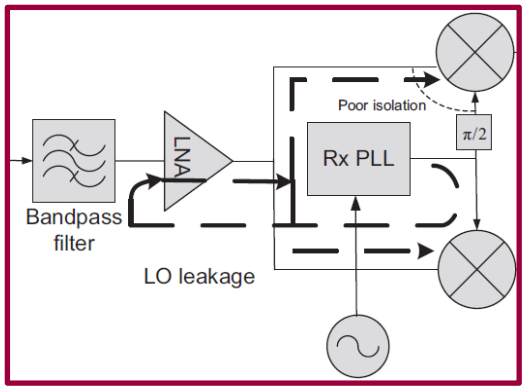
- Most radios use the classic super-heterodyne receiver architecture
  - The RF signal from antenna is mixed with local oscillator to produce an intermediate frequency (IF) signal
  - The IF signal is a fixed lower-frequency signal which is then filtered and demodulated.
  - **(+)** Excellent selectivity and sensitivity
  - **(+)** DC offset is not a major concern in super-heterodyne receivers
  - **(-)** integration, complexity and power consumption



AFE ARCHITECTURE

- Most SDR uses a direct-conversion receiver architecture
  - Eliminates the intermediate frequency (IF) by translating the band of interest directly to baseband
  - LO frequency is set to the same frequency as the desired RF signal
  - **(+)** High integration level, low cost
  - **(+)** Small size, low power consumption
  - **(-)** Requires very high-Q voltage-controlled oscillator
  - **(-)** Effect of LO leakage on direct-conversion receiver performance
  - **(-)** Impact of strong interferer or blocker on receiver performance
  - **(-)** IQ imbalance, flicker noise





EFFECTS OF NONLINEARITY

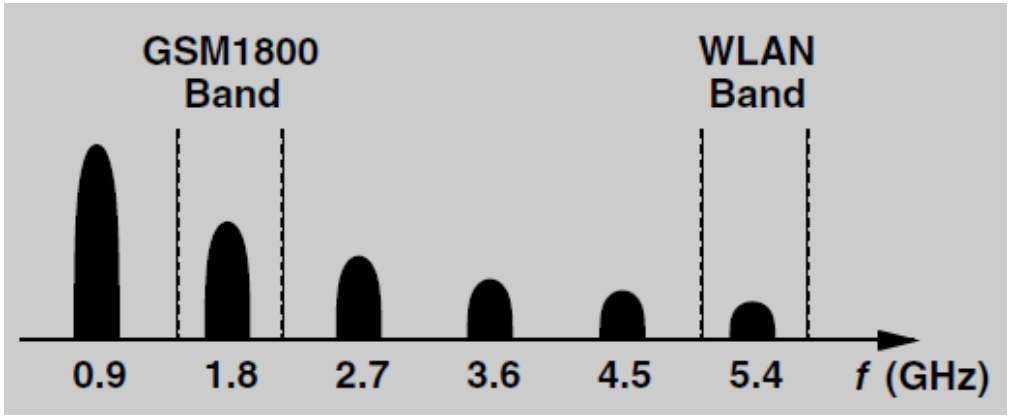
- RF circuits can be approximated by a linear model for small-signal operation
- Nonlinearities lead to interesting phenomena that are not predicted by small-signal models
- Let take the following model:

$$y(t) \approx \alpha_1 x(t) + \alpha_2 x^2(t) + \alpha_3 x^3(t)$$

- **(1) Harmonic distortion**:

$$x(t) = A\cos\omega t \Rightarrow y(t) = \frac{\alpha_2 A^2}{2} + \left(\alpha_1 A + \frac{3\alpha_3 A^3}{4}\right)\cos\omega t + \frac{\alpha_2 A^2}{2}\cos 2\omega t + \frac{\alpha_3 A^3}{4}\cos 3\omega t$$

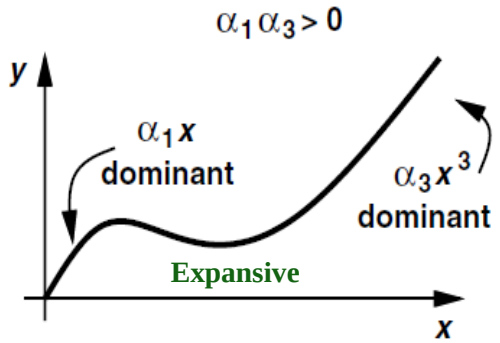
DC quantity — fundamental — Second harmonic — Third harmonic

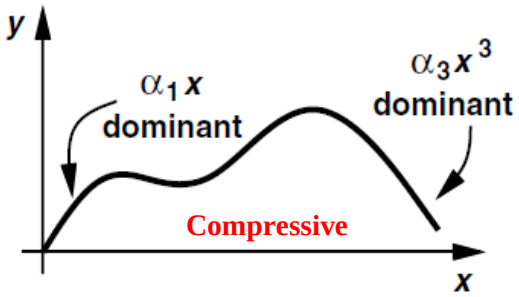


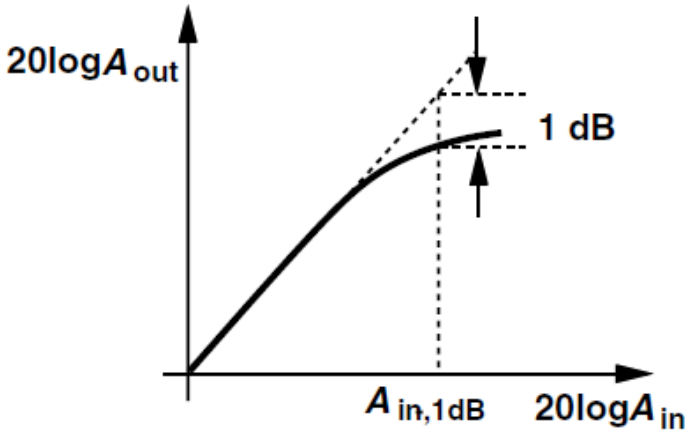
EFFECTS OF NONLINEARITY

$$y(t) = \frac{\alpha_2 A^2}{2} + \left(\alpha_1 A + \frac{3\alpha_3 A^3}{4}\right)\cos\omega t + \frac{\alpha_2 A^2}{2}\cos 2\omega t + \frac{\alpha_3 A^3}{4}\cos 3\omega t$$

- **(2) Gain compression**:
  - The gain experienced by $Acos(\omega t)$ is equal to $\alpha_1 + 3\alpha_3 A^2/4$, and, hence varies appreciably as *A* becomes larger







$$20\log\left|\alpha_1 + \frac{3}{4}\alpha_3 A_{in,1dB}^2\right| = 20\log|\alpha_1| - 1\ \text{dB}$$

$$A_{in,1dB} = \sqrt{0.145\left|\frac{\alpha_1}{\alpha_3}\right|}$$

EFFECTS OF NONLINEARITY

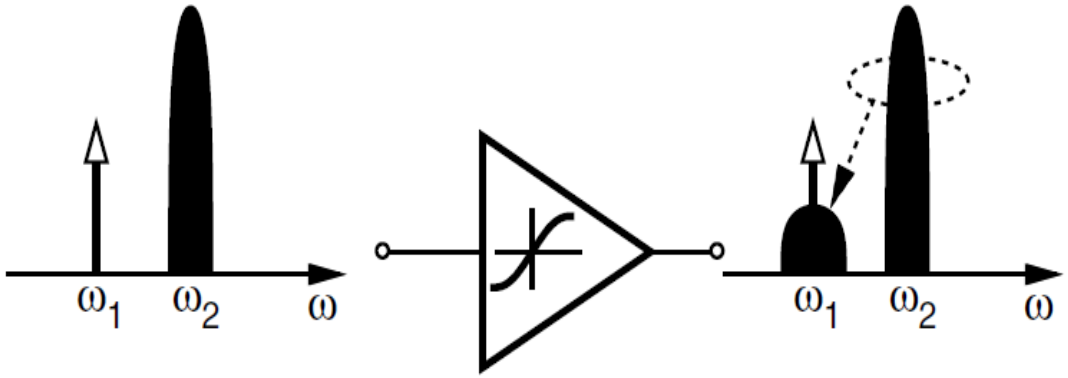
- **(3) Cross compression**:
  - Occurs when a weak desired signal and a strong interferer pass through a nonlinear circuit
  - The interferer is an amplitude modulated signal

$$A_2(1 + m\cos\omega_m t)\cos\omega_2 t$$

  - Then

$$y(t) = \left[\alpha_1 + \frac{3}{2}\alpha_3 A_2^2\left(1 + \frac{m^2}{2} + \frac{m^2}{2}\cos 2\omega_m t + 2m\cos\omega_m t\right)\right] A_1\cos\omega_1 t + \cdots$$

  - ➔ The desired signal at the output of circuit suffers from amplitude modulation at ω_m and $2\omega_m$



EFFECTS OF NONLINEARITY

- **(4) Intermodulation**:
  - Third Intercept Point **IP3** is a common method of intermodulation characterization using two-tone test

$$x(t) = A[cos(\omega_1 t) + cos(\omega_2 t)]$$

$$y(t) = \alpha_0 + \alpha_1 x(t) + \alpha_2 x^2(t) + \alpha_3 x^3(t)$$

$$y(t) = \underbrace{(\alpha_0 + A^2\alpha_2) + \left(A\alpha_1 + \frac{3}{4}A^3\alpha_3\right)[cos(\omega_1 t) + cos(\omega_2 t)]}_{\text{DC + Fundamental Components}}$$

Output bias point; DC offset; Desired gain

$$+\underbrace{\frac{A^2\alpha_2}{2}[cos(2\omega_1 t) + cos(2\omega_2 t)] + \frac{A^3\alpha_3}{4}[cos(3\omega_1 t) + cos(3\omega_2 t)]}_{\text{Harmonics terms, most of them attenuated by tuned stages}}$$

Filtered by tuned stage

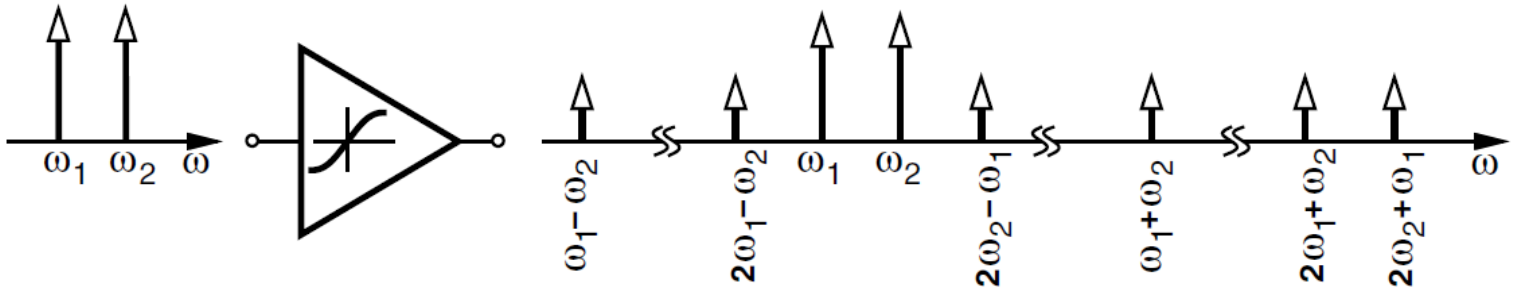
$$\underbrace{+A^2\alpha_2[cos(\omega_1 t + \omega_2 t) + cos(\omega_1 t - \omega_2 t)]}_{\text{Second order intermodulation (IM2)}}$$

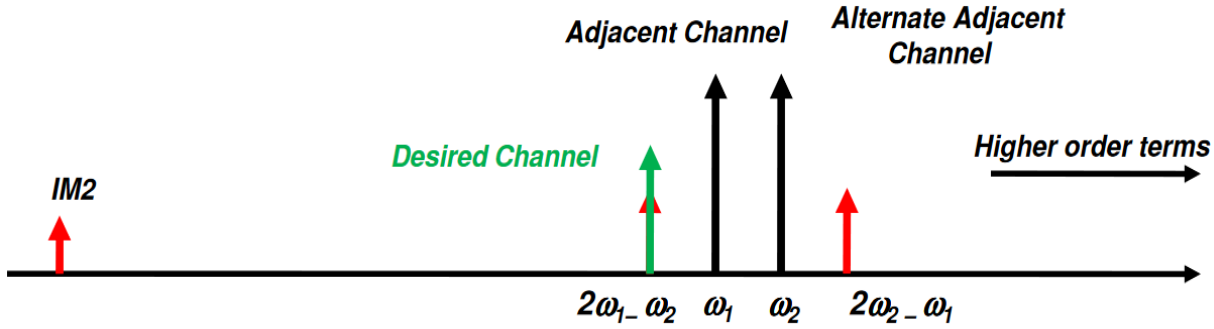
Filtered by tuned stage

$$+\underbrace{\frac{3A^3\alpha_3}{4}[cos(\omega_1 t + 2\omega_2 t) + cos(2\omega_1 t + \omega_2 t) + cos(\omega_1 t - 2\omega_2 t) + cos(2\omega_1 t - \omega_2 t)]}_{\text{Third order intermodulation (IM3): some terms are in-band (fall in channel ➔ Problematic)}}$$

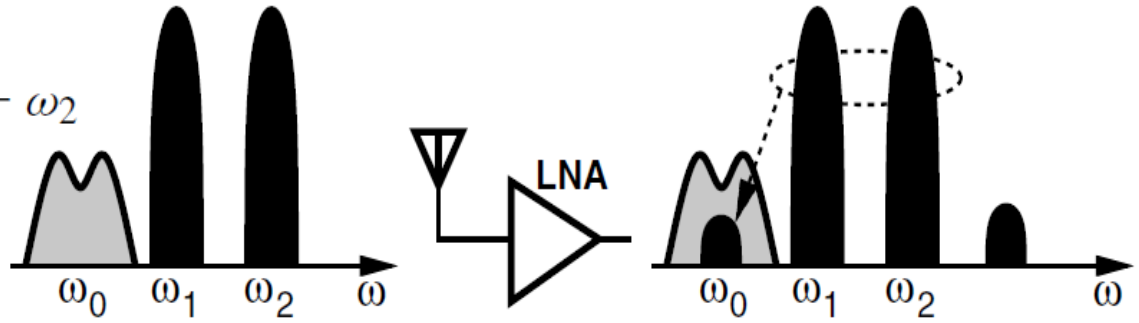
EFFECTS OF NONLINEARITY

- **(4) Intermodulation**:





$\omega_0 = 2\omega_1 - \omega_2$

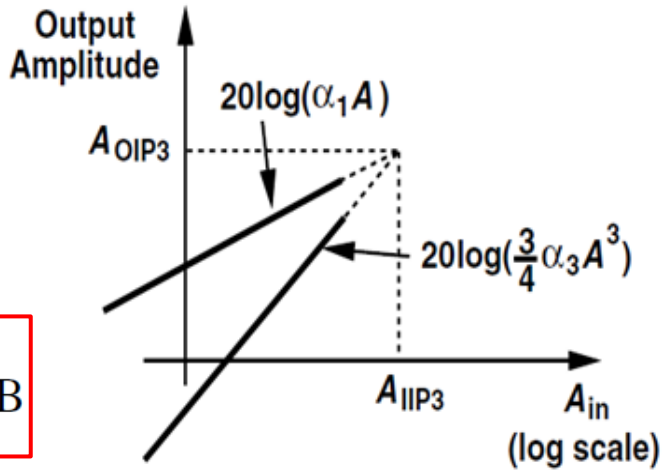


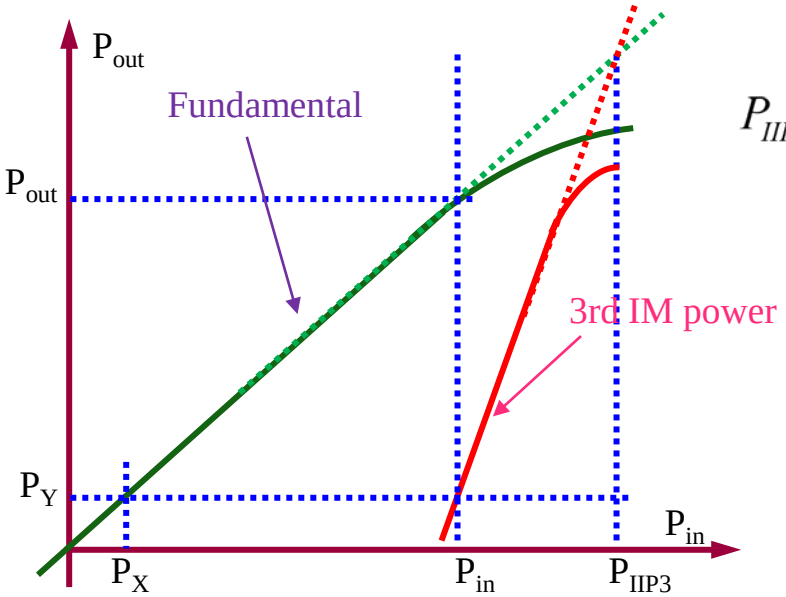
EFFECTS OF NONLINEARITY

- **(4) Intermodulation**:
  - To determine the IIP3, we simply equate the fundamental and the IM amplitudes

$$|\alpha_1 A_{IIP3}| = \left|\frac{3}{4}\alpha_3 A_{IIP3}^3\right|$$

$$A_{IIP3} = \sqrt{\frac{4}{3}\left|\frac{\alpha_1}{\alpha_3}\right|} \Rightarrow \frac{A_{IIP3}}{A_{1dB}} = \sqrt{\frac{4}{0.435}} \approx 9.6\,\text{dB}$$





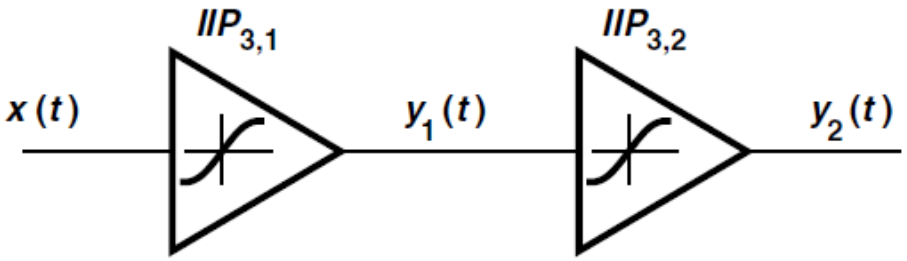
$$P_{IIP3} = P_{in} + \frac{P_{out} - P_Y}{2} = P_{in} + \frac{(P_{in}+G)-(P_X+G)}{2} = \frac{3P_{in} - P_X}{2}$$

If P_X is the minimum receivable input power, and **if** the corresponding output P_Y is not to be exceeded by IM3, **then** P_{in} must stay below the following expression

$$P_{in} = \frac{2P_{IIP3} + P_X}{3}$$

CASCADED NONLINEAR CIRCUIT

- **(4) Intermodulation**:
  - Consider two nonlinear stages in cascade



$$y_1(t) = \alpha_1 x(t) + \alpha_2 x^2(t) + \alpha_3 x^3(t)$$

$$y_2(t) = \beta_1 y_1(t) + \beta_2 y_1^2(t) + \beta_3 y_1^3(t)$$

$$y_2(t) = \alpha_1\beta_1 x(t) + (\alpha_3\beta_1 + 2\alpha_1\alpha_2\beta_2 + \alpha_1^3\beta_3)x^3(t) + \cdots$$

  - Then,

$$A_{IP3} = \sqrt{\frac{4}{3}\left|\frac{\alpha_1\beta_1}{\alpha_3\beta_1 + 2\alpha_1\alpha_2\beta_2 + \alpha_1^3\beta_3}\right|} \quad \Rightarrow \quad \frac{1}{A_{IP3}^2} \approx \frac{1}{A_{IP3,1}^2} + \frac{\alpha_1^2}{A_{IP3,2}^2}$$

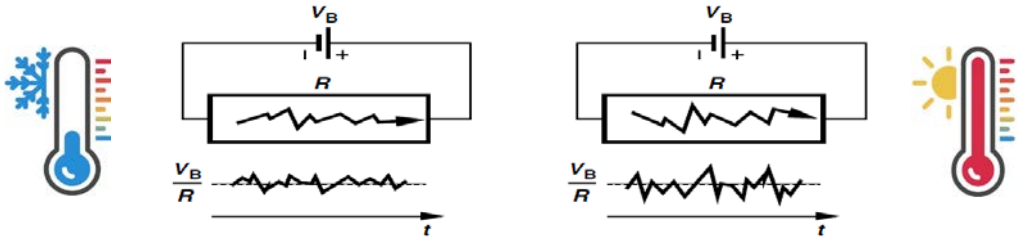
  - General case, more than two stages

$$\frac{1}{A_{IP3}^2} \approx \frac{1}{A_{IP3,1}^2} + \frac{\alpha_1^2}{A_{IP3,2}^2} + \frac{\alpha_1^2\beta_1^2}{A_{IP3,3}^2} + \cdots$$

NOISE FIGURE

Thermal noise:

- The noise can be seen as a random process



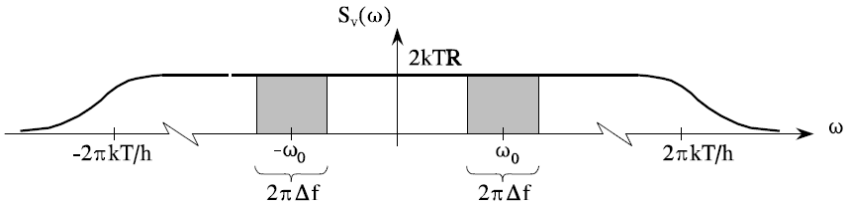
- Thermal motion of the electrons in a resistor causes a fluctuating noise voltage across a given circuit
- The voltage has a Gaussian distribution around a mean value of zero
- The two-side power spectral density is given by

$$S_n(f) = 2R\frac{Hf}{e^{HfKT}-1} \overset{f \ll KT/H}{\approx} 2KTR$$

K: Boltzman's constant
T: Absolute temperature
R: resistor
H: Planck's constant

- Around room temperature 300°K, we have KT/H=6.44 THz.
- In a frequency band of Δf around any frequency$f_0 \ll KT/H$ bound, the thermal noise provides a fluctuation voltage v_n that has zero mean and variance given by

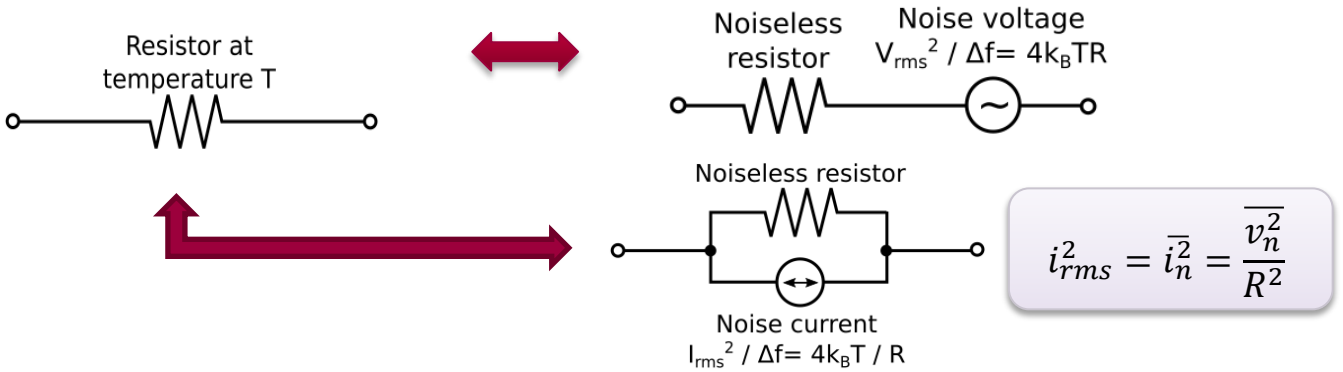
$$\overline{v_n^2} = 2\int_{f_0-\Delta f/2}^{f_0+\Delta f/2} S_n(f)df = 4KTR\Delta f$$



NOISE FIGURE

Thermal noise:

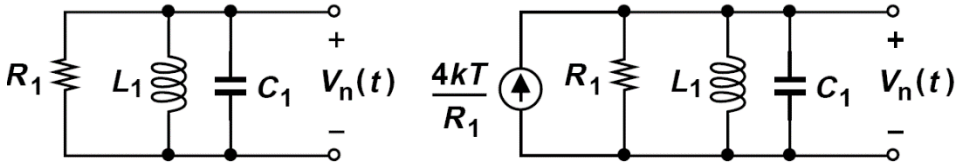
- A resistor model

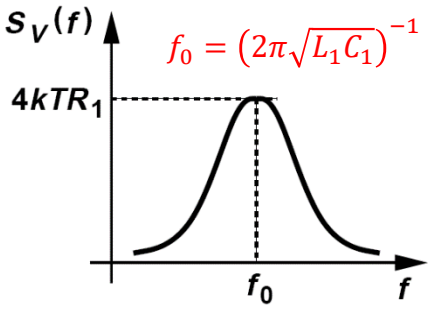


$$i_{rms}^2 = \overline{i_n^2} = \frac{\overline{v_n^2}}{R^2}$$

Device noise:

- Example the PSD of the noise voltage across RLC structure





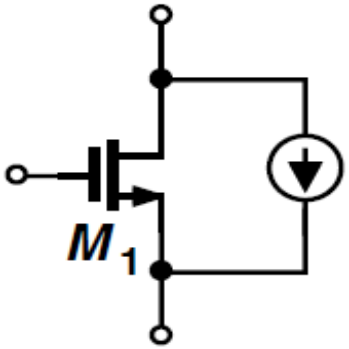
$$\overline{v_n^2} = \overline{i_{n1}^2}|Z_T|^2$$

NOISE FIGURE

Thermal noise of MOSFETS:

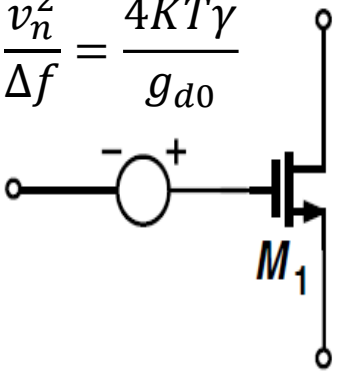
- The thermal noise of MOS transistors operating in **the saturation region** is approximated by a current source tied between the source and drain

Excess noise coefficient ∈ [2/3,2]



$$\frac{\overline{I_n^2}}{\Delta f} = 4KT\gamma g_{d0}$$

$$\frac{\overline{v_n^2}}{\Delta f} = \frac{4KT\gamma}{g_{d0}}$$

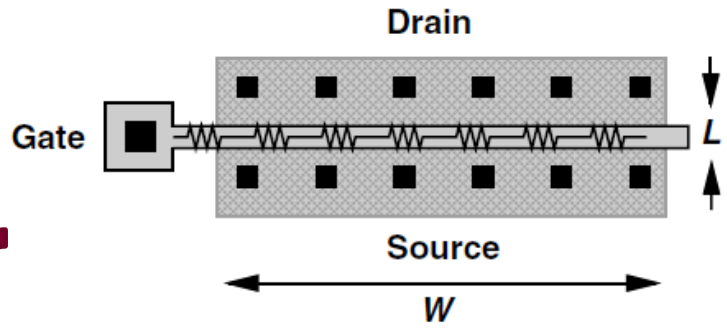


$$g_{d0} = \frac{\mu C_{ox} W}{L}\left(V_{gs} - V_T\right) \approx g_m$$

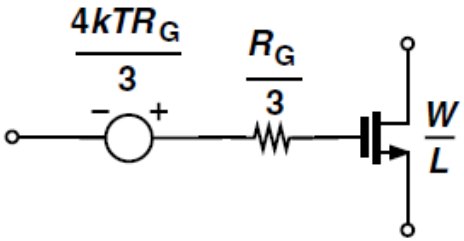
NOISE FIGURE

Thermal noise of MOSFETS:

- The thermal noise from the gate resistance of MOSFETS transistors



$$R_G = \frac{W}{L} R$$



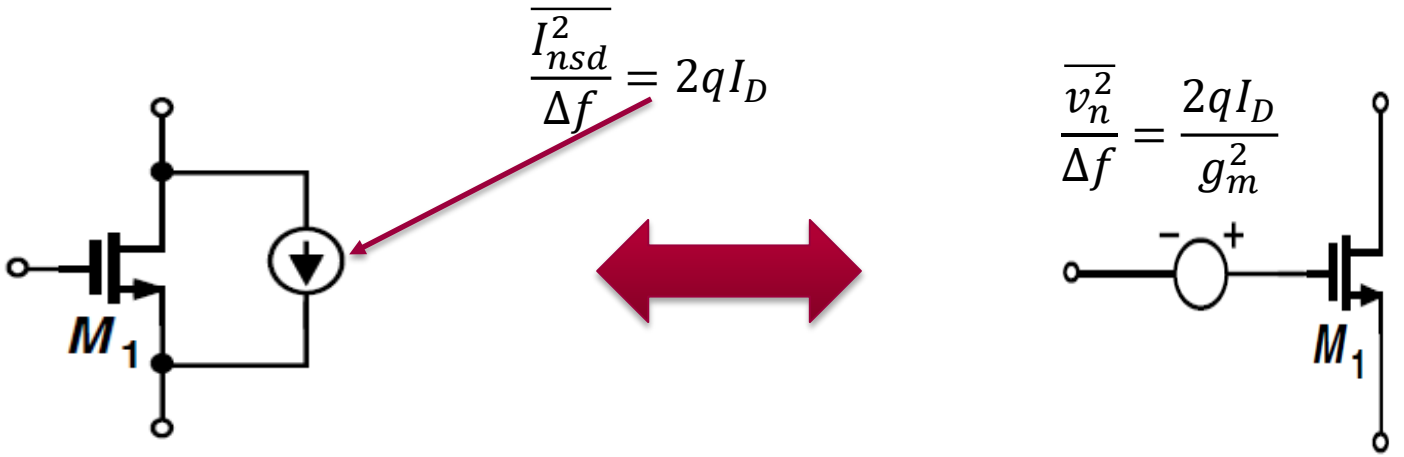
- In a good circuit design, the noise of gate resistance must be much less than that of the channel (between **drain-source**)

$$\frac{4KTR_G}{3} \ll \frac{4KT\gamma}{g_{d0}}$$

NOISE FIGURE

Noise in MOSFETS:

- The shot noise is modeled as a noise current source in parallel with the junction
- The shot noise due to the current following in the channel (**drain-source**) is given by



- The shot noise du to the current following in the gate is given by



NOISE FIGURE

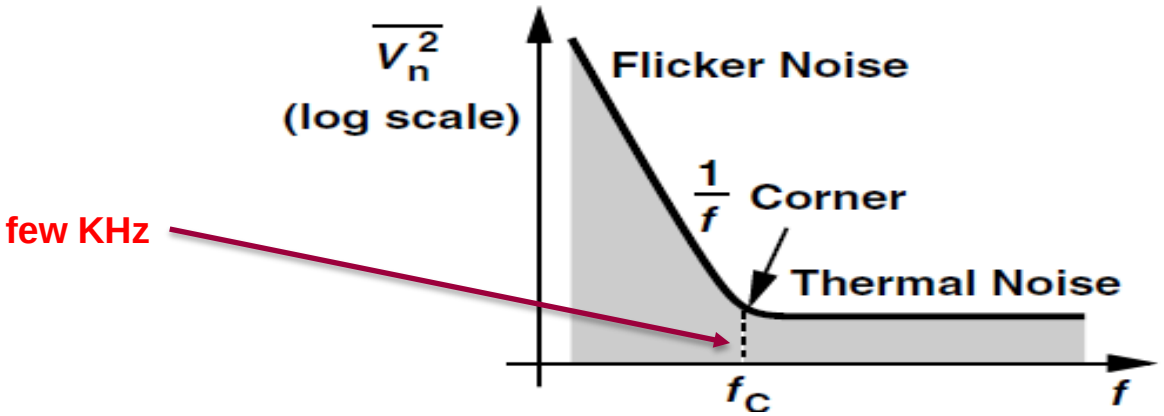
Flicker noise:

- MOS devices also suffer from "**flicker**" noise modeled by a voltage in series with the gate. This flicker noise exhibits the following PSD

$$\overline{v_n^2} = \frac{\alpha}{WLC_{ox}} * \frac{1}{f}$$

- The flicker noise PSD intercepts the thermal noise PSD at some frequency called "$1/f$ noise corner frequency" f_c

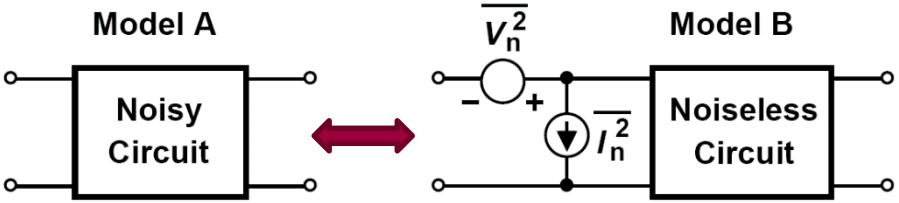
$$4KT\gamma g_{d0} = \frac{\alpha}{WLC_{ox}} \times \frac{1}{f_c} \times g_m^2 \quad \Rightarrow \quad f_c = \frac{\alpha g_m}{WLC_{ox}} * \frac{1}{4KT\gamma}$$



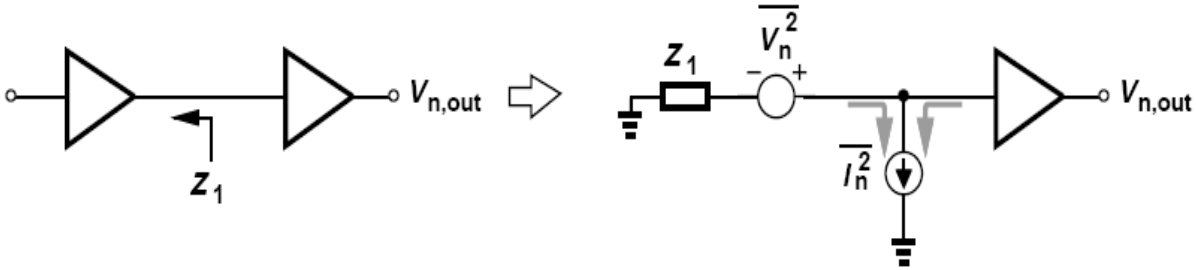
NOISE FIGURE

Representation of noise in circuits:

- In analog design, the input-referred noise is modeled by a series voltage source and a parallel current source



- Voltage source: the input-referred noise is obtained by shorting the input port of models A and B, and equating their output noise voltage.
- Current source: the input-referred noise is obtained by leaving the input ports open and equating the output noise voltage.
- Let's the following cascade structure

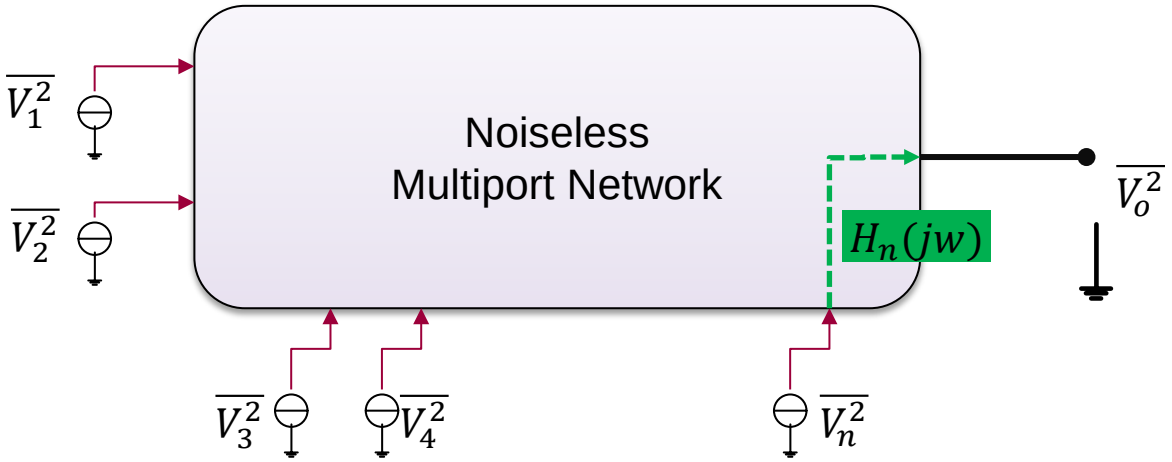


- The noise current flows through Z_1, generating a noise voltage at the input that depends on $|Z_1|$.
- The output noise, $V_{n,out}$, also depends on $|Z_1|$.

NOISE FIGURE

Calculation of noise in circuit

- Let $V_1, \ldots, V_n$ are mutually independent.



- The output-referred noise is given by

$$\overline{v_o^2} = \sum_{k=1}^{N} \overline{v_k^2} \,.\, H_k^2(jw)$$

- The input-referred noise is given by

$$\overline{v_k^2} = \frac{\overline{v_O^2}}{H_k^2(jw)}$$

NOISE FIGURE

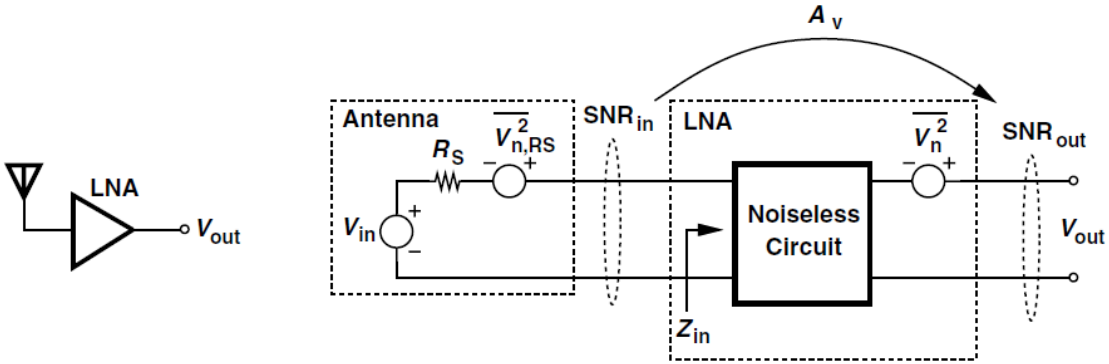
Definition:

- In circuit, we are interested in the signal-to-noise ratio (SNR). If the circuit contains no noise ➔ *input SNR* is equal to *output SNR*
- We define the noise figure of a given circuit as

$$\mathrm{NF} = \frac{SNR_{in}}{SNR_{out}}$$

$$NF|_{dB} = 10log\left(\frac{SNR_{in}}{SNR_{out}}\right)$$

- Example: antenna followed by LNA



The input SNR is

$$SNR_{in} = \frac{|\alpha|^2 V_{in}^2}{|\alpha|^2 \overline{V_{RS}^2}}$$

The output SNR is

$$SNR_{out} = \frac{V_{in}^2 |\alpha|^2 A_v^2}{\overline{V_{RS}^2}|\alpha|^2 A_v^2 + \overline{V_n^2}}$$

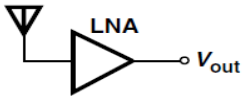
$$\alpha = Z_{in}/(Z_{in} + R_s).$$

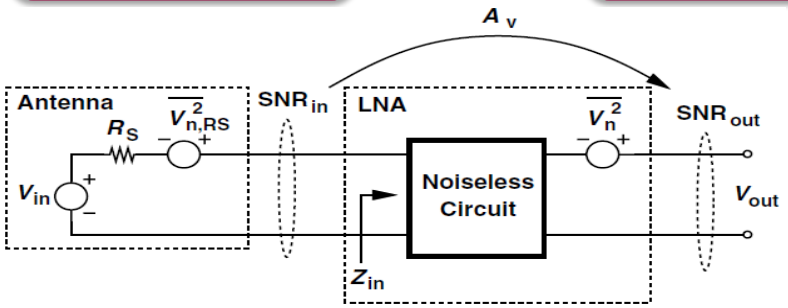
NOISE FIGURE

Calculation of NF:

$$\mathrm{NF} = \frac{SNR_{in}}{SNR_{out}}$$

$$NF|_{dB} = 10log\left(\frac{SNR_{in}}{SNR_{out}}\right)$$





$$SNR_{in} = \frac{|\alpha|^2 V_{in}^2}{|\alpha|^2 \overline{V_{RS}^2}}$$

$$SNR_{out} = \frac{V_{in}^2 |\alpha|^2 A_v^2}{\overline{V_{RS}^2}|\alpha|^2 A_v^2 + \overline{V_n^2}}$$

$$\mathrm{NF} = \frac{1}{\overline{V_{RS}^2}} \cdot \frac{\overline{V_{RS}^2}|\alpha|^2 A_v^2 + \overline{V_n^2}}{|\alpha|^2 A_v^2}$$

→ Total noise measured at the output

$$\mathrm{NF} = 1 + \frac{\overline{V_n^2}}{|\alpha|^2 A_v^2} \cdot \frac{1}{\overline{V_{RS}^2}}$$

Noise figure in cascade of stages:

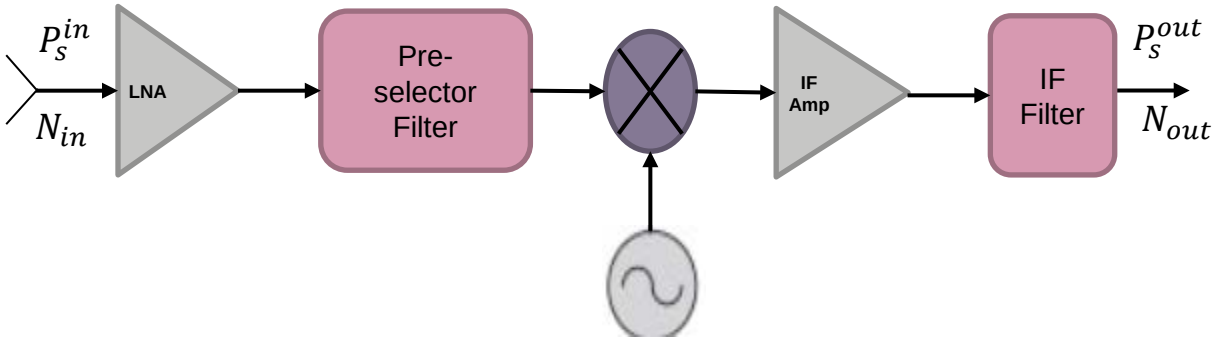
- The Friis formula for noise factor (***m* stages**)

$$NF_{tot} = 1 + (NF_1 - 1) + \frac{NF_2 - 1}{A_{P1}} + \cdots + \frac{NF_m - 1}{A_{P1} \cdots A_{P(m-1)}}$$

https://aaronscher.com/Gain_NF_calculator/gain_NF_calculator.html

RX GAIN AND NOISE FIGURE

- We can determine the overall gain and noise figure for super-heterodyne receiver



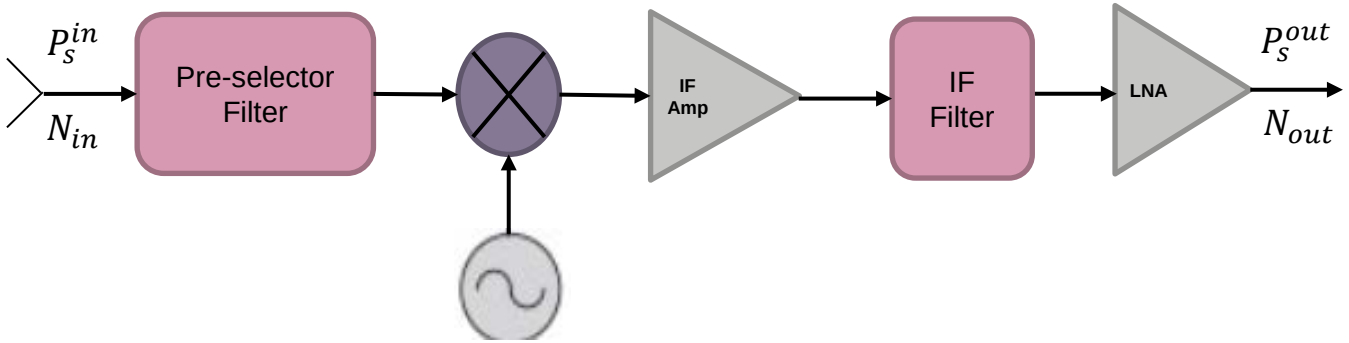
- **Antenna**: we assume that the antenna temperature is $T_A = 290°K$ → $N_{in} = -174 \frac{dBm}{Hz}$
- **Low Noise Amplifier**: say this device has gain $G_1 = 10$ and noise figure $F_1 = 1.5$
- **Pre-selector Filter**: say this device has an insertion loss of 1dB → $G_2 \approx 0.8$ and $F_2 \approx 1.26$
- **Mixer**: say this device has a conversion loss of 6 dB → $G_3 \approx 0.25$ and $F_3 \approx 4$
- **IF Amplifier**: say this device has a gain of 30 dB and a noise figure of 6 dB → $G_4 \approx 10^3$ and $F_4 \approx 4$
- **IF Filter**: say this device has an insertion loss of 2dB → $G_5 \approx 0.63$ and $F_5 \approx 1.58$

- → **The gain of the receiver is**: $G = G_1 \times G_2 \times G_3 \times G_5 \times G_5 = 1260 = 31dB$
- → **The noise figure of the receiver is**: $\mathrm{F} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \frac{F_4 - 1}{G_1 G_2 G_3} + \frac{F_5 - 1}{G_1 G_2 G_3 G_4} = 2$
- → The noise at the output of the IF Filter is

$$P_n^{out} = FGKTB \approx 10^{-17} B \ Watts$$

RX GAIN AND NOISE FIGURE

- Let's see what happens if we move the LNA



- **The gain of the receiver is**: $G = G_1 \times G_2 \times G_3 \times G_5 \times G_5 = 1260$ (*didn't change*)
- **The noise figure of the receiver is**: $\mathrm{F} = F_2 + \frac{F_3 - 1}{G_2} + \frac{F_4 - 1}{G_2 G_3} + \frac{F_5 - 1}{G_2 G_3 G_4} + \frac{F_1 - 1}{G_2 G_3 G_4 G_5} = 20$
- The noise at the output of the LNA amplifier is

$$P_n^{out} = FGKTB \approx 10^{-16} B \ Watts$$

10 times more noise than before !!!!!

This example shows how important the LNA amplifier is for low-noise receiver design

SENSITIVITY, DR AND SFDR

- **Sensitivity:** minimum signal level that can be detected with desired SNR

$$SNR_{in} = \frac{P_{sig}}{KTB}$$

Received signal power

  - We want to determine $P_{sig,min}$ for a desired $SNR_{out,min}$

$$NF = \frac{SNR_{in}}{SNR_{out}} = \frac{\frac{P_{sig}}{KTB}}{SNR_{out}} \Rightarrow P_{sig} = KTB \times SNR_{out} \times F$$

  - Expressed in decibels (dBm or dB)

$$P_{sen\ |dBm} = P_{in,\min\ |dBm} = \underbrace{10log(KT)_{|dbm/Hz} + NF_{dB} + 10log(B)}_{\text{Noise Floor Level}} + SNR_{\min\ |dB}$$

  - $P_{in,\min}$ is a function of bandwidth B ➔ narrowband channel may appear to be sensitive compared to another with a wideband channel.

- **Dynamic range:**

$$DR \equiv \frac{\text{Max input level tolerated}}{\text{Min input level meeting SNR requirements}}$$

SENSITIVITY, DR AND SFDR

- **SFDR:** spurious free dynamic range
  - Upper bound defined by IM behavior
  - Lower bound defined by sensitivity

- **Upper bound:** max input level for which IM3 products do not exceed the noise floor
  - Recall that

$$P_{IIP3} = P_{in} + \frac{P_{out} - P_{IM,out}}{2}$$

$$P_{IIP3} = P_{in} + \frac{P_{in} - P_{IM,in}}{2}$$

$$NFL = 10log(KT) + NF_{dB} + 10log(B)$$

  - Then

$$\left.\begin{aligned} P_{in,max} &= \frac{2P_{IIP3} + NFL}{3} \\ P_{in,min} &= NFL + SNR_{\min} \end{aligned}\right\} \Rightarrow \boxed{SFDR = P_{in,max} - P_{in,min} = \frac{2(P_{IIP3} - NFL)}{3} - SNR_{\min}}$$

CentraleSupélec

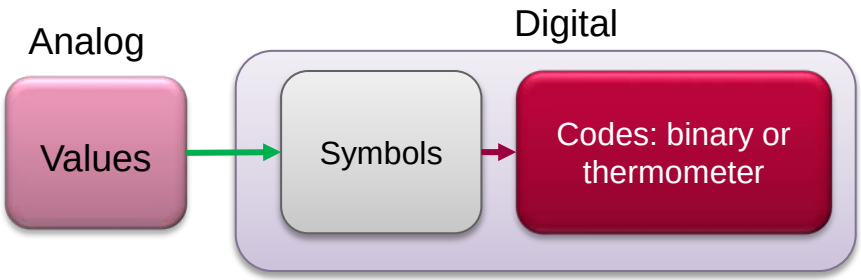
PART IV:

MIXED-SIGNAL FRONT-END (M-SFE)

ANALOG-TO-DIGITAL CONVERTERS

- Analogue is loosely defined
  - Refers to a quantity that is a continuous function of time
  - The resolution (granularity of the value) of an analogue quantity is quite large

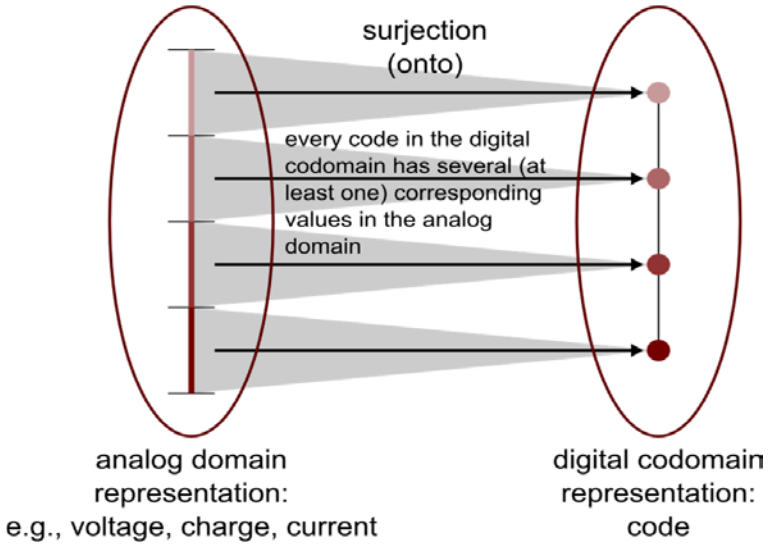
- ADC usually consists of the following steps:
  - Time quantization: continuous-valued samples
  - Amplitude quantization: discrete-valued samples
  - Providing information on the time quantization → Bandwidth limitation
  - Finding and providing an unambiguous representation of each quantized sample
    - Sample is initially unknown to the ADC
    - Comparison and classification required



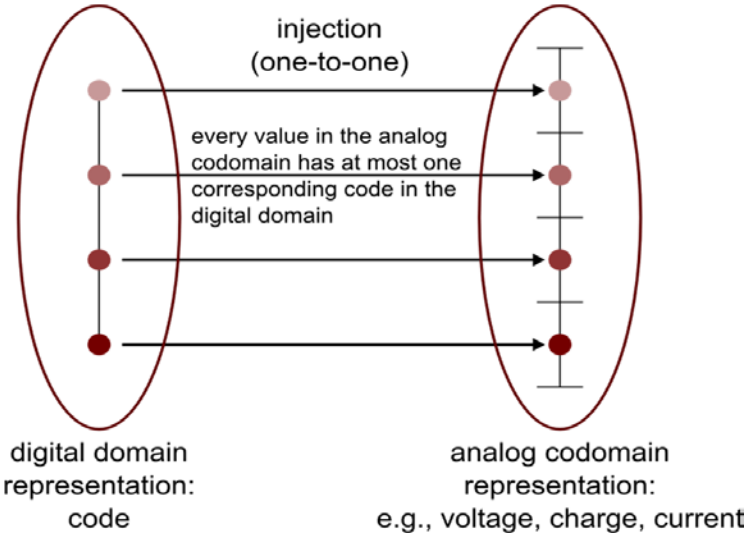
Main task of ADC converters

1. Mapping algorithms (assignment of values to symbols)
2. Compare unknown value to series of known reference values (searching algorithms)

DIGITAL DOMAIN ←→ ANALOGUE DOMAIN



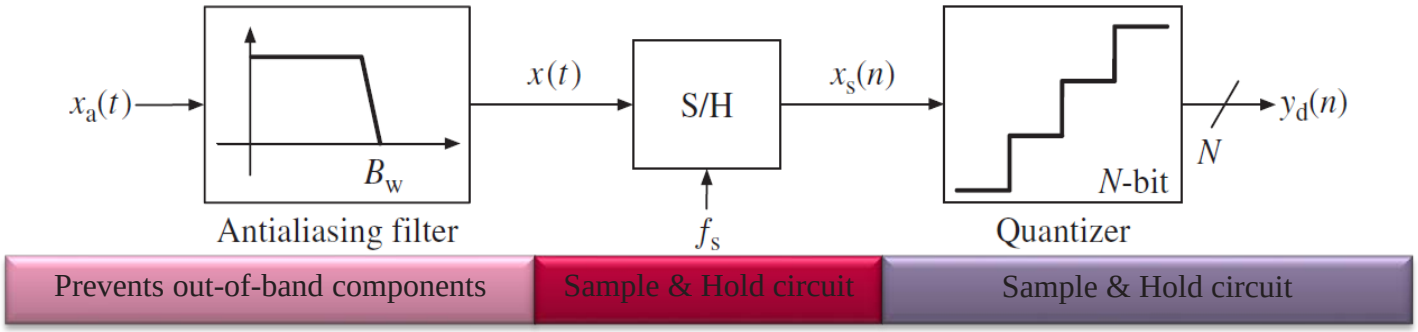
- Digitization maps a larger (possibly infinite) set of values to a smaller (finite) one.
- Automatic mapping of unknown quantities relies on algorithms for search and classification



- A finite set of digital codes can be uniquely mapped to the analogue domain
- No ambiguity arises from DAC

BASICS OF A/D CONVERSION

- ADCs are circuits that perform the transformation of analog signals (which are continuous in time and in amplitude) into digital signals (which are discrete in both time and amplitude).

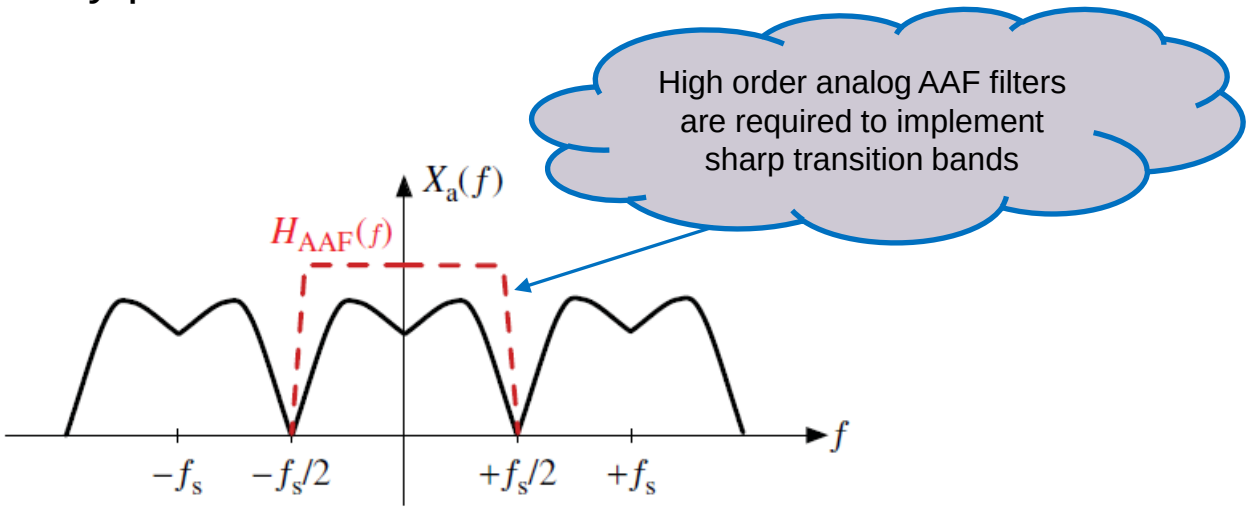


- Sampling:
  - According to the Nyquist theorem, to prevent information loss, *x(t)* must be sampled at a minimum rate of $f_N = 2B_w$ (Nyquist frequency)
  - *Nyquist-rate* ADCs means that $f_s = f_N$
  - *Oversampling* ADCs means that $f_s > f_N$. We define the oversampling ratio as

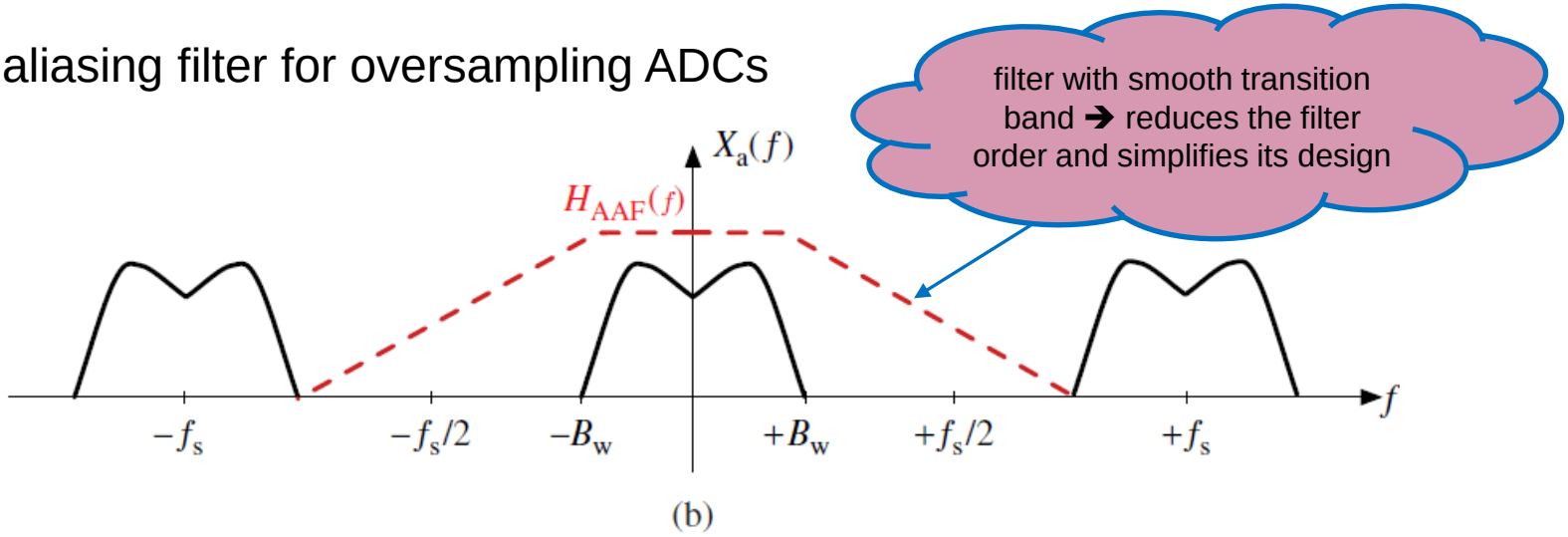
$$OSR = \frac{f_s}{f_N} = \frac{f_s}{2B_w}$$

BASICS OF A/D CONVERSION

- Antialiasing filter for Nyquist-rate ADCs



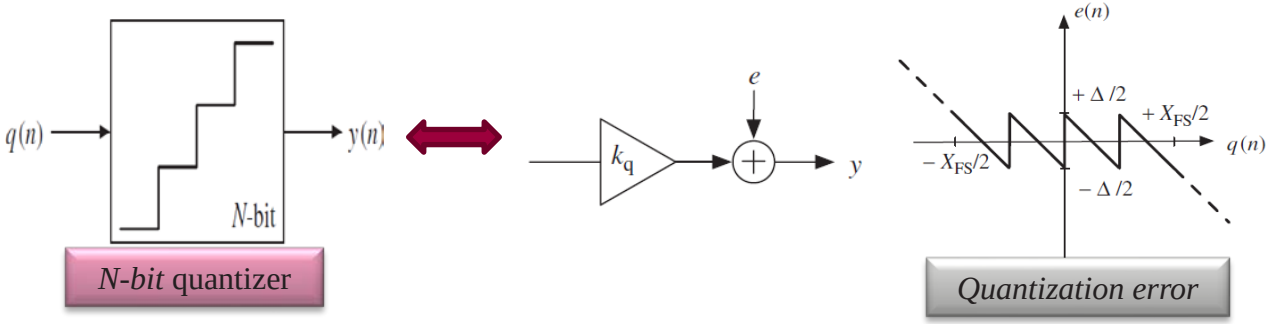
- Antialiasing filter for oversampling ADCs



BASICS OF A/D CONVERSION

Quantization White Noise Model

- An ideal quantizer is often modeled using the linear scheme



Full-Scale output range

$$\Delta = \frac{Y_{FS}}{2^N - 1}$$

- if *q(n)* is assumed to change randomly from sample to sample within the range [−Δ/2,+Δ/2], *e(n)* will also be uncorrelated from sample to sample.

- The quantization error is a *uniform probability distribution* in the range [−Δ/2,+Δ/2].

- The power (variance) of *e(n)* can be computed as :

Probability Density Function

$$\overline{e^2} = \sigma_e^2 = \int_{-\infty}^{+\infty} e^2 h(e) de = \frac{1}{\Delta}\int_{-\Delta/2}^{+\Delta/2} e^2 de = \frac{\Delta^2}{12}$$

$$y(n) = k_q q(n) + e(n)$$

BASICS OF A/D CONVERSION

Quantization White Noise Model

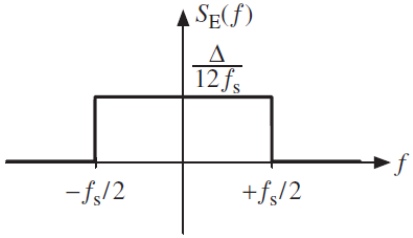
- The power of the *e(n)* will also be uniformly distributed in the range $\left[\frac{-f_s}{2}, \frac{+f_s}{2}\right]$.

Power Spectral Density

$$\overline{e^2} = \int_{-\infty}^{+\infty} S_E(f)df = S_E \int_{\frac{-f_s}{2}}^{\frac{+f_s}{2}} df = \frac{\Delta^2}{12}$$

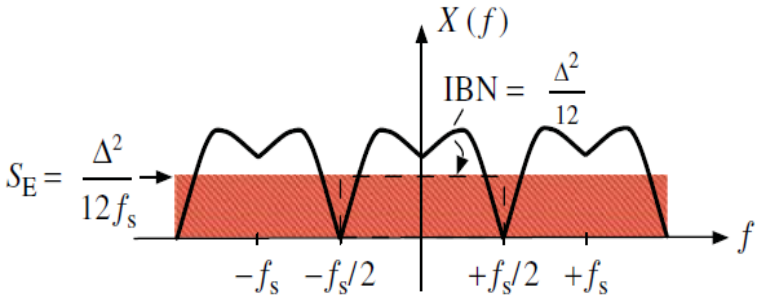
- The PSD of the quantization error is given by:

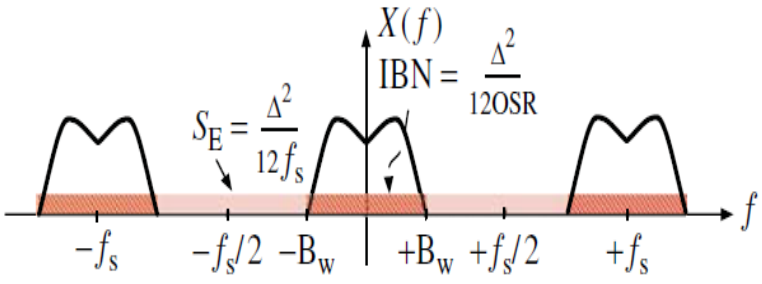
$$S_E = \frac{\overline{e^2}}{f_s} = \frac{\Delta^2}{12 f_s}$$



- We define the *in-band noise power* (IBN) as

$$IBN = \int_{-B_w}^{+B_w} S_E(f)df = S_E \int_{-B_w}^{+B_w} \frac{\Delta^2}{12 f_s} df = \frac{\Delta^2}{12 OSR} \qquad OSR = \frac{f_s}{2B_w}$$





BASICS OF A/D CONVERSION

- Dynamic Range of an ideal ADC
  - The *DR* is defined as the ratio of the output power at the frequency of an input sinusoid with maximum amplitude to the in-band quantization noise

$$DR(dB) = 10log_{10}\left(\frac{P_{sig.out.max}}{IBN}\right)$$

  - The output power is:

$$P_{sig.out.max} = \frac{\left(2^N \frac{\Delta}{2}\right)^2}{2} = 2^{2N-3}\Delta^2$$

  - The in-band noise power is:

$$IBN = \frac{\Delta^2}{12OSR}$$

  - The *DR* of an ideal oversampling ADC yields

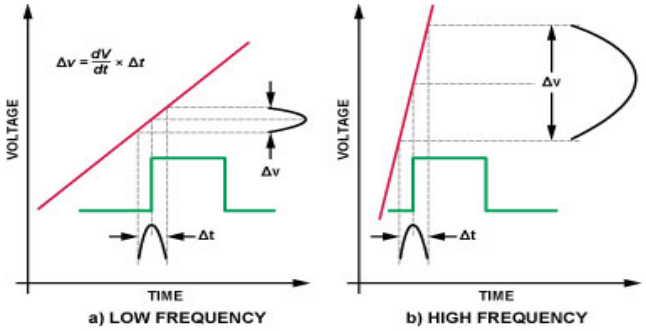


$$DR(dB) = 6.02N + 1.76 + 10log_{10}(OSR)$$

An OSR of 4 is similar to having one extra bit

REAL S&H CIRCUIT: CLOCK JITTER

- The ADC has a sample-and-hold function that is clocked by a sample clock ➔ **Jitter** on the sample clock causes the sampling instants to vary from the ideal sample time.



SNR of a perfect ADC with N bits resolution

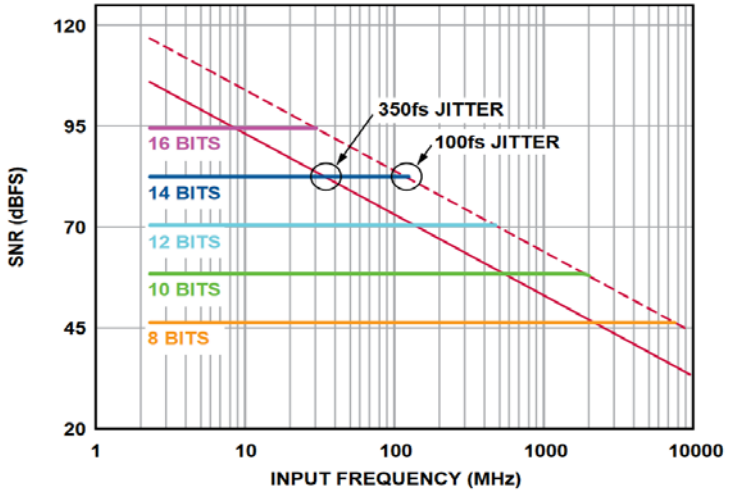
$$SNR_{bits}(dB) = 6.02N + 1.76$$

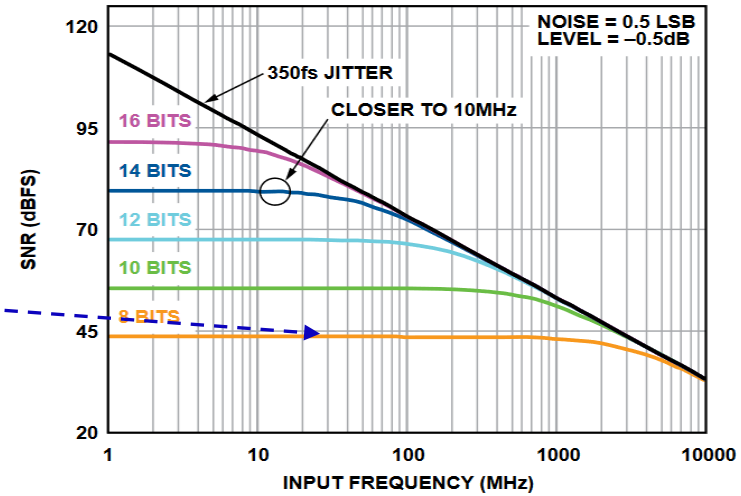
SNR of a perfect ADC having infinite resolution

$$SNR_{ideal}(dB) = 20\ log_{10}\left[\frac{1}{2\pi f_a t_{jrms}}\right]$$

$$SNR = -20\log_{10}\sqrt{(2\pi f_a\ t_{j\ rms})^2 + \frac{2}{3}\left(\frac{1+\varepsilon}{2^N}\right)^2 + \left(\frac{2\sqrt{2}\,V_{NOISE\,rms}}{2^N}\right)^2}$$

SAMPLE-CLOCK JITTER | QUANTIZATION NOISE, DNL | EFFECTIVE INPUT NOISE

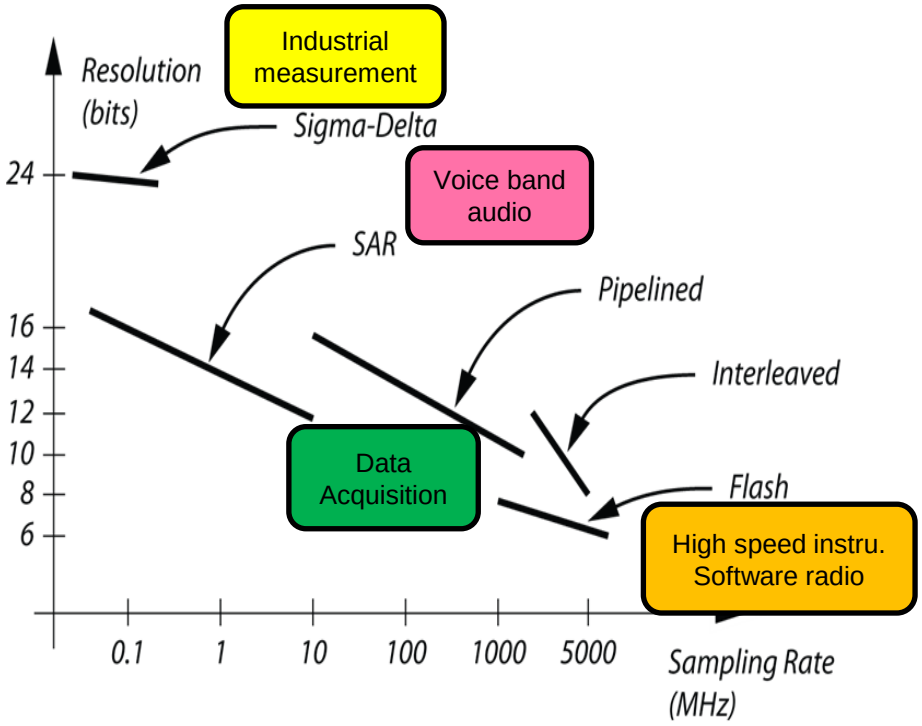




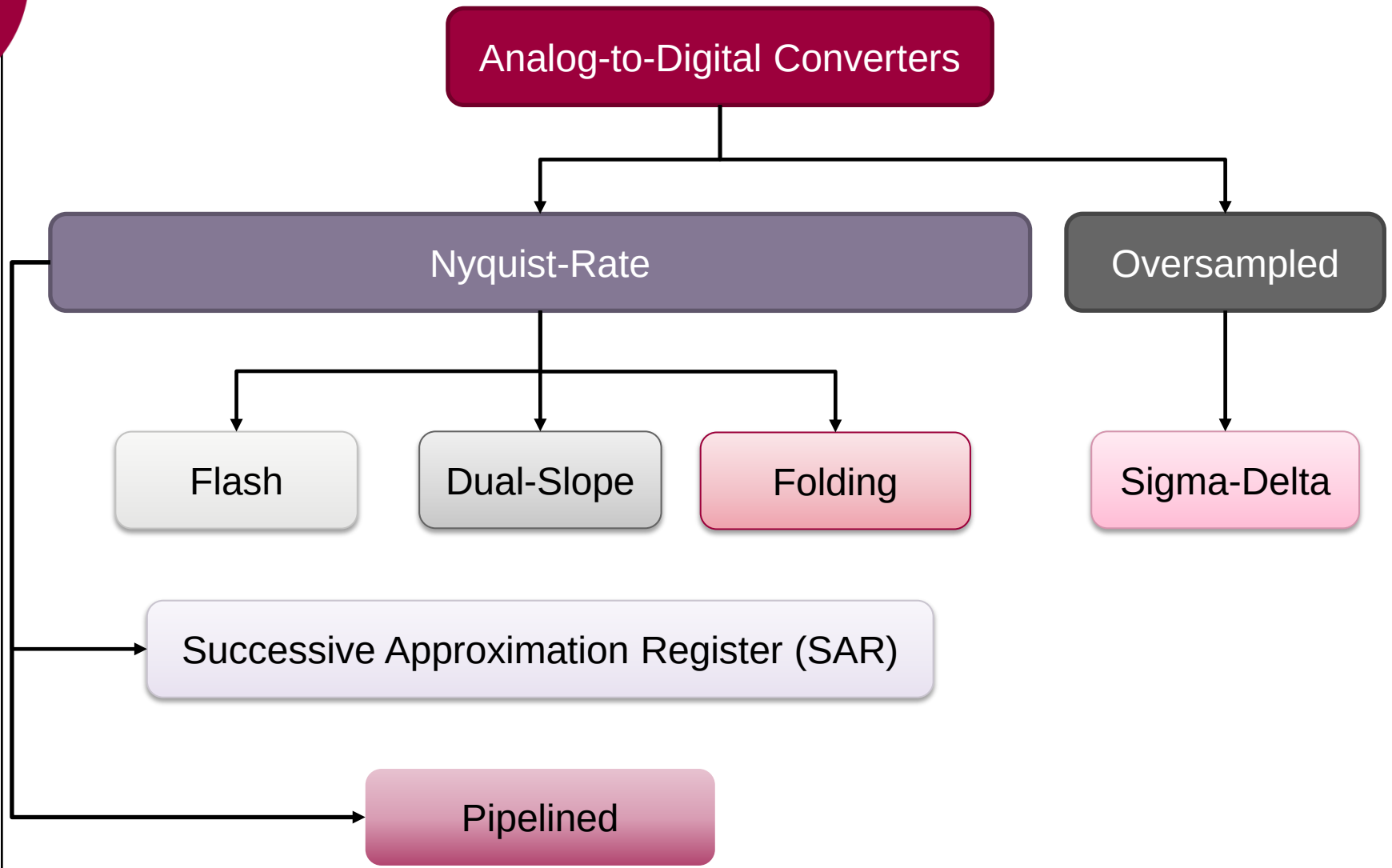
ADC: TRADEOFFS AMONG PARAMETERS

- High speed and high resolution are hard to achieve at the same time.
- Upper limit exists for speed-resolution product
- Different architectures, based on different search strategies, accommodate different speed-resolution requirements

Resolution (in bits)	Range	± Range	Quantization Error	Maximum Sampling Rate
24	16,777,216	± 8,388,608	± 0.000003%	200 KHz
16	65,536	± 32,768	± 0.0008%	250 MHz
14	16,384	± 8,192	± 0.003%	400 MHz
12	4,096	± 2,048	± 0.012%	1.8 GHz
10	1,024	± 512	± 0.05%	2.2 GHz
8	256	± 128	± 0.2%	3 GHz
6	64	± 32	± 0.8%	6 GHz



ADC ARCHITECTURES

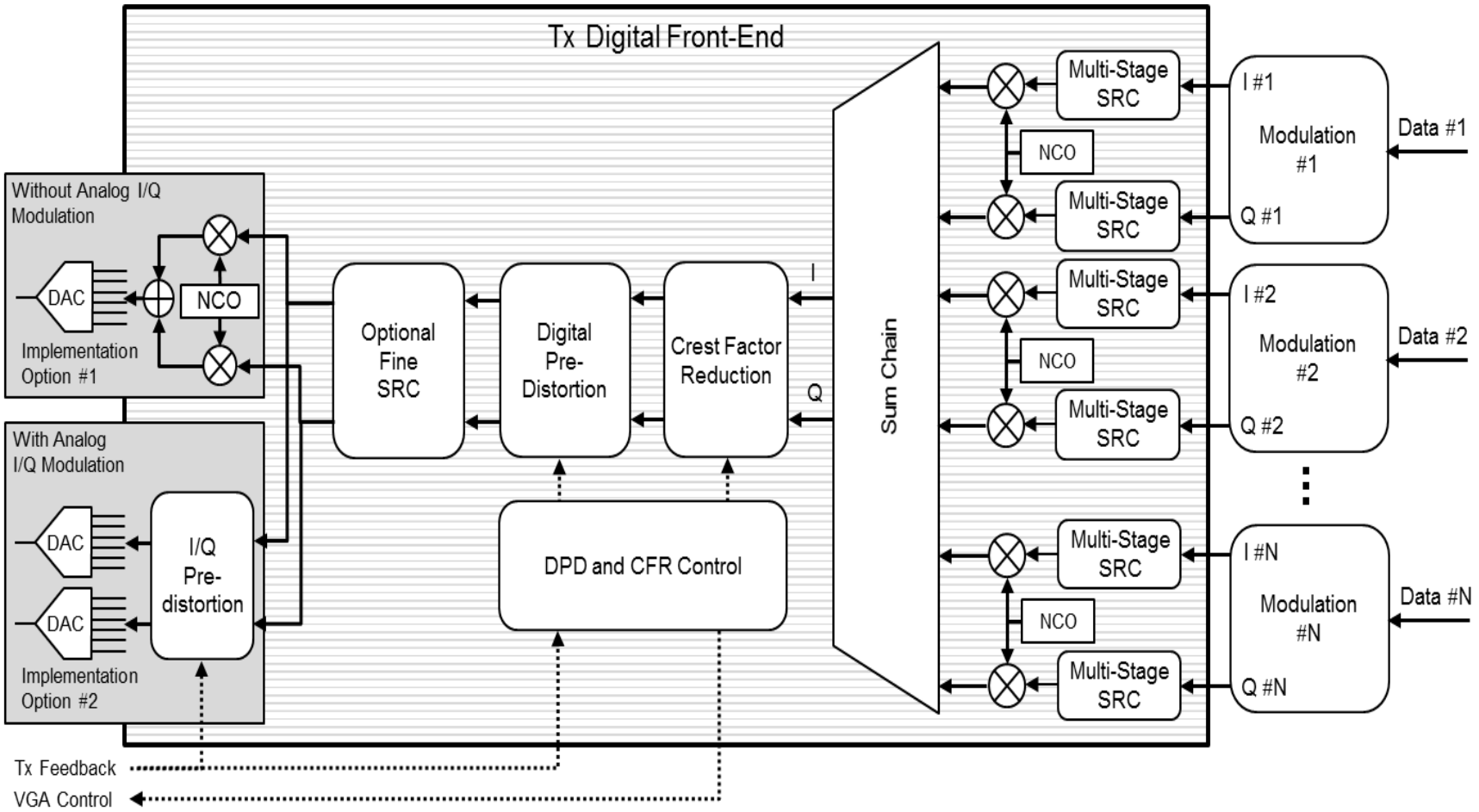


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PART V:

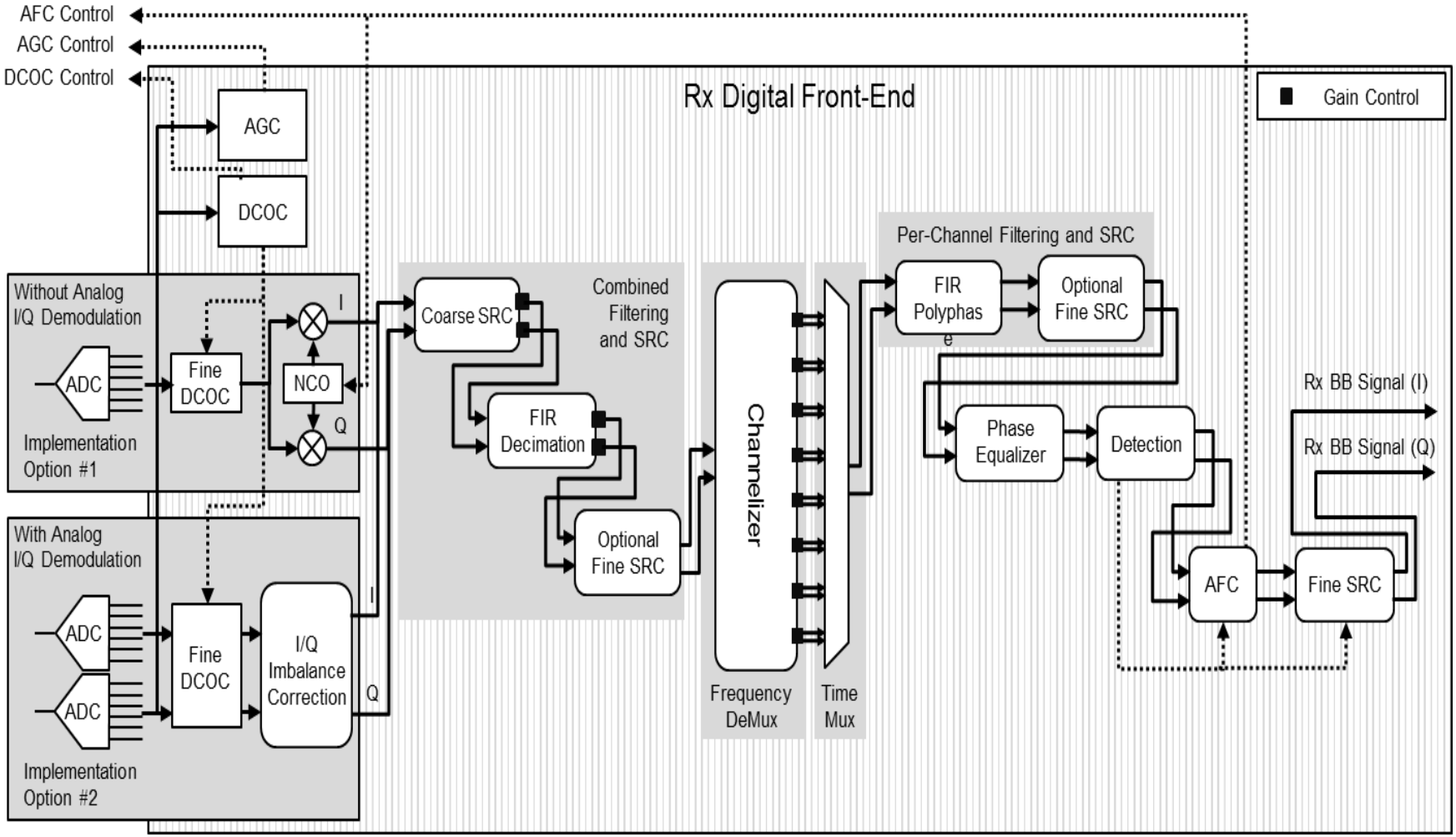
DIGITAL FRONT-END (DFE)

TX GENERIC DFE ARCHITECTURE



http://www.theses.fr/fr/2019CSUP0001

RX GENERIC DFE ARCHITECTURE

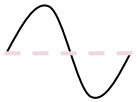


http://www.theses.fr/fr/2019CSUP0001

SRC MAIN IDEA

F_S: Sampling Frequency (Hz or samples/second)

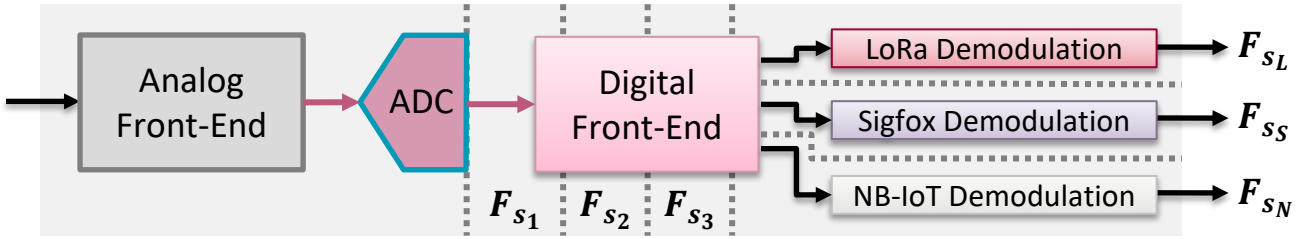
Time domain



Analog signal

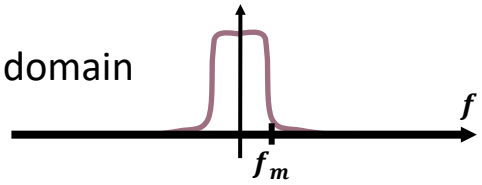


Digital signal

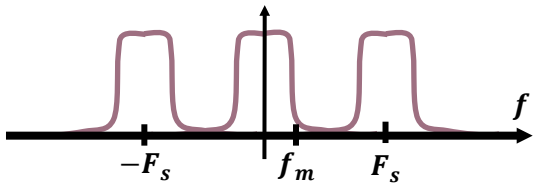


Frequency domain

Analog signal



Digital signal

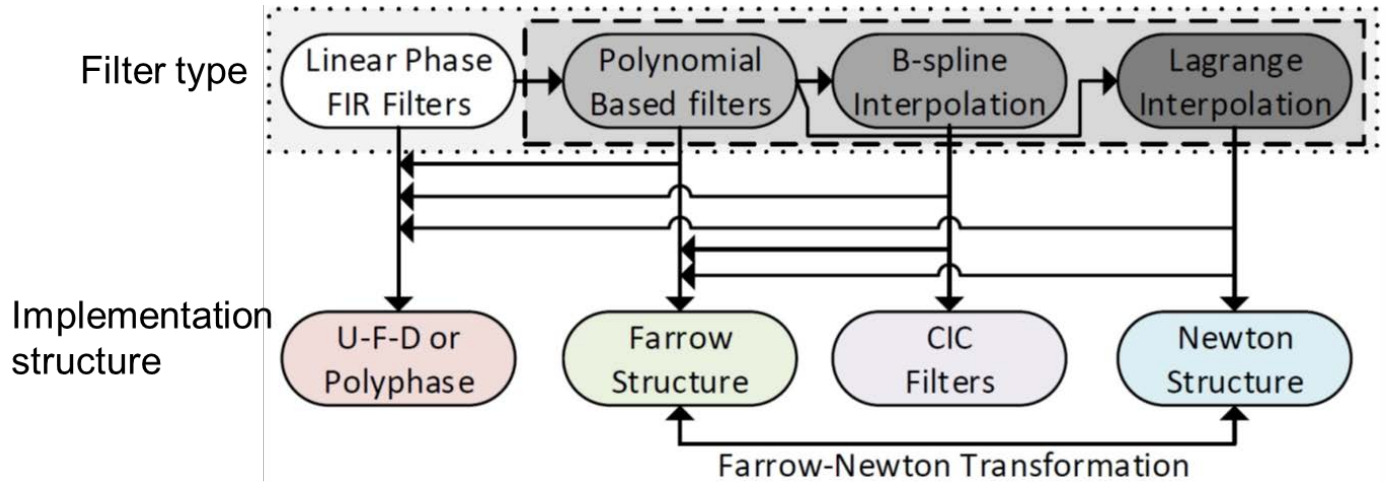


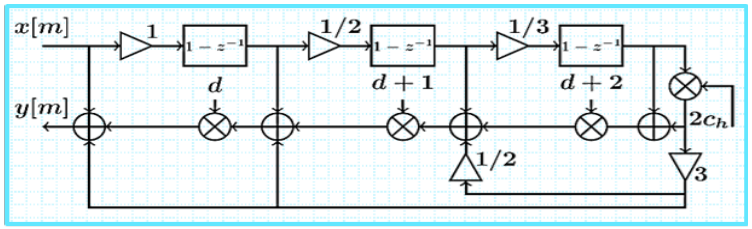
Nyquist-Shannon Sampling Theorem

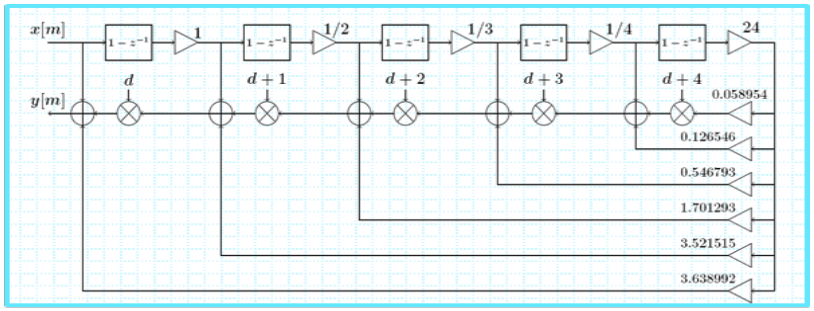
$$F_s \geq 2f_m$$

http://www.theses.fr/fr/2019CSUP0001

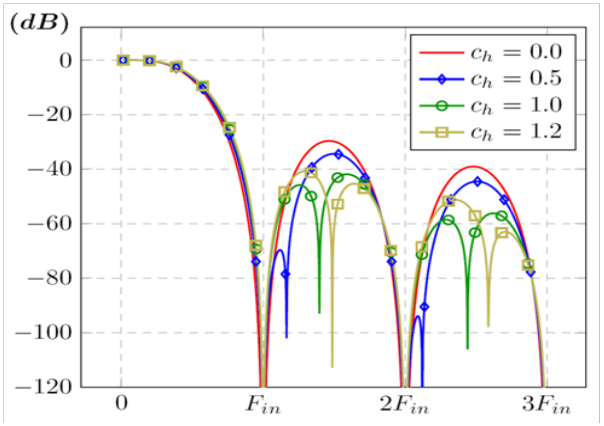
SRC: UNIFIED VISION



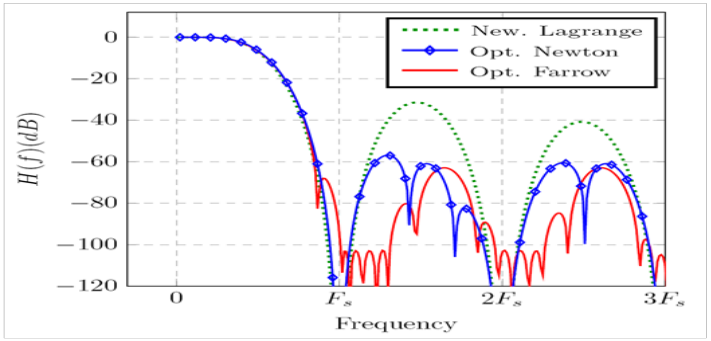




Lagrange-Hermite

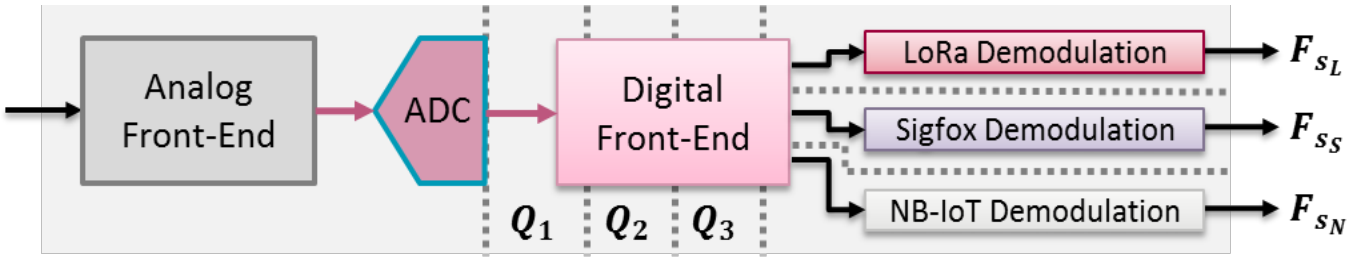


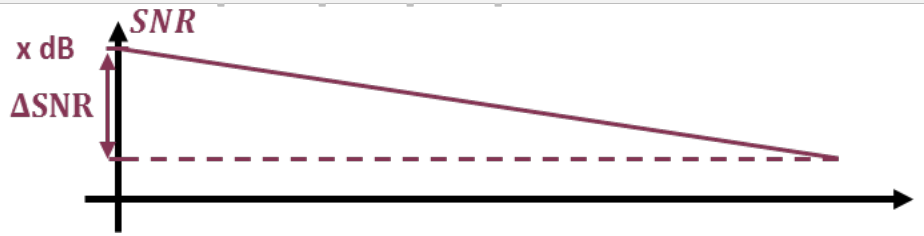
Newton Structure

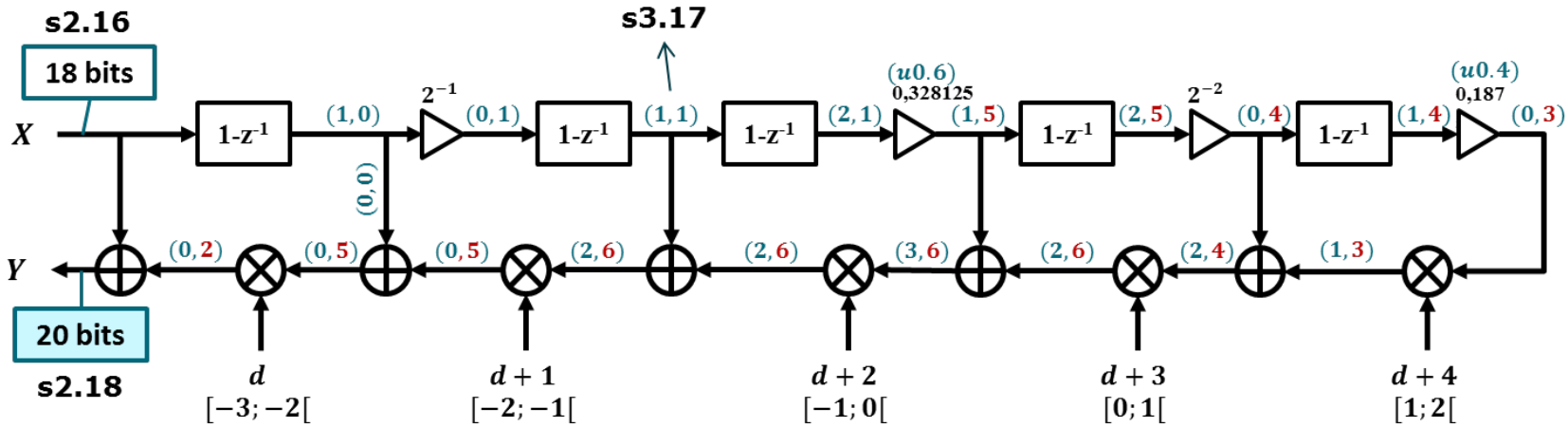


https://link.springer.com/article/10.1007/s11265-020-01575-6

SRC: QUANTIZATION





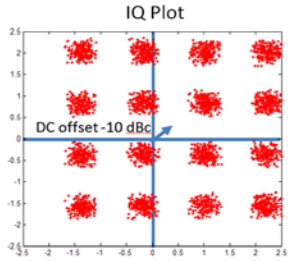


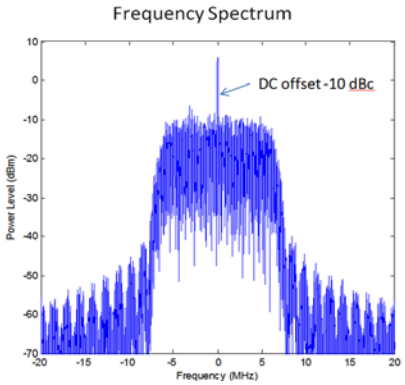
Interpolation	69 bit Quantization	18 bit Quantization	20 bit Quantization
Lagrange 5	$\Delta SNR = 0$ dB	$\Delta SNR \approx -18,5$dB (3,1 bit)	$\Delta SNR \approx -0,6$ dB (0,1 bit)

https://link.springer.com/article/10.1007/s11265-020-01575-6

DC OFFSET / IQ IMBALANCE

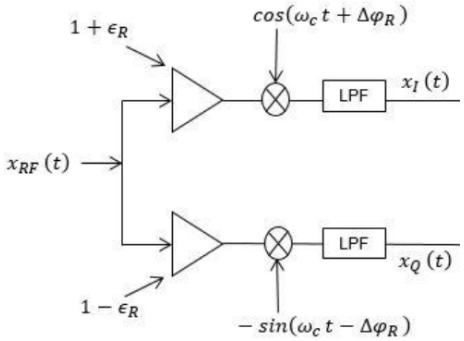
- DC offset is an offsetting of a signal from zero

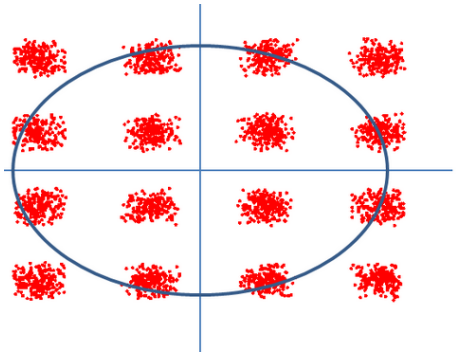


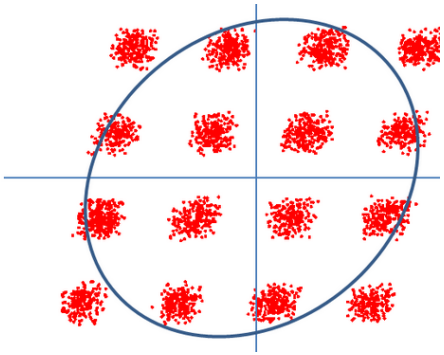


  - USRP calibration utilities: *>>uhd_cal_tx_dc_offset --verbose --args=<optional device args>*

- IQ imbalance is due to the local oscillator







  - USRP calibration utilities: *>>uhd_cal_tx_iq_balance --verbose --args=<optional device args>*
    >>uhd_cal_rx_iq_balance --verbose --args=<optional device args>

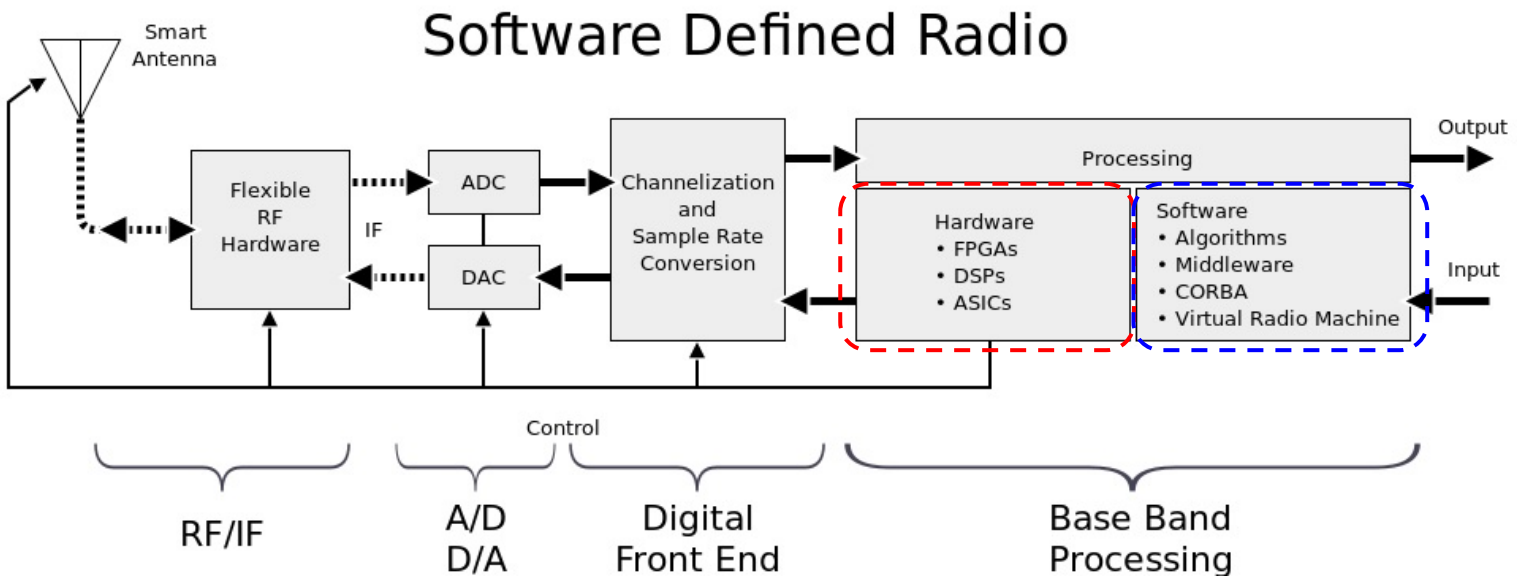
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PART VI:

DIGITAL BASE-BAND (DBB)

DBB: PROCESSING

- Hardware Processing / Software Processing



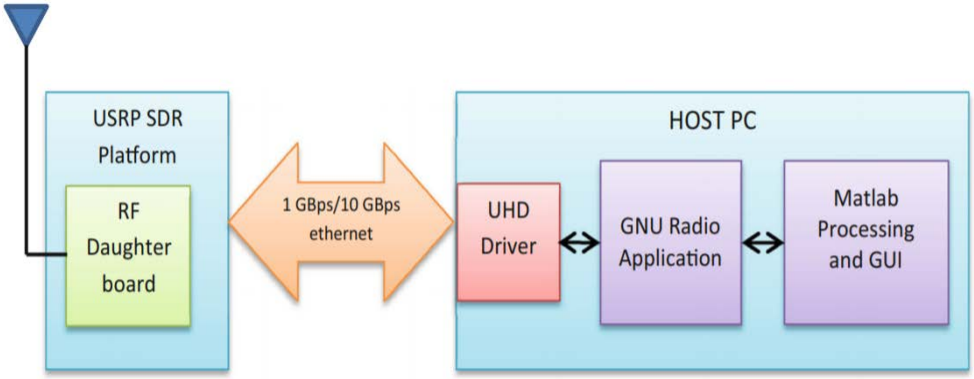
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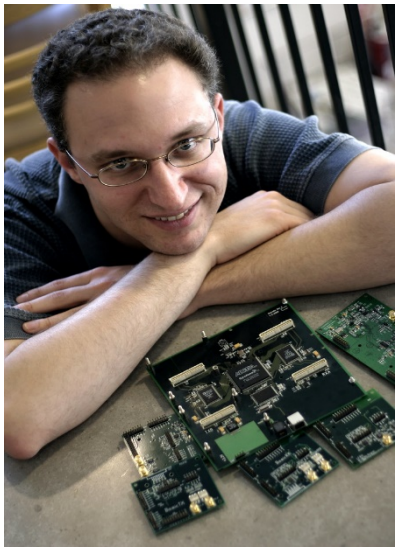
PART VII:

SDR HARDWARE PLATFORMS

USRP PLATFORM

- Universal Software Radio Peripheral





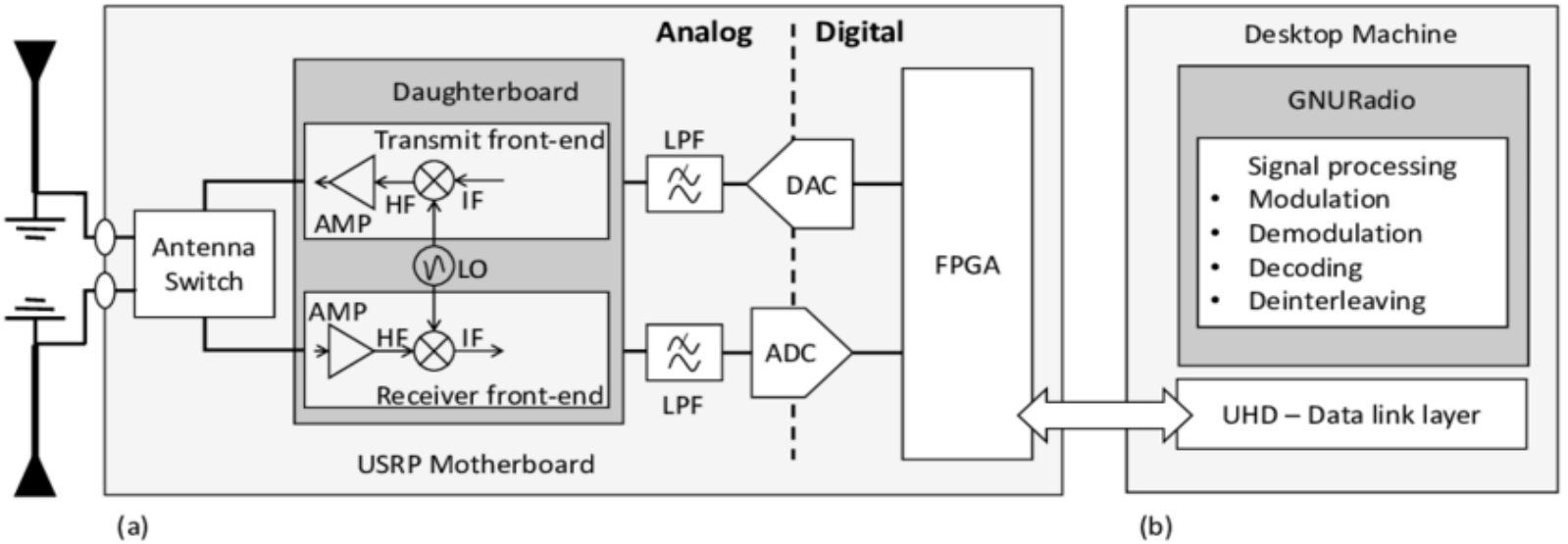
Matt Ettus 2012
(Ettus Research 2004)

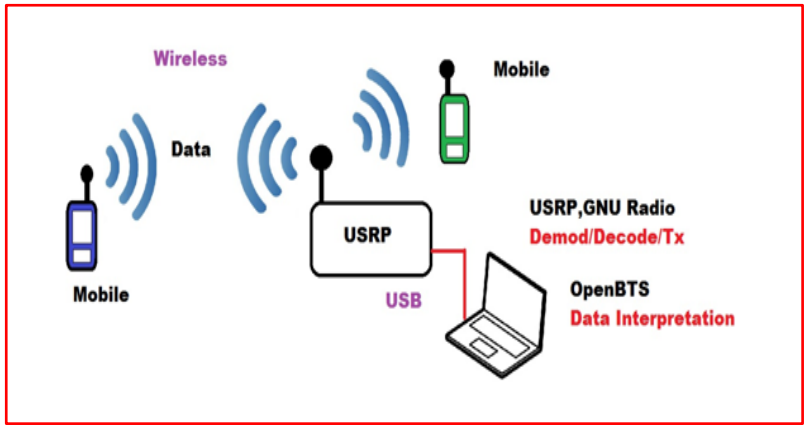


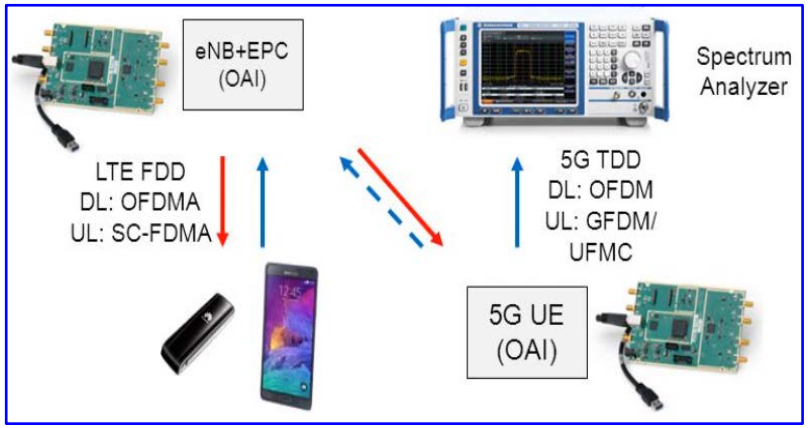
NI acquired Ettus Research (2010)

USRP PLATFORM

- Ettus research Founded in 2004.
- Compatibility with GNU Radio.







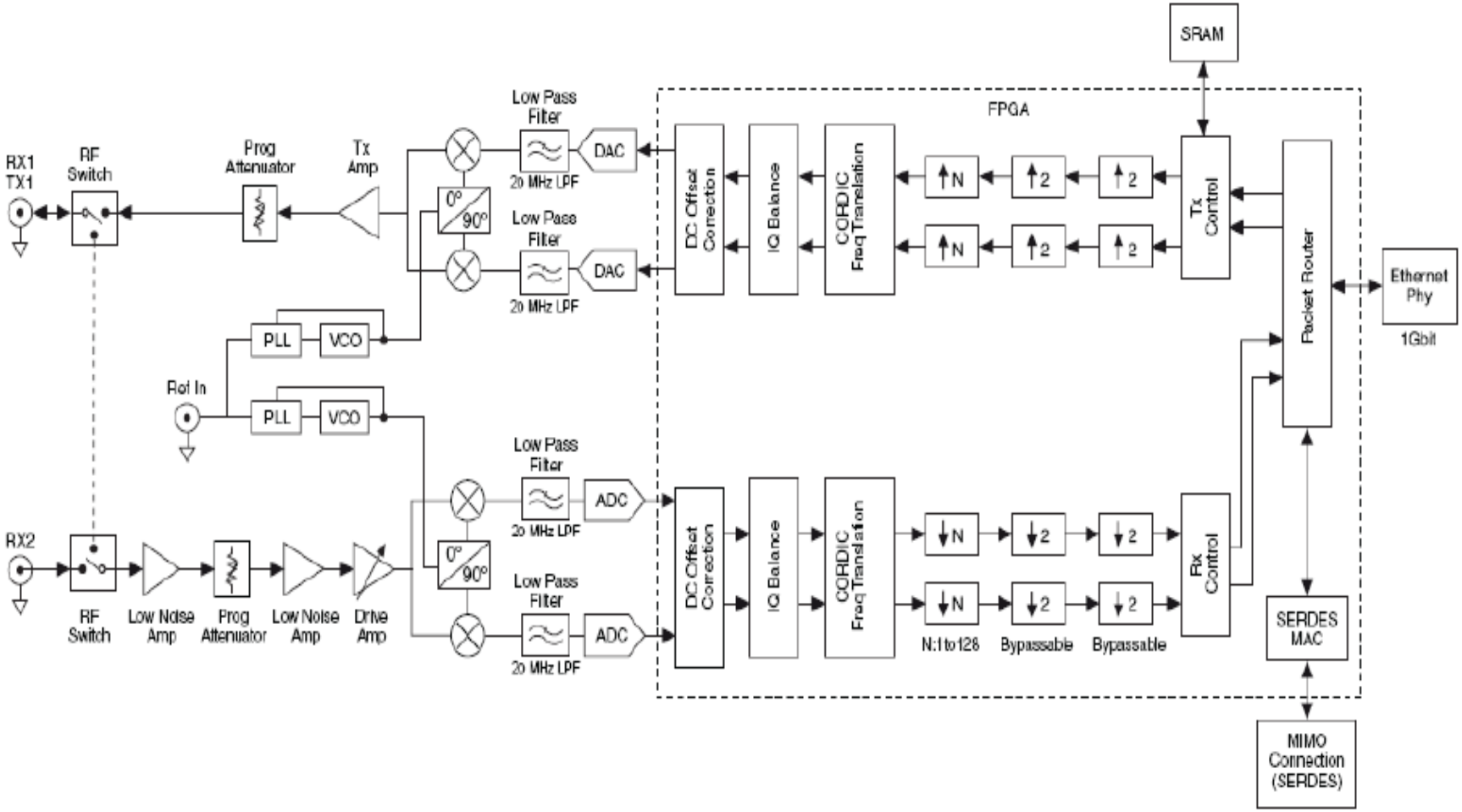
USRP PLATFORM

- USRP families
  - B-series (B200, B210, B200 mini): USB 3.0 host interface
  - N-series (N200, N210, N300/N310, N320/N321): 1 Gb Ethernet host interface
  - X-series (X300, X310): 1 and 10 Gb Ethernet host interface
  - E-series (E310, E312, E313, E320): Embedded stand-alone SDR with ARM CPU

- USRP Daughterboards (for N-series and X-series only)
  - BasicTX and BasicRX: no tuner, 1 MHz to 250 MHz
  - LFTX and LFRX: no tuner, 0 Hz to 30 MHz
  - WBX: 50 MHz to 2200 MHz, 40 or 120 MHz capture bandwidth, 1 Tx and 1 Rx
  - SBX: 400 MHz to 4400 MHz, 40 or 120 MHz capture bandwidth, 1 Tx and 1 Rx
  - CBX: 1200 MHz to 6000 MHz, 40 or 120 MHz capture bandwidth, 1 Tx and 1 Rx
  - UBX: 10 MHz to 6000 MHz, 40 or 160 MHz capture bandwidth, 1 Tx and 1 Rx
  - TwinRX: 10 MHz to 6000 MHz, 80 MHz capture bandwidth, Dual Rx Only, No Tx

USRP ARCHITECTURE

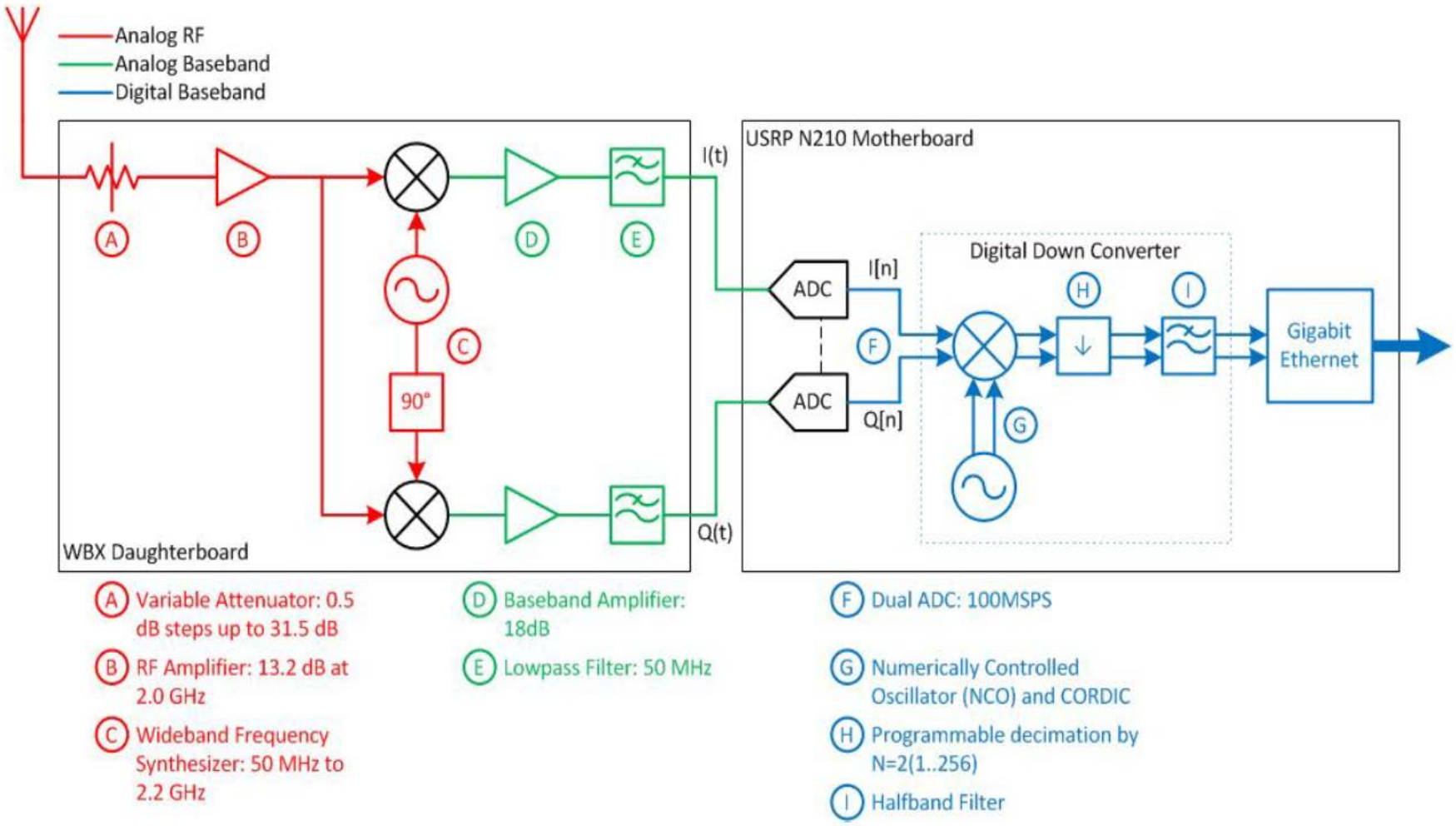
- The second-generation USRP architecture



https://www.ettus.com/

USRP ARCHITECTURE

- RX architecture and parameters



USRP MODEL COMPARISON






	Bus B2xx	Embedded E3xx	Networked N2xx	High Performance X3xx
Frequency (Hz)	70 M – 6 G	70 M – 6 G	DC-30M & 10M–6G	DC-30M & 10M–6G
Bandwidth	56MHz (32 MHz in 2x2)	56MHz (32 MHz in 2x2)	40 MHz	160 MHz
Channels	2 Tx, 2 Rx	2 Tx, 2 Rx w/ filter banks	1 Tx, 1Rx	2 Tx, 2 Rx
RF Performance	Good	Good	Better	Best
Architecture	Integrated RF	Integrated RF	RF Daughterboard	RF Daughterboards
Communication	USB	Embedded	1GbE	10GbE or PCIe
MIMO Capability	2x2	2x2	Up to 2x2	2x2 to 256x256
LabVIEW Support	Yes	No	Yes	Yes
NI Version	USRP-290x	None	USRP-292x USRP-293x	USRP-294x USRP295x
S/W Ecosystem	GNU Radio C++ MatLab Xilinx ISE	GNU Radio C++ Xilinx Vivado C Coder HDL Coder	GNU Radio C++ MatLab Xilinx ISE	GNU Radio C++ MatLab Xilinx Vivado Simulink C Coder HDL Coder

*Same Daughterboard in each slot

https://www.ettus.com/

USRP MODEL B200/B210



USRP B200

- Xilinx Spartan 6 XC6SLX75 FPGA
- Analog Devices AD9364 RFIC direct-conversion transceiver
- Frequency range: 70 MHz - 6 GHz
- Up to 56 MHz of instantaneous bandwidth
- Full duplex, SISO (1 Tx & 1 Rx)
- Fast and convenient bus-powered USB 3.0 connectivity
- Optional Board Mounted GPSDO

USRP B210

- Xilinx Spartan 6 XC6SLX150 FPGA
- Analog Devices AD9361 RFIC direct-conversion transceiver
- Frequency range: 70 MHz - 6 GHz
- Up to 56 MHz of instantaneous bandwidth (61.44MS/s quadrature)
- Full duplex, MIMO (2 Tx & 2 Rx)
- Fast and convenient bus-powered USB 3.0 connectivity
- Optional Board Mounted GPSDO

https://www.ettus.com/

USRP MODEL B200MIN



- USRP B200mini
  - Xilinx Spartan-6 XC6SLX75 FPGA
  - Analog Devices AD9364 RFIC direct-conversion transceiver
  - Frequency range: 70 MHz to 6 GHz
  - Up to 56 MHz of instantaneous bandwidth
  - Full duplex, SISO (1 Tx & 1 Rx)
  - Power from the USB 3.0 bus
  - Power Output >10dBm
  - IIP3 (@ typ NF) -20dBm
  - Typical Noise Figure <8dB
  - Maximum Input Power: 0 dBm
- Optional configuration: B200mini-i
  - Industrial-grade XC6SLX75 FPGA
- Optional configuration: B205mini-i
  - Industrial-grade XC6SLX15 FPGA

Slice Logic Utilization:				
Number of Slice Registers:	15949	out of	93296	17%
Number of Slice LUTs:	19963	out of	46648	42%
Number used as Logic:	16140	out of	46648	34%
Number used as Memory:	3823	out of	11072	34%
Number used as RAM:	972			
Number used as SRL:	2851			

Slice Logic Utilization:				
Number of Slice Registers:	15949	out of	184304	8%
Number of Slice LUTs:	19963	out of	92152	21%
Number used as Logic:	16140	out of	92152	17%
Number used as Memory:	3823	out of	21680	17%
Number used as RAM:	972			
Number used as SRL:	2851			

https://www.ettus.com/

USRP MODEL E3XX



- E3xx specifications
  - Wide frequency range: 70 MHz to 6 GHz
  - Based on Analog Devices AD9361 Transceiver
  - Up to 56 MHz of instantaneous bandwidth
  - 2x2 MIMO transceiver
  - RX and TX filter banks
  - Xilinx Zynq 7020 FPGA
  - ARM Cortex A9 866 MHz dual-core CPU
  - Up to 10 MS/s sample data transfer rate to ARM processor
  - Synchronization with PPS time reference
  - Integrated GPS receiver (**not** GPSDO)
  - Rich set of peripherals such as host USB and 1 Gigabit Ethernet

	USRP N210	USRP X300	USRP X310
FPGA	Spartan3 XC3SD3400A	Kintex 7-325T	Kintex 7 -410T
Logic Cells	53k	328k	406k
Memory	2,268 Kb	16,020 Kb	28,620 Kb
Multipliers	126	840	1540
Clock Rate	100 MHz	200 MHz	200 MHz
Streaming Bandwidth per Channel (16-bit)	25 MS/s	200 MS/s	200 MS/s

https://www.ettus.com/

USRP MODEL E3XX



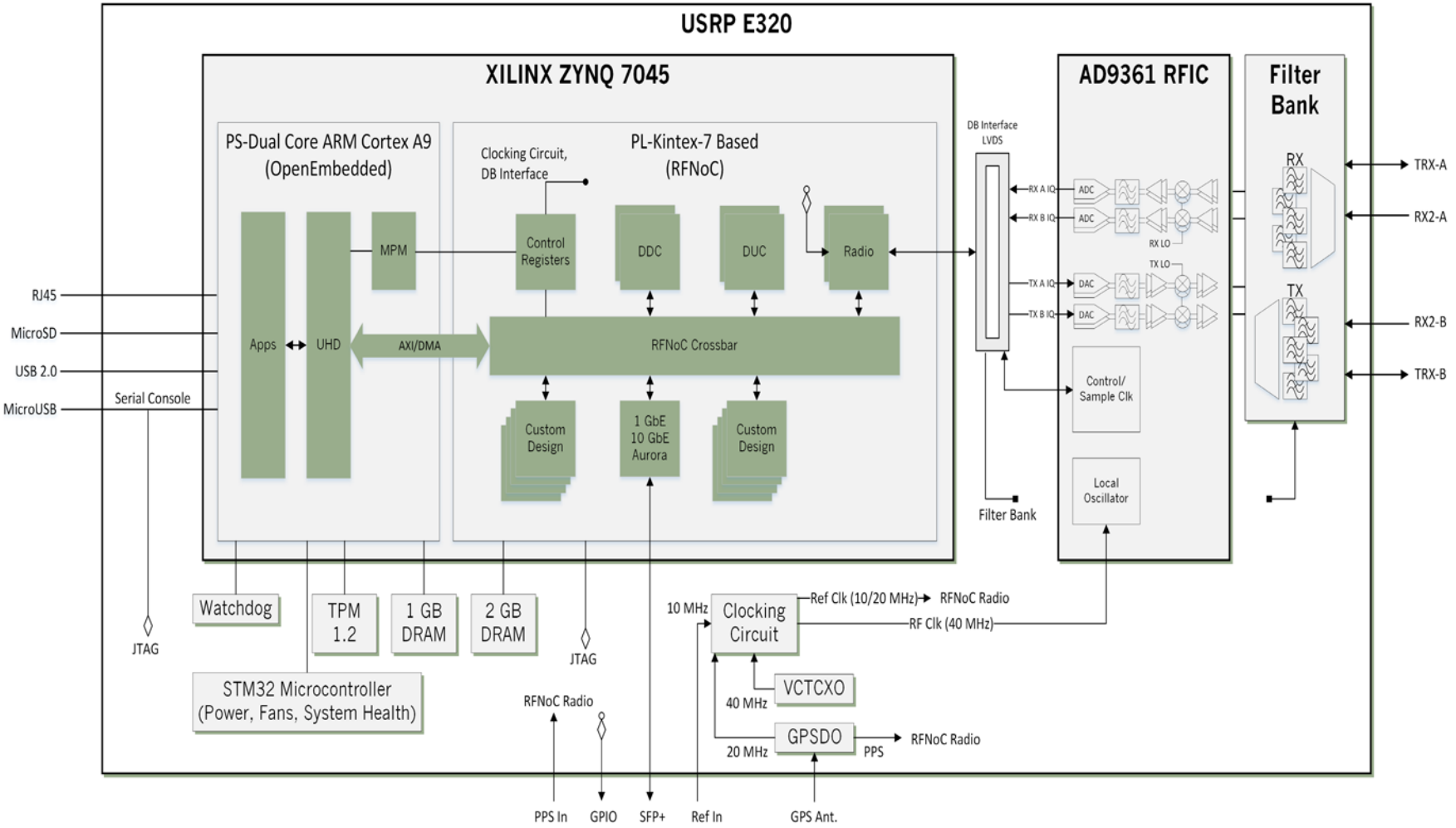
- E3xx specifications
  - Wide frequency range: 70 MHz to 6 GHz
  - Based on Analog Devices AD9361 Transceiver
  - Up to 56 MHz of instantaneous bandwidth
  - 2x2 MIMO transceiver
  - RX and TX filter banks
  - Xilinx Zynq-7045 SOC
  - Dual-core ARM Cortex-A9 866 MHz CPU with 1 GB DDR3 RAM
  - 7 Series FPGA with 2 GB DDR3 RAM
  - Up to 10 MS/s sample data transfer rate to ARM processor
  - Synchronization with PPS time reference
  - Integrated GPS receiver (**not** GPSDO)
  - Rich set of peripherals such as host USB and 1 Gigabit Ethernet
  - Stand-alone (embedded) or host-based (network streaming) operation

https://www.ettus.com/

USRP MODEL E3XX



- E320 block diagram



https://www.ettus.com/

USRP MODEL N2XX



- N2xx specifications
  - Xilinx Spartan 3A-DSP XC3SD1800A (N200)/ XC3SD3400A (N210)
  - 50 MHz (8-bit samples) / 25 MHz (16-bit samples) RF bandwidth
  - Frequency range: DC - 6 GHz with suitable daughterboard
  - Gigabit Ethernet connectivity
  - Ability to lock to external 5 or 10 MHz clock reference
  - Optional internal GPS locked reference oscillator

POWER		
DC Input	6	V
Current Consumption	1.3	A
w/ WBX Daughterboard	2.3	A
CONVERSION PERFORMANCE AND CLOCKS		
ADC Sample Rate	100	MS/s
ADC Resolution	14	bits
ADC Wideband SFDR	88	dBc
DAC Sample Rate	400	MS/s
DAC Resolution	16	bits
DAC Wideband SFDR	80	dBc
Host Sample Rate (8b/16b)	50/25	MS/s
Frequency Accuracy	2.5	ppm
w/ GPSDO Reference	0.01	ppm

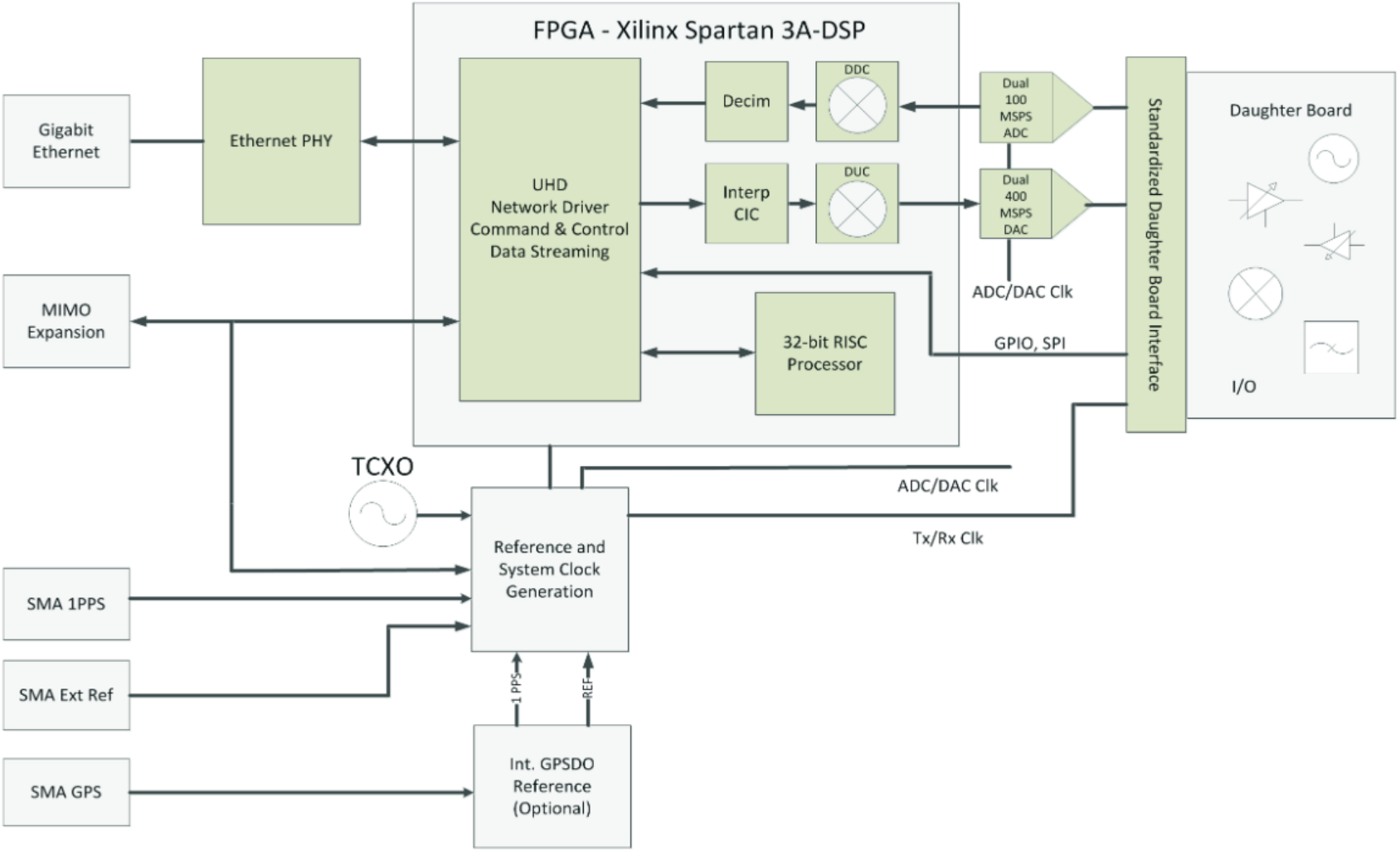
RF PERFORMANCE (W/ WBX)		
SSB/LO Suppression	35/50	dBc
Phase Noise (1.8 Ghz)		
10 kHz	-80	dBc/Hz
100 kHz	-100	dBc/Hz
1 MHz	-137	dBc/Hz
Power Output	15	dBm
IIP3	0	dBm
Receive Noise Figure	5	dB
PHYSICAL		
Operating Temperature	0 to 55°	C
Dimensions (l x w x h)	22x16x5	cm
Weight	1.2	kg

https://www.ettus.com/

USRP MODEL N2XX



- N200 block diagram



https://www.ettus.com/

USRP MODEL X300/X310



X300/X310 specifications

- Xilinx Kintex-7 XC7K325T FPGA (X300) / XC7K410T FPGA (X310)
- Frequency range: DC to 6 GHz with daughterboard
- Up to 160 MHz bandwidth per channel
- Two wide-bandwidth RF daughterboard slots
- Optional internal GPSDO
- Multiple high-speed interfaces
  - Single 1 Gbps Ethernet
  - Single 10 Gbps Ethernet
  - Dual 10 Gbps Ethernet
  - PCIe Gen 2

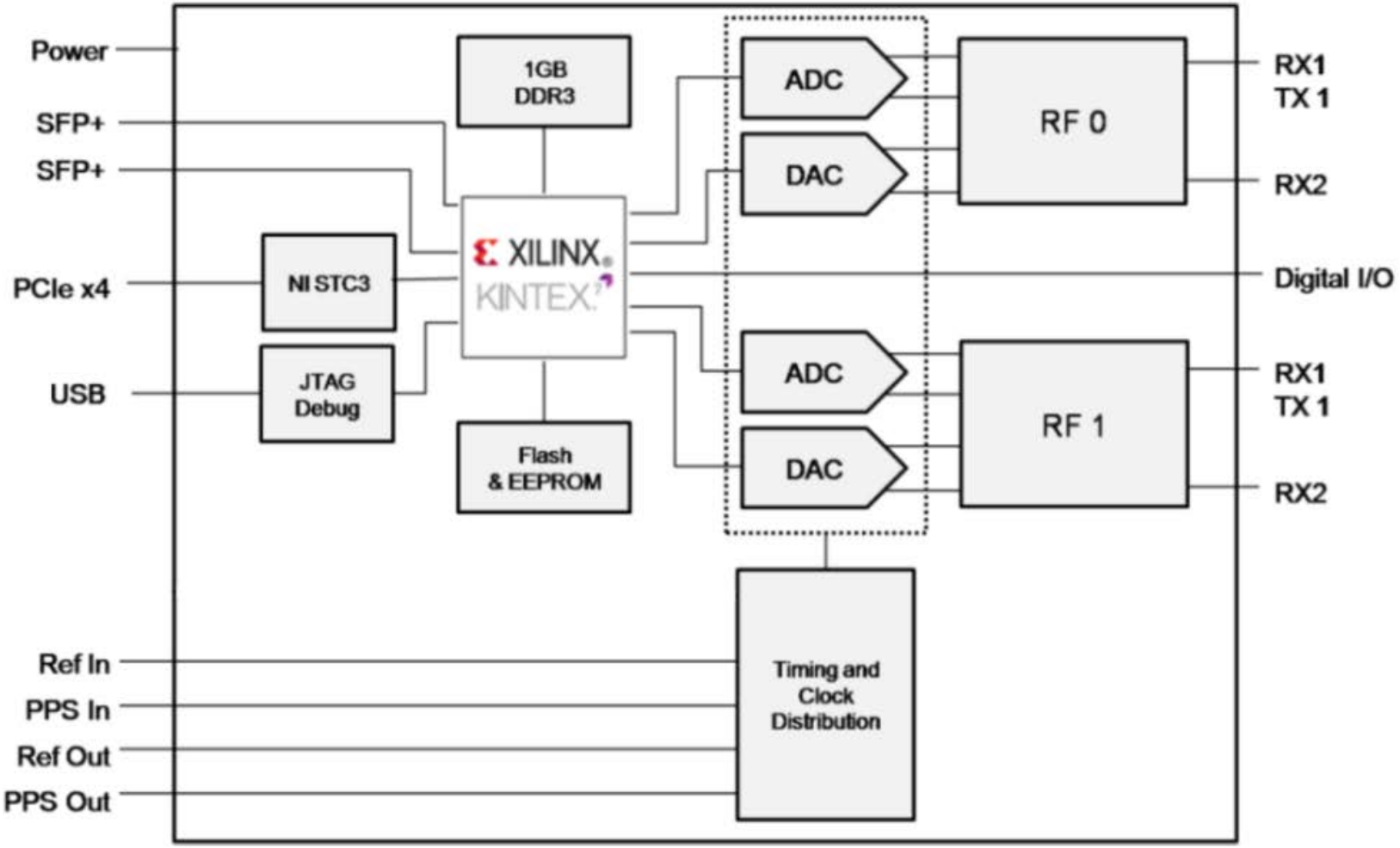
	USRP N210	USRP X300	USRP X310
FPGA	Spartan3 XC3SD3400A	Kintex 7-325T	Kintex 7 -410T
Logic Cells	53k	328k	406k
Memory	2,268 Kb	16,020 Kb	28,620 Kb
Multipliers	126	840	1540
Clock Rate	100 MHz	200 MHz	200 MHz
Streaming Bandwidth per Channel (16-bit)	25 MS/s	200 MS/s	200 MS/s

https://www.ettus.com/

USRP MODEL X300/X310



- X310 block diagram



https://www.ettus.com/

ETTUS VERSUS NI SDR PRODUCT & SUPPORT

- NI versus Ettus research SDR model number

Bus Series	
NI-USRP	ER-USRP
NI-2900	B200
NI-2901	B210

Network Series	
NI-USRP	ER-USRP
NI-2920	N210 + WBX
NI-2921	N210 + XCVR2450
NI-2922	N210 + SBX
NI-2930	N210 + WBX + GPSDO
NI-2932	N210 + SBX + GPSDO

X Series	
NI-USRP	ER-USRP
NI-2940R	X310 + WBX (x2)
NI-2942R	X310 + SBX (x2)
NI-2943R	X310 + CBX (x2)
NI-2944R	X310 + UBX (x2)
NI-2945R	X310 + TwinRX (x2)

X Series	
NI-USRP	ER-USRP
NI-2950R	X310 + WBX (x2) + GPSDO
NI-2952R	X310 + SBX (x2) + GPSDO
NI-2953R	X310 + CBX (x2) + GPSDO
NI-2954R	X310 + UBX (x2) + GPSDO
NI-2955R	X310 + TwinRX (x2) + GPSDO

https://www.ni.com/nl-nl/innovations/white-papers/19/what-is-the-difference-between-ni-and-ettus-usrps.html

SAMPLING RATE

- Integer decimation of the Master Clock Rate (MCR)
  - Even decimation rate preferred
  - Odd decimation rate allowed but with CIC filter roll-off attenuation
  - For N200, N210:
    - MCR is 100 MHz
    - Decimation rates from 1 to 512
    - Sample rates: 100, 50, 25, 16.67, 12.5, 10, 8.33 MHz, … 195.31 KHz
  - For X300, X310:
    - MCR is 200 MHz
    - Decimation rates from 1 to 1024
    - Sample rates: 200, 100, 50, 33.33, 25, 20, 16.67, 14.29, 12.5 MHz, … 195.31 KHz
    - MCR 184.32 MHz also supported
  - For N300, N310:
    - MCR are 122.88, 125, 153.6 MHz
    - Decimation rates from 1 to 1024
  - For B200, B210, B200mini, B205mini, E310, E312, E320:
    - All based on AD9361
    - MCR can be anything between 1 MHz and 61.44 MHz (30.76 MHz in 2x2)
    - Decimation rates between 1 and 1024

ETTUS VERSUS NI SDR PRODUCT & SUPPORT

- Product and support comparison

	Ettus Research™ (A National Instruments Brand)	NATIONAL INSTRUMENTS
Hardware	 • Sold Modularly as Kits • More Motherboard and Daughterboard Combinations • Not CE Certified as an Assembled Unit • Stand Alone (N310) and Embedded (E31X) Options	 • Sold Pre-assembled and Tested as an Assembly • CE Certified • Subset of All Radio/Front-End Combos • All NI USRPs Have an Ettus USRP Hardware Equivalent
Software	GNURadio, python, MathWorks, XILINX, VIVADO • Primarily Used with UHD (USRP Hardware Driver) • Provided Open Source • FPGA Modification Framework: RFNoC (VHDL) • Supports Third-Party Tools: VHDL Coder, MATLAB®, Simulink®, Python	LabVIEW • Primarily Used with NI-USRP Driver/LabVIEW • Unified FPGA/ CPU Design Flow • Algorithmic Design Flow • Ability to Import/Export 3rd Party IP (MATLAB, etc.) • Wireless Application Frameworks: 5G, LTE, 802.11, MIMO
Support	• Open Community and Support@ettus.com • Extensive Online Knowledgebase at ettus.com	Supported by NI Standard Service Program

https://www.ni.com/nl-nl/innovations/white-papers/19/what-is-the-difference-between-ni-and-ettus-usrps.html

OTHER PLATFORMS



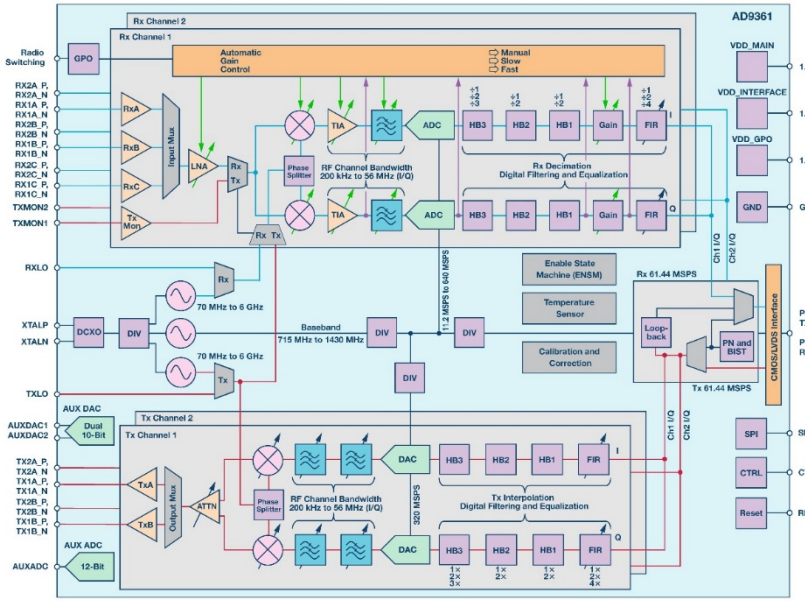
BladeRF 2.0 micro families:

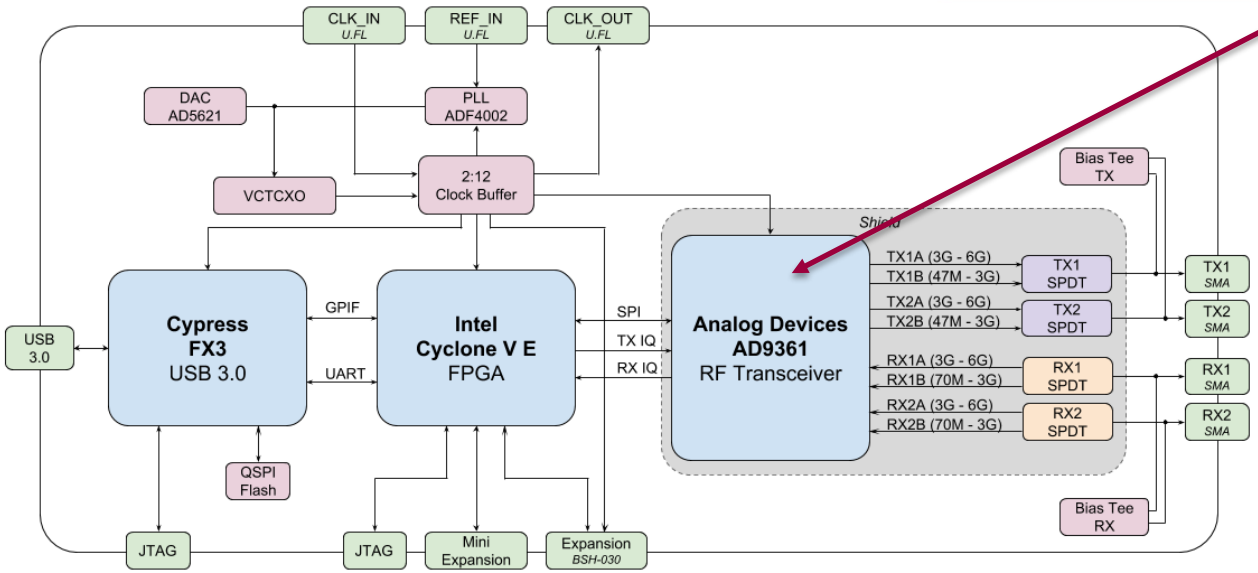
- Fully bus-powered USB 3.0
- ADC/DAC Sample Rate = 61.44 Msps
- ADC/DAC Resolution = 12 bits
- RF Tuning Range (RX) = 70MHz-6GHz
- RF Tuning Range (TX) = 47MHz-6GHz
- FPGA logic elements = 49kLE (xA4) / 301kLE (xA9)
- FPGA Variable-precision DSP blocks = 66 (xA4) / 342 (xA9)
- 2x2 MIMO configurable with SMB cable, expandable up to 4x4
- 56 MHz filtered bandwidth (IBW)
- Automatic gain control (AGC)
- Automatic IQ and DC offset correction
- Support:
  - GNU Radio via gr-osmosdr
  - Pothos via SoapySDR
  - MathWorks MATLAB & Simulink via libbladeRF bindings.

https://www.nuand.com/bladerf-2-0-micro/

OTHER PLATFORMS

- **BladeRF 2.0 micro block diagram:**





https://www.nuand.com/bladerf-2-0-micro/

OTHER PLATFORMS

- **HackRF one specifications:**

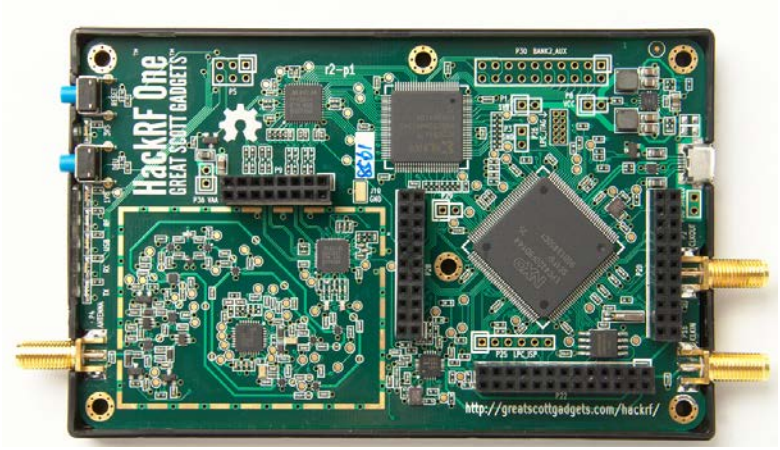


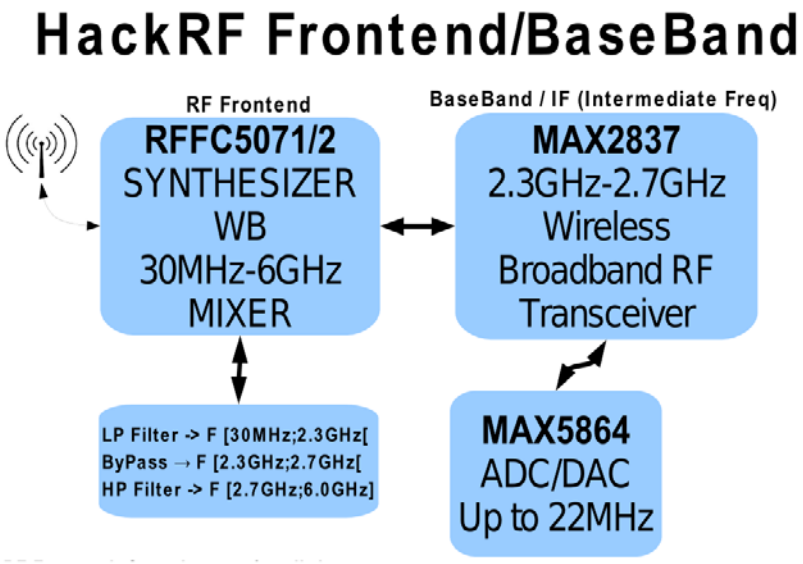
1 MHz to 6 GHz operating frequency	half-duplex transceiver
up to 20 million samples per second	8-bit quadrature samples (8-bit I and 8-bit Q)
compatible with GNU Radio, SDR#, and more	software-configurable RX and TX gain and baseband filter
software-controlled antenna port power (50 mA at 3.3 V)	SMA female antenna connector
SMA female clock input and output for synchronisation	convenient buttons for programming
internal pin headers for expansion	internal pin headers for expansion
USB-powered	open source hardware

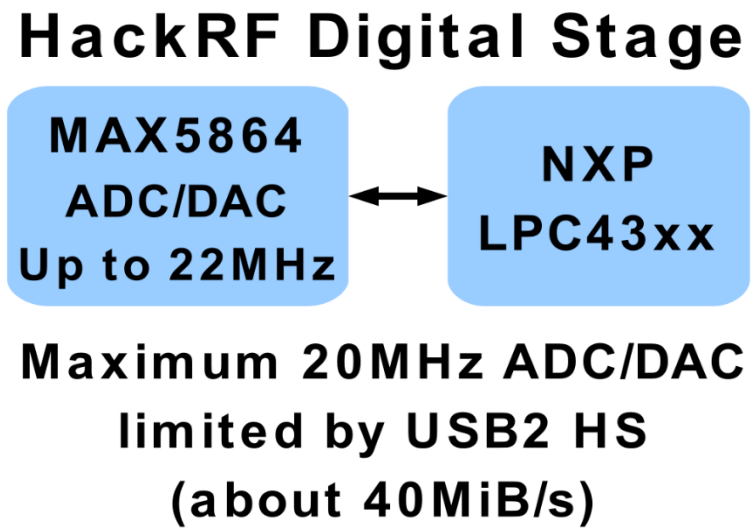
https://github.com/mossmann/hackrf/

OTHER PLATFORMS

- **HackRF block diagram:**



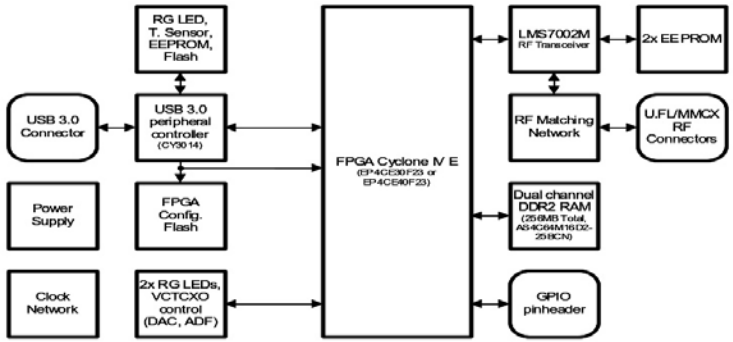




https://github.com/mossmann/hackrf/

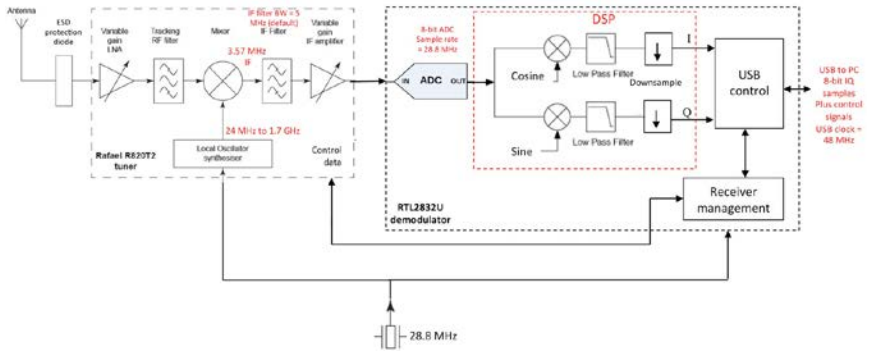
OTHER PLATFORMS

LimeSDR:





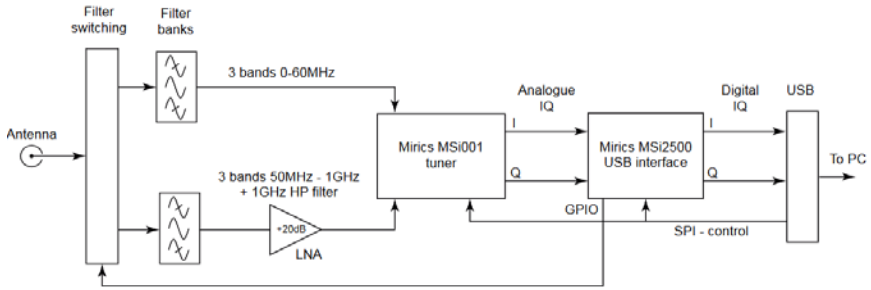
RTL-SDR:







SDRplay:





https://www.sdrplay.com/radiouserreview.pdf

LOW-COST SDR PLATFORMS

Lowest measured noise figure:

Device	Lowest Noise Figure (dB)					
	145 MHz	437 MHz	1.28 GHz	2.25 GHz	2.425 GHz	5.83 GHz
RTL-SDR	5.8	5.4	11.9	-	-	-
Airspy Mini	7.1	7.4	10.7	-	-	-
SDRplay RSPduo	2.1	2.5	5.4	-	-	-
LimeSDR Mini	11.1	13.8	15.6	-	-	-
BladeRF 2.0 micro	5.0	4.9	7.4	9.0	9.5	15.8
USRP B210	5.5	4.4	5.7	4.8	6.3	9.6
PlutoSDR	6.2	6.1	4.1	4.7	5.0	9.3

Interface, frequency range & bandwidth

Front end	Interface	Frequency range	Bandwidth	ADC resolution	IC	Mode
ADALM-Pluto	USB 2.0	325 MHz - 3.8 GHz	61.44 MHz	12 bit I/Q	Analog Devices AD9363	Full duplex
Airspy Mini	USB 2.0	24 MHz - 1.8 GHz	6 MHz	12 bit I/Q	Rafael Micro R820T2	Receive only
BladeRF	USB 3.0	300 MHz - 3.8 GHz	40 MHz	12 bit I/Q	LimeMicro LMS6002D	Full duplex
FreeSRP	USB 3.0	70 MHz - 6 GHz	61.44 MHz	12 bit I/Q	Analog Devices AD9364	Full duplex
HackRF	USB 2.0	1 MHz - 6 GHz	20 MHz	8 bit I/Q	Maxim MAX2837 / 5864	Half duplex
LimeSDR	USB 3.0/PCIe	100 kHz – 3.8 GHz	61.44 MHz	12 bit I/Q	Lime Micro LMS7002M	Full duplex
RTL-SDR	USB 2.0	24 MHz - 1.766 GHz	2.4 MHz	8 bit I/Q	Realtek RTL2832U	Receive only
SDRplay	USB 2.0	1 kHz - 2 GHz	6 (10) MHz	14 (8) bit I/Q	Mirics Msi001	Receive only
Sidekiq	Mini PCIe	70 MHz - 6 GHz	50 MHz	12 bit IQ	Analog Devices AD9361*	Full duplex
SiGe GN3S	USB 2.0	1.57542 GHz	8 MHz	4 bit I/Q	SiGe SE4110L	Receive only
USRP N210	Gigabit Ethernet	0 - 6 GHz	25 MHz (50 MHz)	14 (8) bit IQ	Analog Devices AD9361	Full duplex
USRP X310	Gigabit Ethernet	0 - 6 GHz	200 MHz	14 bit IQ	Analog Devices AD9361	Full duplex
XTRX	Mini PCIe	100 kHz – 3.8 GHz	160 MHz	14 bit IQ	Lime Micro LMS7002M	Full duplex

** Indicates best guess of internal chipset.*

https://www.gpsworld.com/innovation-the-continued-evolution-of-the-gnss-software-defined-radio/

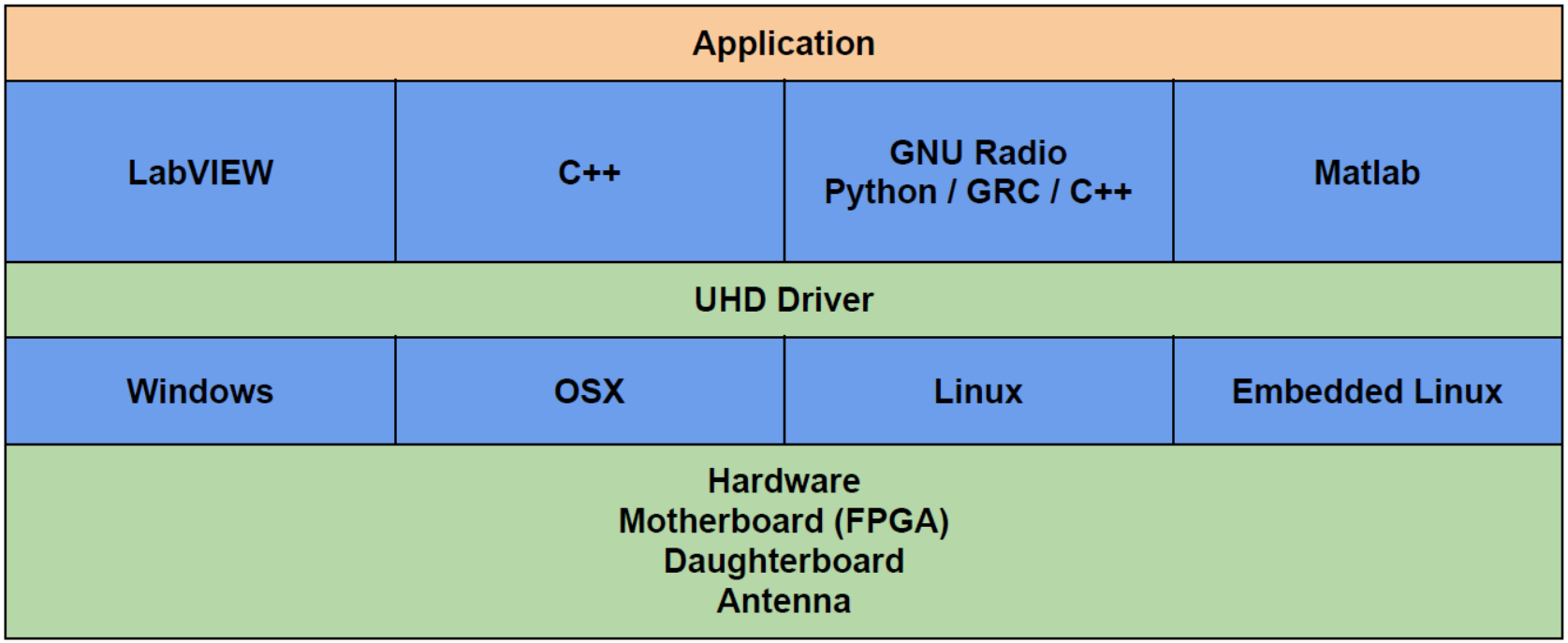
CentraleSupélec

PART VIII:

SDR SOFTWARE PLATFORMS

USRP HARDWARE DRIVER (UHD)

- Provides a single, common interface to all USRP devices
- Host-side software driver running in user-space
- Open-source and hosted on GitHub
- Cross-platform (Windows, macOS, Linux)
- Four components: host-side software, FPGA, firmware, MPM (module peripheral manager)
- UHD and RFNoC are free software and open-source projects



UHD UTILITIES

- >> uhd_find_devices (under linux)
  - Uses broadcast packets for discovery.
  - Often blocked by routers, switches, firewalls.

```
user@host:~$ uhd_find_devices
linux; GNU C++ version 4.8.4; Boost_105400; UHD_003.009.002-0-gf18abe54

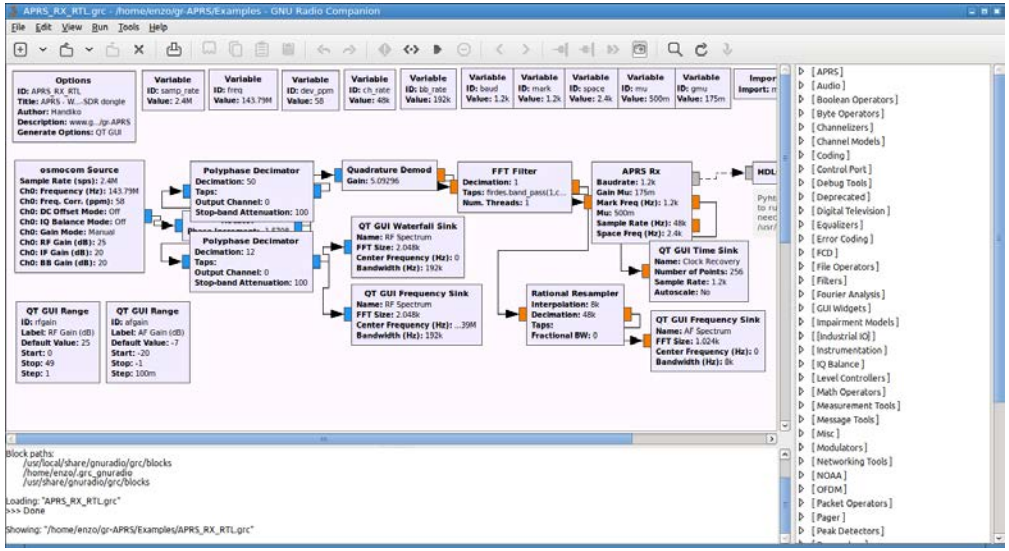
--------------------------------------------------
-- UHD Device 0
--------------------------------------------------
Device Address:
    type: usrp2
    addr: 192.168.10.2
    name:
    serial: F38688
```

- >> uhd_usrp_probe (under linux)
  - Report detailed information on UHD-supported Software Radio Peripherals attached by USB, network, or embedded configuration.

```
user@host:~$ uhd_usrp_probe
linux; GNU C++ version 4.8.4; Boost_105400; UHD_003.009.002

-- Opening a USRP2/N-Series device...
-- Current recv frame size: 1472 bytes
-- Current send frame size: 1472 bytes
  _____________________________________________________
 /
|       Device: USRP2 / N-Series Device
|     _________________________________________________
|    /
|   |       Mboard: N210r4
|   |   hardware: 2577
|   |   mac-addr: 00:80:2f:0a:d5:bd
|   |   ip-addr: 192.168.10.2
|   |   subnet: 255.255.255.255
|   |   gateway: 255.255.255.255
|   |   gpsdo: none
|   |   serial: F38688
|   |   FW Version: 12.4
|   |   FPGA Version: 11.1
|   |
|   |   Time sources: none, external, _external_, mimo
|   |   Clock sources: internal, external, mimo
|   |   Sensors: mimo_locked, ref_locked
|   |    _____________________________________________
|   |   /
|   |  |       RX DSP: 0
|   |  |   Freq range: -50.000 to 50.000 MHz
|   |    _____________________________________________
|   |   /
|   |  |       RX DSP: 1
|   |  |   Freq range: -50.000 to 50.000 MHz
|   |    _____________________________________________
|   |   /
|   |  |       RX Dboard: A
|   |  |   ID: WBX, WBX + Simple GDB (0x0053)
|   |  |   Serial: 7f708b81
```

GNU RADIO



- Founded by Eric Blossom in 2001
- Project leader Ben Hilburn
- Block-based dataflow architecture
- Each block runs in its own thread
- Open-source framework for SDR and signal processing
- Data flows through a graph called a Flowgraph
- Blocks are nodes in a Flowgraph, and perform operations and signal processing
- Signals normalized between -1.0 and +1.0
- Similar in concept to MathWorks Simulink
- Running C++ and Python under-the-hood
- Can write code directly, or use the GNU Radio Companion (GRC) graphical tool
- Hosted on GitHub at https://github.com/gnuradio/gnuradio
- Homepage is http://gnuradio.org/

UHD / GNU RADIO INSTALLATION

- Detailed step-by-step instructions for the following operating systems:
  - Ubuntu 18.04, 18.10, 19.04
  - Fedora 27, 28, 29
  - Windows 7, 8, 10
  - OS X

- https://kb.ettus.com/Building_and_Installing_the_USRP_Open-Source_Toolchain_(UHD_and_GNU_Radio)_on_Linux
- https://kb.ettus.com/Building_and_Installing_the_USRP_Open_Source_Toolchain_(UHD_and_GNU_Radio)_on_Windows
- https://kb.ettus.com/Building_and_Installing_the_USRP_Open-Source_Toolchain_(UHD_and_GNU_Radio)_on_OS_X

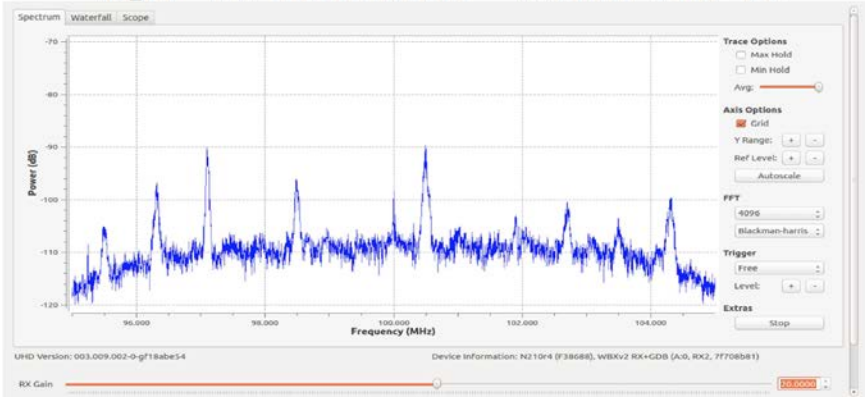
USING GNU RADIO

- Dial tone example:

```
from gnuradio import analog
from gnuradio import audio
from gnuradio import blocks
from gnuradio import eng_notation
from gnuradio import gr
from gnuradio.eng_option import eng_option
from gnuradio.filter import firdes
from optparse import OptionParser
class dial_tone_basic(gr.top_block):
    def __init__(self):
        gr.top_block.__init__(self, "Dial Tone Basic")
        ##################################################
        # Variables
        ##################################################
        self.samp_rate = samp_rate = 32000
        ##################################################
        # Blocks
        ##################################################
        self.blocks_add_xx = blocks.add_vff(1)
        self.audio_sink = audio.sink(32000, '', True)
        self.analog_sig_source_x_1 = analog.sig_source_f(samp_rate, analog.GR_COS_WAVE, 440, .4, 0)
        self.analog_sig_source_x_0 = analog.sig_source_f(samp_rate, analog.GR_COS_WAVE, 350, .4, 0)
        self.analog_noise_source_x_0 = analog.noise_source_f(analog.GR_GAUSSIAN, .005, -42)
        ##################################################
        # Connections
        ##################################################
        self.connect((self.analog_noise_source_x_0, 0), (self.blocks_add_xx, 2))
        self.connect((self.analog_sig_source_x_0, 0), (self.blocks_add_xx, 0))
        self.connect((self.analog_sig_source_x_1, 0), (self.blocks_add_xx, 1))
        self.connect((self.blocks_add_xx, 0), (self.audio_sink, 0))
    def get_samp_rate(self):
        return self.samp_rate
    def set_samp_rate(self, samp_rate):
        self.samp_rate = samp_rate
        self.analog_sig_source_x_1.set_sampling_freq(self.samp_rate)
        self.analog_sig_source_x_0.set_sampling_freq(self.samp_rate)
def main(top_block_cls=dial_tone_basic, options=None):
    tb = top_block_cls()
    tb.start()
    try:
        raw_input('Press Enter to quit: ')
    except EOFError:
        pass
    tb.stop()
    tb.wait()
if __name__ == '__main__':
    main()
```

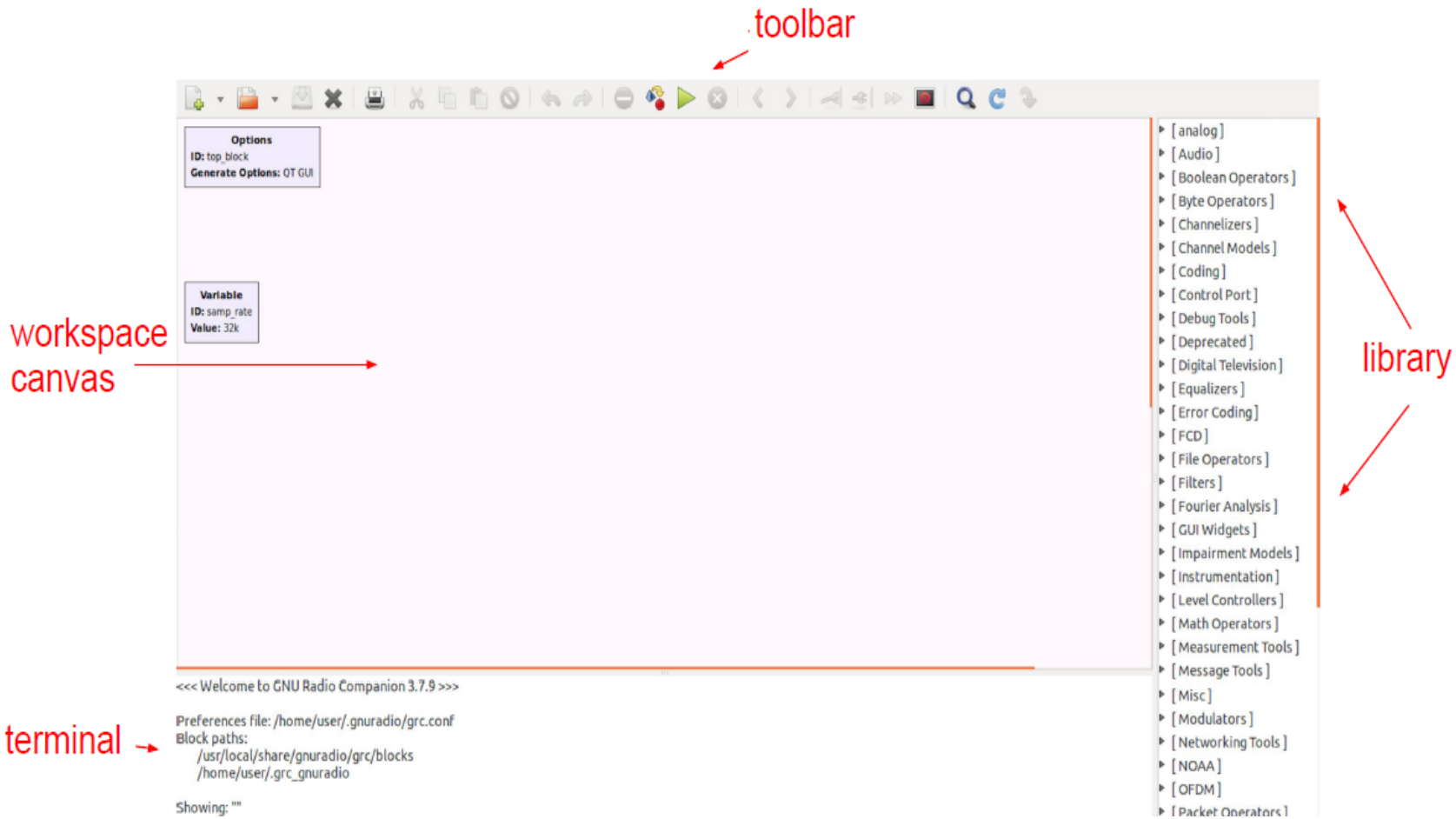
- Verifying USRP

```
uhd_fft --args "addr=192.168.10.2" --freq 100e6 -s 10e6 -g 20
```



USING GNURADIO-COMPANION

- At a command prompt, type: >> *gnuradio-companion*

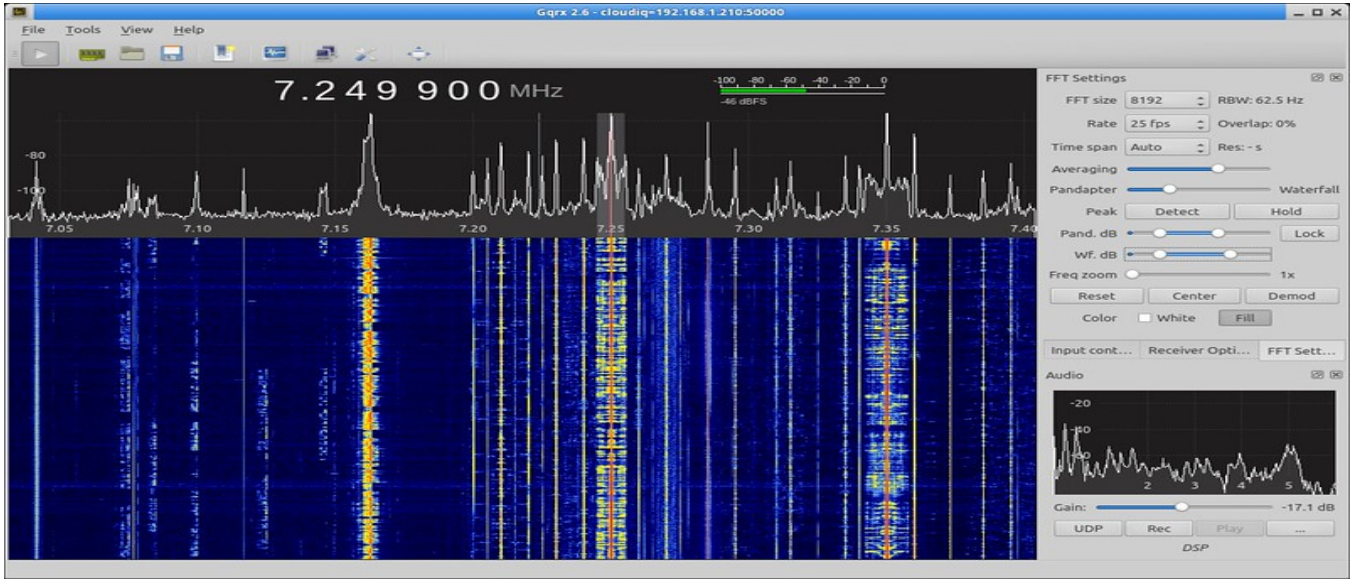


OUT-OF-TREE (OOT) MODULES

- An OOT module is a GNU Radio component that does not live within the GNU Radio source tree, and is not included with the GNU Radio distribution
- OOT modules allow third-parties to extend GNU Radio with their functions and blocks
- Comprehensive GNU Radio Archive Network (CGRAN)
  - Directory of open-source OOT modules
  - Not a hosting site
  - Most OOT modules are hosted on GitHub
  - **http://www.cgran.org/**
- **gr_modtool**
  - The Swiss army knife of module editing/creating
  - https://gnuradio.org/redmine/projects/gnuradio/wiki/OutOfTreeModules

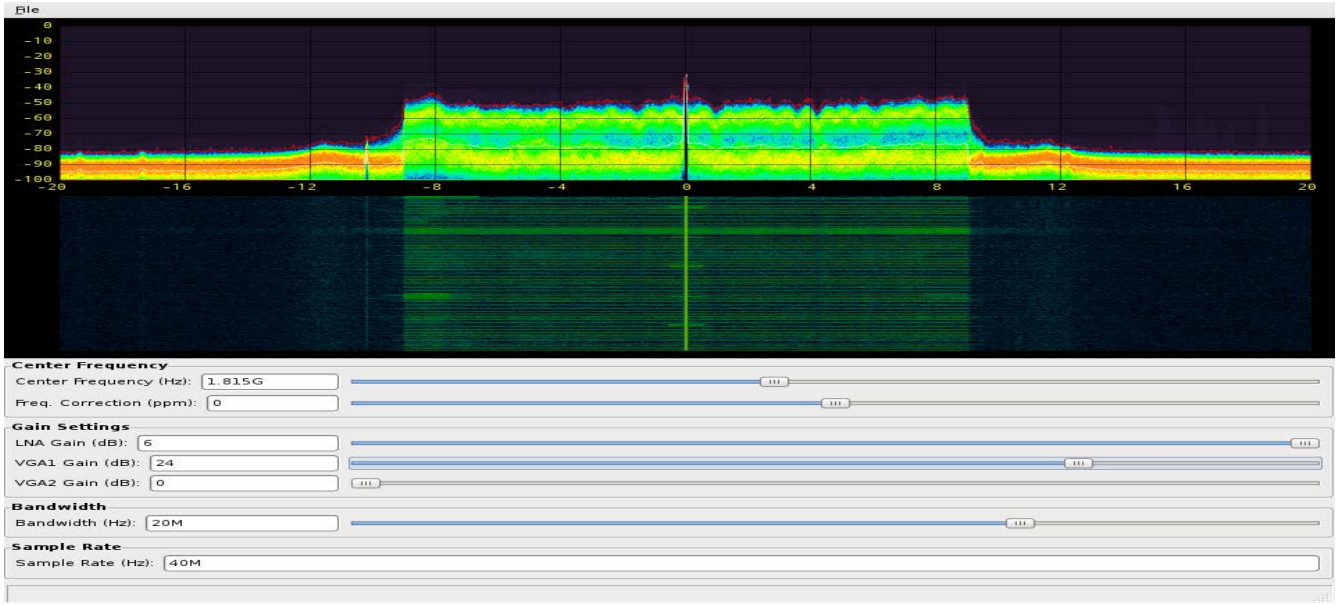
GQRX

- A free open-source SDR receiver built on GNU Radio and QT
- Features:
  - Real-time FFT plot and waterfall
  - Demodulators for AM, SSB, NBFM (mono), WBFM (stereo)
  - Record and playback to/from IQ file
  - Basic remote control through TCP socket connection
- Created by Alexandru Csete in Denmark
- https://github.com/csete/gqrx



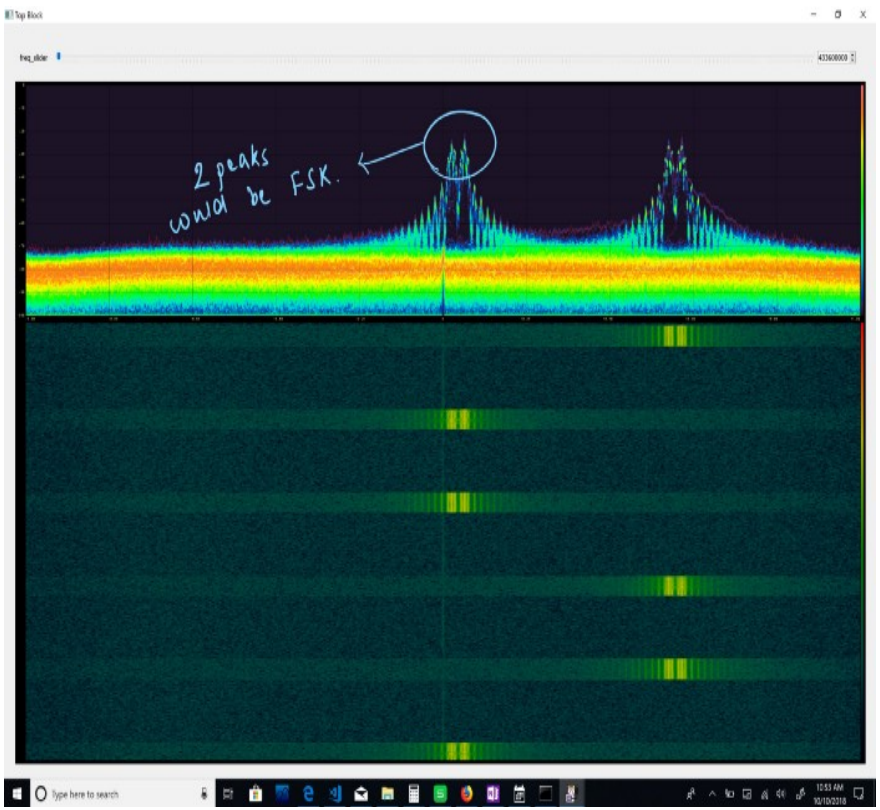
GR-FOSPHOR

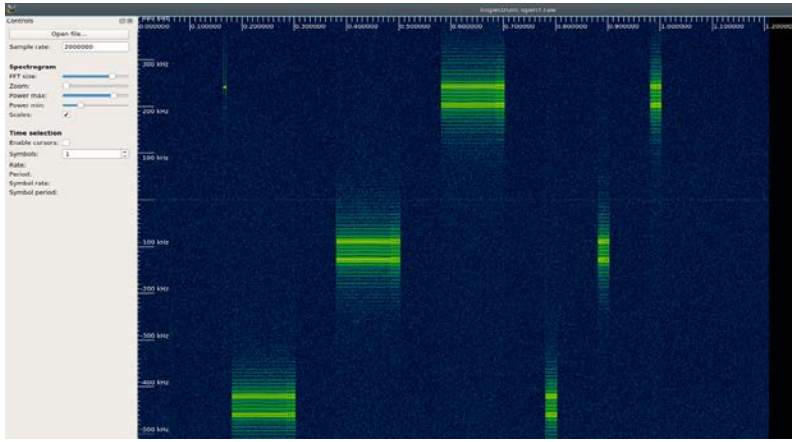
- Open-source, GPU-accelerated FFT and Waterfall display tool
- GNU Radio block for RTSA-like spectrum visualization
- Uses OpenCL and OpenGL for acceleration
- High frame rate, great for visualizing fast-changing signals
- RFNoC variant for X3xx, E3xx USRPs - FPGA accelerated FFT/Waterfall
- Created by Sylvain Munaut, **https://github.com/smunaut**
- Homepage **http://sdr.osmocom.org/trac/wiki/fosphor**
- Video demo **https://www.youtube.com/watch?v=mjD-l3GAghU**

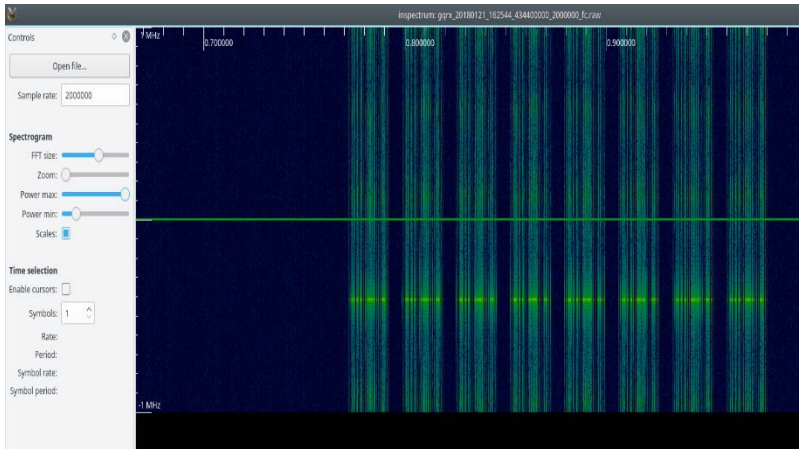


INSPECTRUM

- Off-line spectrum analysis tool
- Spectrogram with zoom
- Large file support
- **https://github.com/miek/inspectrum**







PROJECT EXAMPLES USING SDR

- FM Radio Receiver
  - http://www.ettus.com/kb/detail/sdr-for-beginners-building-an-fm-receiver-with-theusrp-and-gnu-radio
- GNSS (GPS, Glonass, Galileo) Receiver
  - https://gnuradio.org/redmine/projects/gnuradio/wiki/GlobalPositioningSystem
- GSM Base Station (OpenBTS)
  - http://openbts.org/
- Radar
  - http://www.ettus.com/application/detail/usrp-technology-for-multiband-passivebi-static-radar
- ZigBee Transceiver
  - http://www.ccs-labs.org/software/gr-ieee802-15-4/
- SrsLTE
  - https://github.com/srsLTE/srsLTE
- OpenAirInterface
  - https://gitlab.eurecom.fr/oai/openairinterface5g
- Further examples:
  - http://cgran.org

REFERENCES

- **Report/Journal/Conference papers**

[1] J. B. Johnson, “Thermal Agitation of Electricity in Conductors,” Phys. Rev., pp. 97-109, July 1928

[2] D. Abbott, B. R. Davis, N. J. Phillips and K. Eshraghian, "Simple derivation of the thermal noise formula using window-limited Fourier transforms and other conundrums," in *IEEE Transactions on Education*, vol. 39, no. 1, pp. 1-13, Feb. 1996,

[3] http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.115.5449&rep=rep1&type=pdf

- **Books**

[1] T. J. Rouphael, *Wireless Receiver Architectures and Design: Antennas, RF, Synthesizers, Mixed Signal, and Digital Signal Processing*, Elsevier Academic Press, 2014.

[2] Q. Yang and L. Huang, *Inside Radio: An Attack and Defense Guide*, Springer, first edition, 2018.